

Gamma Constants and Scaling Limits for Rounding Extinction

Cloitre's power-rounding conjecture, smooth-weight universality,
and the limits of regular variation

Part II: quantitative thresholds, inverses, and global mass profiles

Part III: microscopic schedules and averaged ceiling laws

Parts I and III: research report prepared for Vladimir Reshetnikov
with OpenAI

Part II: Report 188 (no author named)

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Abstract

For $m > 0$, start from an integer $n \geq 0$ and repeatedly replace the current value by the largest multiple of k^m below it, at stages $k = 2, 3, \dots$. We prove that the first zero occurs at

$$\tau_m(n) \sim \left[m \Gamma\left(\frac{m}{m+1}\right)^{m+1} n \right]^{1/(m+1)}.$$

This establishes the real-parameter asymptotic conjecture recorded by Benoit Cloitre in OEIS A082528 and identifies the constants conjectured numerically for A082527 and A082528. The case $m = 1$ recovers the classical Tchoukaillon asymptotic and is not a new claim. The proof reverses the rounding process, establishes a deterministic scaling limit despite a discontinuous ceiling function, and solves the limit through an explicit Gamma product. We also prove the same limit for positive weights satisfying $k(w_k/w_{k-1} - 1) \rightarrow m$, give elementary finite-product certificates for the constant, and show by an exact block construction that ordinary regular variation alone does not determine it. Exact integer computations, including rational exponents, accompany the proof. Quantitative error terms, the bounded-error conjecture in the linear case, and more general microscopic rounding schedules remain open within this report.

Status. Complete mathematical proofs are supplied here; the report is AI-assisted, unrefereed, and not formally verified. The historical search was targeted, not exhaustive.

Editorial note (7 October 2026, batch 110). Since batch 110 the report has three Parts. Part I is the report described above. Part II ([section 11](#) onwards) is built from Report 188 of a session bundle, written without sight of this report, on the same pure power process: it proves the leading law of [theorem 1.1](#) again by a second route and adds the remainder $T_m(K) = L_m K^{m+1} + O_m(K^{(m+1)^2/(m+2)})$, uniform on compact m -sets, with an explicit all-index enclosure, a global minimal-mass profile with a rate and a law for positive terminal quotients; this answers Research question 1. Part III ([section 24](#) onwards) is built from the manuscript *Microscopic Schedules in Rounding Extinction*, which starts from Research question 4: for nondecreasing weights with bounded scaled increments whose joint empirical law of increment and activity converges, it proves the extinction law with an averaged ceiling drift, finite Gamma products, a jump of the constant from $1/(2\pi)$ to $\pi/8$ under perturbations

of the weights smaller than every inverse power of the index, a schedule with no limiting constant, the exact range $(0, 1/(m+1))$ of the constants, and recovery of the law from the drift. It answers that question for this class and the sparse-exception question in its bounded form. The two Parts continue the two directions named at the end of [section 10](#). Their labels carry the prefixes **rex:qt:** and **rex:sch:**; each keeps its source's letters, with a table of readings. Dated notes in [sections 1, 2, 5, 6, 8](#) and [10](#) record what has changed. Parts II and III are AI-assisted, unrefereed and not formalized.

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Part I

Gamma constants and scaling limits for rounding extinction

1 The problem and the principal results

Repeated rounding is a simple deterministic operation with unexpectedly structured large-scale behavior. Fix a real number $m > 0$ and an integer $n \geq 0$. Define

$$x_1 = n, \quad x_k = k^m \left\lfloor \frac{x_{k-1}}{k^m} \right\rfloor \quad (k \geq 2), \quad \tau_m(n) = \min\{k \geq 1 : x_k = 0\}. \quad (1.1)$$

The value x_k is nonnegative and does not increase with k . The stopping time is finite: once $k^m > n$, the next rounding gives zero. We use the convention $\tau_m(0) = 1$. For $n > 0$, one always has $\tau_m(n) \geq 2$. When m is not an integer, the intermediate values can be real; the quotient x_k/k^m is nevertheless an integer for every $k \geq 2$.

The sequences for $m = 1, 2, 3$ are respectively [A073047](#), [A082527](#), and [A082528](#). Cloitre's entry for the cubic case also asks for the asymptotic law for every real $m > 0$ [[1](#), [2](#), [3](#)]. The following theorem determines that law and its constant.

Theorem 1.1 (Power-rounding extinction). *For every fixed real $m > 0$, set*

$$c_m = m \Gamma\left(\frac{m}{m+1}\right)^{m+1}, \quad L_m = \frac{1}{c_m}. \quad (1.2)$$

Then, as the integer n tends to infinity,

$$\tau_m(n) \sim (c_m n)^{1/(m+1)}. \quad (1.3)$$

Equivalently, if $T_m(K)$ is the smallest integer initial value that survives through stage K , then

$$T_m(K) \sim L_m K^{m+1}. \quad (1.4)$$

In particular,

$$\tau_2(n) \sim \left(2\Gamma(2/3)^3 n\right)^{1/3}, \quad (1.5)$$

$$\tau_3(n) \sim \left(3\Gamma(3/4)^4 n\right)^{1/4}. \quad (1.6)$$

The constants are

$$c_2 = 4.965917162430455473\dots, \quad c_3 = 6.764822981008760234\dots \quad (1.7)$$

These identify the values $4.96\dots$ and $6.76\dots$ conjectured numerically in the two OEIS entries. The specialization $c_1 = \pi$ is the classical Tchoukaillon constant, whose prior history is discussed in [section 2](#).

The same proof identifies more than a stopping index. Define the crossing times

$$t_0 = 1, \quad t_j = \prod_{r=1}^j \left(1 - \frac{1}{(m+1)r}\right) \quad (j \geq 1), \quad (1.8)$$

and the continuous function $y_m : (0, 1] \rightarrow [0, \infty)$ by

$$y_m(t) = \frac{(m+1)j-1}{m} t_{j-1} - jt, \quad t_j \leq t \leq t_{j-1}. \quad (1.9)$$

The definitions agree at each shared endpoint. Set

$$F_m(t) = c_m t^m y_m(t) \quad (0 < t \leq 1), \quad F_m(0) = 1. \quad (1.10)$$

Theorem 1.2 (The full extinction trajectory). *Let $J = \tau_m(n)$ for $n > 0$, and put*

$$X_n(t) = \frac{x_{\max(1, \lfloor Jt \rfloor)}}{n}, \quad 0 \leq t \leq 1.$$

Then

$$\sup_{0 \leq t \leq 1} |X_n(t) - F_m(t)| \rightarrow 0. \quad (1.11)$$

Thus the fraction of the initial value remaining, plotted against the fraction of the extinction time elapsed, has an explicit deterministic limiting shape.

Our extension to more general weights uses a local condition on their successive ratios. This condition is stronger than specifying only their leading growth.

Theorem 1.3 (Smooth-weight universality). *Suppose $w_1 = 1$, $w_k > 0$, and*

$$k \left(\frac{w_k}{w_{k-1}} - 1 \right) \rightarrow m > 0. \quad (1.12)$$

Replace k^m in [equation \(1.1\)](#) by w_k , and denote the corresponding threshold and stopping time by $T_w(K)$ and $\tau_w(n)$. Then

$$T_w(K) \sim L_m K w_K, \quad L_m \tau_w(n) w_{\tau_w(n)} \sim n. \quad (1.13)$$

With $J = \tau_w(n)$, the normalized forward trajectory converges uniformly to the same F_m in [equation \(1.10\)](#).

Theorem 1.3 allows logarithmic factors, shifts, and many small perturbations. Its hypothesis cannot simply be replaced by ordinary regular variation: [section 8](#) constructs weights $w_k \sim (k/r)^m$ for which the threshold constant is L_m/r .

1.1 What has and has not been resolved

Question or claim	Status in this report
A082528, every real $m > 0$	The stated leading-equivalence conjecture is proved; c_m is explicit.
A082527 and A082528 constants	The square and cube constants are determined by equations (1.5) and (1.6) .
A073047 leading equivalent	Recovered as a classical specialization, with no novelty claim.
A073047 bounded error	The stronger assertion $\tau_1(n) = \sqrt{\pi n} + O(1)$ is not proved here.
Full trajectories and smooth weights	Proved in theorems 1.2 and 1.3 .
Quantitative discrete errors	No rate is inferred from the compactness argument; these are explicit research questions.
Historical priority	No matching general power theorem was found in the targeted source search; exhaustive priority is not certified.

All limits involving m in the first three theorems hold for fixed $m > 0$. They make no assertion uniform in a moving parameter $m = m(n)$.

[write] **Added 7 October 2026, batch 110.** The table describes Part I. Two of its rows have since been advanced by the later Parts. “Quantitative discrete errors”: Part II proves $T_m(K) = L_m K^{m+1} + O_m(K^{(m+1)^2/(m+2)})$ and $\tau_m(n) = (c_m n)^{1/(m+1)} + O_m(n^{1/((m+1)(m+2))})$, uniformly on compact m -sets, with an explicit enclosure at every index ([theorems 12.1](#) and [17.1](#)); the bounded error of A073047 remains unproved. “Full trajectories and smooth weights”: Part III proves the extinction law for every nondecreasing weight schedule whose bounded scaled increments have a convergent joint law with their activity, which contains [theorem 1.3](#) for nondecreasing weights ([theorem 27.2](#), [remark 24.1](#)). The leading law of [theorem 1.1](#) is not new in Part II; it is proved there by a second route.

2 Historical context and the ProveIt starting point

2.1 The OEIS questions

Cloitre introduced the square and cube stopping sequences in April 2003. The entry for A082528 explicitly asks whether the first zero for real powers k^m has order $(c(m)n)^{1/(m+1)}$. A082527 records the numerical constant $4.96\dots$, and A082528 records $6.76\dots$ [[2](#), [3](#)]. The entries were inspected during the source audit for this report. Their shared site-footer time is not treated as an individual sequence revision date.

The square entry contains a minor arithmetic slip in its example for $n = 10$: the correct intermediate value is $4\lfloor 10/4 \rfloor = 8$. The reported stopping index 3 is correct. The definitions, programs, and initial terms agree with [equation \(1.1\)](#); the slip has no effect on the theorem.

2.2 The classical linear sieve

When $m = 1$, the backward threshold is A002491. Starting with K , one successively rounds upward to a multiple of $K - 1, K - 2, \dots, 1$. Its first values are

$$1, 2, 4, 6, 10, 12, 18, 22, 30, 34, \dots$$

This is a standard description of the Tchoukaillon or Mancala sieve [[4](#)]. Erdős and Jabotinsky proved

$$T_1(K) = \frac{K^2}{\pi} + O(K^{4/3}) \tag{2.1}$$

and proposed the sharper $O(K)$ error [[5](#)]. Their paper already analyzes successive ranges of equal integer increments. The use of such ranges and a product limit in the present proof belongs to that established method.

There is a relevant caution about error terms. Broline and Loeb later claimed the sharper threshold estimate [[6](#)]. Knuth’s 2021 comment recorded in A002491 identifies an accumulation-of-errors problem in that argument and explains why its stated estimates do not establish the claimed improvement [[4](#)]. We use neither that sharpening nor any consequence of it. The classical [equation \(2.1\)](#) would imply

$$\tau_1(n) = \sqrt{\pi n} + O(n^{1/6}),$$

whereas the $O(1)$ error recorded as presumptive in A073047 is a stronger assertion.

Brown’s exposition obtains the ordinary constant from successive parabolic pieces and a Wallis product [[7](#)]. A 2023 answer by Thomas Lehéricy studies a different generalization, rounding upward to the r th admissible multiple, and obtains a ratio of Gamma functions [[8](#)]. Its use of fixed increment regimes and a final truncation estimate is methodologically relevant.

Its parameter changes the rounding rule; the parameter m here changes the moduli themselves. The two statements concern different recurrences.

We also checked Li’s 2026 preprint on the two-dimensional Tchoukaillon array [9]. Its main theorem concerns a uniform quadratic approximation to array entries. It does not state the real-power extinction theorem proved here. None of our proofs depends on that preprint or on its discussion of sharp edge estimates.

2.3 Repository audit and choice of target

The research began with the current ProveIt repository [12]. Targeted searches and directory inspection showed extensive recent work on OEIS asymptotics, including Bell-scale growth, derivative expansion supports, and balanced-word enumeration. Those established results were taken into account when choosing a question with a clear remaining mathematical claim.

The repository search was indexed at revision

`ce37e13f4aa16819c87c3ccc611362d15758f2ac`.

Exact searches for A082527, A082528, A073047, A002491, and Tchoukaillon returned no matching treatment. We also inspected the 59-entry `oeis-sequence-asymptotics` directory and the 58-entry `series-and-transseries` directory at that revision. This is a bounded nonduplication check; it is not a claim to have read every file, binary, branch, or historical version of the repository.

The present article extends the repository’s research program through a new target. It is mathematically self-contained and assumes none of its unrefereed reports as a theorem. The substantive contribution claimed here is the real-parameter power theorem, its explicit constant, and the trajectory and weight extensions. The classical linear case and the increment-regime method are credited to the preceding literature.

[write] Added 3 October 2026, batch 85. This report is a single manuscript, printed in full: manuscript 05 of batch 85 of ProveIt’s incoming-reports intake, delivered as the archive `Rounding_Extinction_OEIS.zip` (wrapper directory `Rounding_Extinction/`) in commit `317c1ce2e`, and placed in commit `713149ded` in the research-report collection, among the OEIS-sequence asymptotics reports, as the report `a082528-rounding-extinction`. Its main file `article.tex` and the `\input` files `body.tex`, `extensions.tex`, `computation.tex`, `questions.tex` and `bibliography.tex` keep their delivered names. The author line and the PDF metadata read “Research report prepared for Vladimir Reshetnikov with OpenAI”: the package names OpenAI as the tool that prepared it and names no human author.

Pin and repository claims. The revision quoted above, `ce37e13f4`, is the ProveIt commit the manuscript was written against. Its two directory counts are true at that commit. At placement the intake repeated the search over the whole tracked tree (all reports, the `Oeis`, number-theory and combinatorics projects, and the Lean and Rocq developments) for A073047, A082527, A082528, A002491, Tchoukaillon, Mancala, Jabotinsky and Broline and found no treatment: this report continues no repository material and has no neighbouring report. No statement here is formalized in Lean or Rocq, and placement in the collection confers no formal status.

Credit for $m = 1$. The linear case, $c_1 = \pi$, $T_1(K) \sim K^2/\pi$ and $\tau_1(n) \sim \sqrt{\pi n}$, is classical: the Tchoukaillon (Mancala) sieve of A002491 and the theorem of Erdős and Jabotinsky [5]. The report claims nothing new for it, nor for the increment-regime method; what it claims is the case $m \neq 1$ (real $m > 0$), the constants of equations (1.5) and (1.6), and the extensions of theorems 1.2 and 1.3 and sections 6 and 8.

OEIS facts checked by the intake. On 3 October 2026 the intake read the entries on `oeis.org` and confirmed the manuscript’s statements about them: A082528 records Cloitre’s conjecture for real $m > 0$ and the constant $6.76\dots$; A082527 records $4.96\dots$, and its example for $n = 10$ indeed prints the intermediate value 2 where $4\lfloor 10/4 \rfloor = 8$ is meant (the stopping index 3 is right); and A002491

carries Knuth’s 2021 comment that the error terms of Broline and Loeb accumulate, so that their argument does not give the claimed $O(K)$ bound. An independent integer implementation written at intake reproduced $T_1(1), \dots, T_1(10)$ as the A002491 prefix and every fixture term of the three stopping sequences, and gave $c_m T_m(3000)/3000^{m+1} = 0.999998, 1.00048, 1.00052$ for $m = 1, 2, 3$. The intake did not search the external literature for an earlier theorem for general m ; the novelty statement remains the manuscript’s, bounded as it says by a targeted search.

Notation. The manuscript reuses some letters with local meanings. Every symbol is printed as delivered; the table fixes each meaning, with the tempting false readings.

symbol	meaning here (and what it is <i>not</i>)
y_K, y_m	y_K is the interpolated backward array at terminal stage K (4.1); y_m is the limiting profile for the exponent m (1.9). The curve y_3 of figure 2 is the $m = 3$ limit, not the array at $K = 3$.
$T_m(K), T_w(K)$	the threshold for the exponent m , or for the weight sequence w ; in $T_3(4) = 88$ the subscript is m and the argument K .
$q_k^{(K)}, Q_k, Q; q, v$	$q_k^{(K)}$ is the backward quotient array (definition 3.1, and (7.5) for weights); $Q_k = x_k/k^m$ (or x_k/w_k) are the forward quotients; Q in lemma 3.2 is any integer quotient. In section 9, $m = p/q$ and the exponent q in (9.2) is that denominator, not a quotient; the quotient there is $v = q_k$.
a, a_k	$a = m/(m+1)$ in sections 5 and 6; $a_k = (k/(k-1))^m - 1$ (3.4) is the relative increment at stage k , not a term of a sequence tending to a . A generic $a > 1$ appears in the proof of corollary 3.3 and in R_a of proposition 8.1, and constants $a < \lambda^{1/(m+1)} < b$ in the proof of corollary 7.3.
b_k, b	$b_k = w_k/w_{k-1} - 1$ is the weighted analogue of a_k (section 7); $b \in \mathbb{R}$ is the logarithmic exponent of corollary 7.2.
t, t_j	$t \in [0, 1]$ is the fraction of the horizon; t_j (1.8) are the crossing times, decreasing from $t_0 = 1$, not the grid points j/K .
$c_m, L_m, C(u)$	c_m is the extinction constant and $L_m = 1/c_m$ the threshold constant ($T_m(K) \sim L_m K^{m+1}$); $C(u) = \max\{1, [u]\}$ (4.5) is the positive ceiling function, not a constant.
f_m, F_m	$f_m = t^m y_m$ tends to L_m at 0; $F_m = c_m f_m$ tends to 1 there.
r	the block length in proposition 8.1 (and $r = 3$ in the checks of section 9); elsewhere a summation or product index, the integer $r \geq 2$ in the proof of lemma 4.1, and the integer q th root in section 9.
$\delta, \delta_u, \delta_m$	δ is a cut-off parameter; δ_u is the unit mass at u in (6.5); δ_m is an error exponent in Research question 1.
w_k, v_k	the weights of theorem 1.3; v_k are the comparison weights after proposition 8.1.
z, ρ_m, J	$z(t) = my(t)/t$, and a complex variable in the proof of proposition 6.3; ρ_m is the loss density of corollary 6.2, not a radius; $J = \tau_m(n)$ (or $\tau_w(n)$) is the stopping index of the trajectory at hand.

The writing step prefixed every label with `rex:` (in `body.tex`, `extensions.tex`, `computation.tex` and `questions.tex`, every `\cref`, `\ref` and `\eqref` updated) and added three dated [write] notes: this one, one in section 9 on the shipped layout, and one in section B on the OEIS draft. It also added cross-reference aliases to the preamble, so that a current L^AT_EX prints “lemma 3.2” rather than “theorem 3.2” as the delivered PDF did, and set the double hyphens of one command line in section 9 as hyphens. No statement, proof or number of the manuscript was altered.

[write] **Added 7 October 2026, batch 110.** The note above describes Part I as it was written in batch 85. Since batch 110 the report has two further Parts, from two later manuscripts (Part II from Report 188, arrival 60f54ea06; Part III from `Periodic_Rounding_Extinction.zip`, arrival e4d5dcf9e; both placed in 8622ca7e5); their provenance is in sections 11.1 and 24.1. Part III’s manuscript was written against this Part (pin 52d8ca404); Report 188 was written without it. They are the only repository material on this subject, and neither changes the statements above about the repository at the pin. Nothing in any Part is formalized.

3 Exact reversal: extinction as a survival threshold

The forward process has a floor at every step and an initial value that changes with the question being asked. Fixing a terminal stage instead gives a convenient triangular recurrence.

Definition 3.1 (Backward quotient array). For an integer $K \geq 1$, define

$$q_K^{(K)} = 1, \quad q_{k-1}^{(K)} = \left\lceil \left(\frac{k}{k-1} \right)^m q_k^{(K)} \right\rceil \quad (2 \leq k \leq K), \quad (3.1)$$

and put $T_m(K) = q_1^{(K)}$. We suppress the superscript K when the terminal stage is fixed.

Lemma 3.2 (Survival duality). *For every integer $n \geq 0$ and $K \geq 1$,*

$$\tau_m(n) > K \iff n \geq T_m(K). \quad (3.2)$$

More generally, a value $k^m Q$ at stage $k \leq K$, with Q a nonnegative integer, survives through stage K if and only if $Q \geq q_k^{(K)}$.

Proof. At stage K , survival means a positive multiple of K^m , so its smallest integer quotient is 1. Suppose the least quotient that survives from stage k has been identified as q_k . An input $(k-1)^m Q$ produces an output at least $k^m q_k$ exactly when

$$\left\lceil \frac{(k-1)^m Q}{k^m} \right\rceil \geq q_k \iff Q \geq \frac{k^m q_k}{(k-1)^m}.$$

The least admissible integer Q is the ceiling in [equation \(3.1\)](#). Backward induction reaches stage 1, where the initial value n is already an integer and $1^m = 1$. $\square$

Corollary 3.3 (Exact inverse and strict increase). *The sequence $T_m(K)$ is strictly increasing and unbounded. For $n > 0$,*

$$\tau_m(n) = 1 + \max\{K \geq 1 : T_m(K) \leq n\}. \quad (3.3)$$

Equivalently, $\tau_m(n) = J$ if and only if $T_m(J-1) \leq n < T_m(J)$.

Proof. For terminal stage $K+1$, the quotient at stage K is $\lceil ((K+1)/K)^m \rceil \geq 2$, whereas it is 1 for terminal stage K . Every map $Q \mapsto \lceil aQ \rceil$ with $a > 1$ is strictly increasing on the integers. Reversing further therefore gives $T_m(K+1) > T_m(K)$. Formula (3.3) follows from [lemma 3.2](#). $\square$

3.1 The two estimates that control all rounding errors

Write

$$a_k = \left(\frac{k}{k-1} \right)^m - 1 = \frac{m}{k} + O_m(k^{-2}). \quad (3.4)$$

Because q_k is an integer,

$$q_{k-1} - q_k = \lceil a_k q_k \rceil \geq 1. \quad (3.5)$$

Consequently,

$$q_k \geq K - k + 1. \quad (3.6)$$

The inequality remains significant after scaling, even though $q_K/K \rightarrow 0$.

Multiplying the ceiling inequality by $(k-1)^m$ gives the second estimate:

$$0 \leq (k-1)^m q_{k-1} - k^m q_k < (k-1)^m. \quad (3.7)$$

Telescoping in either direction yields

$$0 \leq T_m(K) - k^m q_k < \sum_{j=1}^{k-1} j^m, \quad (k \geq 2), \quad (3.8)$$

$$k^m q_k \leq K^m + \sum_{j=k}^{K-1} j^m \leq K^m + \frac{K^{m+1}}{m+1}. \quad (3.9)$$

For $k = 1$, the difference in [equation \(3.8\)](#) is exactly zero. The last inequality follows by comparing the increasing function x^m with its integral over $[1, K]$.

[Equation \(3.8\)](#) is the key matching estimate: after a macroscopic part of the backward trajectory has been computed, the entire remaining part changes its weighted value by at most an explicit sum. No assumption on the distribution of fractional parts is involved.

Example 3.4 (A cubic threshold). For $m = 3$ and $K = 4$, the backward quotients are

$$q_4 = 1, \quad q_3 = 3, \quad q_2 = 11, \quad q_1 = 88.$$

Thus $T_3(4) = 88$. The initial value 88 rounds through 88, 88, 81, 64, 0, so its stopping time is 5. The value 87 rounds through 87, 80, 54, 0 and stops at 4. This illustrates the strict inequality $\tau_m(n) > K$ in the survival convention.

4 A rigorous scaling limit for the backward array

For fixed K , define y_K by linear interpolation through

$$y_K(k/K) = q_k^{(K)}/K, \quad 1 \leq k \leq K. \quad (4.1)$$

Its domain is $[1/K, 1]$. The limit will be considered on compact subintervals of $(0, 1]$; the singular endpoint 0 is handled afterward using [equation \(3.8\)](#).

Lemma 4.1 (Compactness away from zero). *For every $0 < \delta < 1$, the restrictions of y_K to $[\delta, 1]$ are uniformly bounded and uniformly Lipschitz for all sufficiently large K . Every sequence of terminal stages tending to infinity has a subsequence whose interpolants converge locally uniformly on $(0, 1]$.*

Proof. For $k \geq \delta K/2$, [equation \(3.9\)](#) gives

$$\frac{q_k}{K} \leq \left(\frac{K}{k}\right)^m \left(\frac{1}{K} + \frac{1}{m+1}\right),$$

which is bounded independently of K . On an interpolation interval $((k-1)/K, k/K)$, its slope is

$$y'_K(t) = q_k - q_{k-1} = -\lceil a_k q_k \rceil. \quad (4.2)$$

The expansion in [equation \(3.4\)](#), together with $q_k = O_\delta(K)$ and $k \geq \delta K/2$, bounds this slope uniformly. The small enlargement to $\delta/2$ handles the single grid interval containing δ . Arzelà–Ascoli on $[1/r, 1]$, followed by a diagonal subsequence over integers $r \geq 2$, gives the stated compactness. $\square$

4.1 Selecting the nonzero terminal solution

Let y be any subsequential limit. Since $y_K(1) = 1/K$,

$$y(1) = 0. \quad (4.3)$$

Equation (3.5) implies, including between grid points,

$$y(s) - y(t) \geq t - s \quad (0 < s < t \leq 1). \quad (4.4)$$

In particular, $y(t) \geq 1 - t > 0$ for $t < 1$.

This is essential. If one wrote only $y' = -\lceil my/t \rceil$ with $y(1) = 0$, the identically zero function would solve that terminal-value problem. The positive integer quotient $q_K = 1$ forces a drift of at least one at each reversed step and rules out that spurious solution. We express this selection by the positive ceiling

$$C(u) = \max\{1, \lceil u \rceil\}, \quad u \geq 0. \quad (4.5)$$

For positive u , it is the ordinary ceiling.

Lemma 4.2 (Passing through the ceiling discontinuities). *Every subsequential limit is locally absolutely continuous and satisfies*

$$y'(t) = -C\left(\frac{my(t)}{t}\right) \quad \text{for almost every } t \in (0, 1), \quad (4.6)$$

together with equations (4.3) and (4.4).

Proof. Local Lipschitz continuity has already been proved. Define $z(t) = my(t)/t$. For $0 < s < t \leq 1$, equation (4.4) gives

$$\frac{y(s)}{s} - \frac{y(t)}{t} \geq y(t) \left(\frac{1}{s} - \frac{1}{t} \right) + \frac{t-s}{s} > 0.$$

Thus z is continuous and strictly decreasing as t increases. It is bounded on $[\delta, 1]$, so it assumes an integer value at only finitely many points in that interval.

Away from these exceptional points and $t = 1$, let $k = \lceil Kt \rceil$. The ratio expansion in equation (3.4) and local uniform convergence show

$$a_k q_k \longrightarrow \frac{my(t)}{t} = z(t).$$

The ceiling is continuous at the noninteger limit, so its value converges to $C(z(t))$. Excluding also the countable union of grid points, equation (4.2) therefore converges to the claimed right-hand side. The slope bounds from lemma 4.1 permit dominated convergence in

$$y_K(t) = \frac{1}{K} - \int_t^1 y'_K(s) ds.$$

It follows that

$$y(t) = \int_t^1 C\left(\frac{my(s)}{s}\right) ds,$$

which proves equation (4.6). $\square$

The argument does not assume uniform convergence of ceiling functions. It proves convergence away from a finite exceptional set on each compact interval and then uses an integrable uniform bound. That distinction is what makes the limiting equation rigorous.

4.2 Solving the increment regimes

Proposition 4.3 (Explicit profile and uniqueness). *The function satisfying equations (4.3), (4.4) and (4.6) is unique. It is precisely y_m in equation (1.9); consequently*

$$y_K \longrightarrow y_m \quad \text{uniformly on every } [\delta, 1], \quad \delta > 0. \quad (4.7)$$

Proof. The function $z(t) = my(t)/t$ decreases strictly from infinity to zero as t increases from zero to one. Its divergence at zero follows from $y(t) \geq 1 - t$. Therefore every positive integer j is attained at a unique crossing time t_j .

On (t_j, t_{j-1}) , one has $j - 1 < z(t) < j$ and hence $y'(t) = -j$. Using $y(t_{j-1}) = (j - 1)t_{j-1}/m$ gives

$$y(t) = \frac{j-1}{m}t_{j-1} + j(t_{j-1} - t).$$

At the next crossing, $y(t_j) = jt_j/m$, so

$$(m+1)jt_j = ((m+1)j-1)t_{j-1}.$$

This is exactly the product recurrence in equation (1.8), and the formula for y is equation (1.9). Since $t_0 = 1$, there is no undetermined constant. Every subsequential limit is the same. If convergence of the whole family failed on some compact interval, a failing subsequence would have a further convergent subsequence by lemma 4.1, contradicting this uniqueness. $\square$

For example, the last portion of every limiting trajectory is $y_m(t) = 1 - t$ on $m/(m+1) \leq t \leq 1$. The next portion has slope -2 , the next slope -3 , and so on. These slopes describe the integer quotient changes; the weighted remaining value is $t^m y_m(t)$.

5 The Gamma constant and the extinction law

5.1 Evaluating the crossing product

Write $a = m/(m+1)$, so $0 < a < 1$. The Gamma functional equation gives

$$t_j = \frac{\Gamma(j+a)}{\Gamma(a)\Gamma(j+1)}. \quad (5.1)$$

The needed asymptotic is

$$t_j \sim \frac{j^{-1/(m+1)}}{\Gamma(m/(m+1))}. \quad (5.2)$$

For completeness, this can be proved without invoking an asymptotic expansion for Γ . The beta integral satisfies

$$B(a, j) = \int_0^1 u^{a-1}(1-u)^{j-1} du = \frac{\Gamma(a)\Gamma(j)}{\Gamma(j+a)}, \quad t_j = \frac{1}{jB(a, j)}.$$

After the substitution $v = ju$,

$$j^a B(a, j) = \int_0^j v^{a-1}(1-v/j)^{j-1} dv \longrightarrow \Gamma(a).$$

Indeed, for $j \geq 2$ the integrand, extended by zero to $[j, \infty)$, is bounded by $v^{a-1}e^{-v/2}$ and converges pointwise to $v^{a-1}e^{-v}$. Dominated convergence proves the limit and equation (5.2).

Lemma 5.1 (The small-time weighted limit). *The function $f_m(t) = t^m y_m(t)$ has a finite limit at zero, and*

$$\lim_{t \downarrow 0} f_m(t) = \frac{1}{m\Gamma(m/(m+1))^{m+1}} = L_m. \quad (5.3)$$

Moreover,

$$-t^m \leq f'_m(t) \leq 0 \quad \text{almost everywhere.} \quad (5.4)$$

Proof. Writing $z = my_m/t$ and applying [equation \(4.6\)](#),

$$f'_m(t) = t^m(z(t) - C(z(t))).$$

For $z \geq 0$, the difference belongs to $[-1, 0]$. This proves [equation \(5.4\)](#). Since t^m is integrable on $(0, 1)$, f_m has a finite limit at zero. At a crossing time,

$$f_m(t_j) = \frac{j}{m} t_j^{m+1} \longrightarrow \frac{1}{m\Gamma(m/(m+1))^{m+1}},$$

by [equation \(5.2\)](#). These crossing times tend to zero, so their limit is the full limit of f_m . $\square$

5.2 Matching to the unscaled initial value

Proof of [theorem 1.1](#). Fix $0 < \delta < 1$ and set $k = \lfloor \delta K \rfloor$. The backward profile theorem implies

$$\frac{k^m q_k}{K^{m+1}} = \left(\frac{k}{K}\right)^m \frac{q_k}{K} \longrightarrow f_m(\delta).$$

The elementary Riemann sum gives

$$\frac{1}{K^{m+1}} \sum_{j=1}^{k-1} j^m \longrightarrow \frac{\delta^{m+1}}{m+1}.$$

Divide [equation \(3.8\)](#) by K^{m+1} to obtain

$$\begin{aligned} f_m(\delta) &\leq \liminf_{K \rightarrow \infty} \frac{T_m(K)}{K^{m+1}} \\ &\leq \limsup_{K \rightarrow \infty} \frac{T_m(K)}{K^{m+1}} \leq f_m(\delta) + \frac{\delta^{m+1}}{m+1}. \end{aligned}$$

Now let $\delta \downarrow 0$ and use [lemma 5.1](#). This proves [equation \(1.4\)](#). The two limits were taken in the specified order; no uniform approximation near the singular endpoint has been assumed.

For $J = \tau_m(n)$, the exact inverse gives $T_m(J-1) \leq n < T_m(J)$. As $n \rightarrow \infty$, necessarily $J \rightarrow \infty$, and both endpoints are asymptotic to $L_m J^{m+1}$. Thus $n \sim L_m J^{m+1}$, which is [equation \(1.3\)](#). $\square$

[write] **Added 7 October 2026, batch 110.** Part II proves [theorem 1.1](#) by a second route, through the monotone coordinate I_p and dyadic telescoping in place of compactness, and obtains the rates $O_m(K^{(m+1)^2/(m+2)})$ for $T_m(K)$ and $O_m(n^{1/((m+1)(m+2))})$ for $\tau_m(n)$, uniformly on compact m -sets ([theorem 12.1](#)). Its profile H_p is f_m , its products P_r are the crossing times t_r , and its [proposition 16.3](#) is the integral of (5.4) at every t ([remark 16.4](#)); its [theorem 18.1](#) adds a rate to (4.7).

5.3 The whole forward trajectory

Proof of theorem 1.2. Let $Q_k = x_k/k^m$ for $k \geq 2$, and set $Q_1 = n$. If $J = \tau_m(n)$, the forward trajectory survives stage $J - 1$ but not stage J . The stagewise form of lemma 3.2 therefore gives

$$q_k^{(J-1)} \leq Q_k < q_k^{(J)} \quad (1 \leq k \leq J-1). \quad (5.5)$$

Both bounding arrays have the same limiting shape, including after replacing the terminal scale $J - 1$ by J . Their local uniform convergence and the continuity of y_m give

$$\frac{Q_{\lfloor Jt \rfloor}}{J} \longrightarrow y_m(t) \quad \text{locally uniformly for } 0 < t < 1.$$

Combining this with $n \sim L_m J^{m+1}$ proves $X_n(t) \rightarrow c_m t^m y_m(t)$ for $0 < t < 1$. At the endpoints, $X_n(0) = 1$ and $X_n(1) = 0$. The limit F_m is continuous on $[0, 1]$ by lemma 5.1.

To promote pointwise convergence to uniform convergence, choose a finite partition of $[0, 1]$ on whose subintervals the oscillation of F_m is less than a prescribed ε . For large n , convergence holds within ε at every partition point. Since each X_n is nonincreasing, its values between consecutive partition points lie between its endpoint values. The resulting error is at most 2ε throughout the interval. This proves equation (1.11). $\square$

The law describes the entire normalized loss curve, even at its early stages where the unweighted quotient $y_m(t)$ diverges. That endpoint is regular after multiplication by t^m .

[write] **Added 7 October 2026, batch 110.** Part II's retention limit (19.8) is this theorem reparametrized by $\lambda = L_m t^{-(m+1)}$, and follows from it (remark 19.2). No rate for (1.11) is known (Research question 7).

6 Finite certificates and the dependence on the exponent

6.1 Finite products bracket the exact constant

The constant in equation (1.2) is expressed by a standard special function, but it also admits rigorous elementary enclosures.

Proposition 6.1 (Product certificates). *For every $m > 0$ and integer $j \geq 1$,*

$$\frac{j}{m} t_j^{m+1} \leq L_m \leq \left(\frac{j}{m} + \frac{1}{m+1} \right) t_j^{m+1}. \quad (6.1)$$

Consequently,

$$\frac{t_j^{-(m+1)}}{j/m + 1/(m+1)} \leq c_m \leq \frac{m}{j} t_j^{-(m+1)}. \quad (6.2)$$

The ratio of the upper bound to the lower bound for c_m is exactly $1 + m/((m+1)j)$. For positive integer m , all endpoints are rational.

Proof. Integrating equation (5.4) between 0 and t_j gives

$$0 \leq L_m - f_m(t_j) \leq \int_0^{t_j} t^m dt = \frac{t_j^{m+1}}{m+1}.$$

Since $f_m(t_j) = (j/m)t_j^{m+1}$, this proves equation (6.1). All quantities are positive, so taking reciprocals proves equation (6.2). The endpoint ratio follows by cancellation. When m is a positive integer, the factors defining t_j and the powers in these formulas are rational. $\square$

These intervals certify the constant without floating-point Gamma evaluation. Their relative gap tends to zero at an explicit rate. They do not by themselves certify how close a finite threshold $T_m(K)$ is to its asymptotic equivalent: that is a different error problem.

[write] **Added 7 October 2026, batch 110.** The same integration at every t , not only at t_j , is Part II's [proposition 16.3](#); Part III's certificate (27.16) extends the bracket to every schedule of its class ([remark 24.1](#)). Part II's [theorem 17.1](#) addresses the different error problem named here: it encloses $T_m(K)$ itself at every index.

There is also an exact finite computation bound. If the backward recurrence has been computed only as far as stage k , then

$$k^m q_k \leq T_m(K) < k^m q_k + \sum_{r=1}^{k-1} r^m \quad (k \geq 2). \quad (6.3)$$

For integer m , this is a certified rational or integer enclosure of a particular threshold, independent of the asymptotic theorem.

6.2 An integral interpretation of the constant

Since $f_m(1) = 0$, [equation \(5.4\)](#) also gives

$$L_m = \int_0^1 t^m \left[C \left(\frac{my_m(t)}{t} \right) - \frac{my_m(t)}{t} \right] dt. \quad (6.4)$$

The term in brackets is a deterministic rounding residual. Formula [equation \(6.4\)](#) shows precisely how the successive increment regimes determine its weighted average. Replacing this residual by a fixed average would discard the dependence of its phase on the current quotient.

The full-trajectory theorem gives a related limit for the distribution of the losses. For a trajectory ending at $J = \tau_m(n)$, define a probability measure on $[0, 1]$ by

$$\mu_n = \sum_{k=2}^J \frac{x_{k-1} - x_k}{n} \delta_{k/J}. \quad (6.5)$$

Here δ_u denotes unit mass at u . The weights sum to one because $x_1 = n$ and $x_J = 0$.

Corollary 6.2 (Limiting distribution of rounding losses). *The measures μ_n converge weakly to the absolutely continuous probability measure with distribution function $1 - F_m(t)$ and density*

$$\rho_m(t) = c_m t^m \left[C \left(\frac{my_m(t)}{t} \right) - \frac{my_m(t)}{t} \right] \quad \text{for almost every } 0 < t < 1. \quad (6.6)$$

In particular, $0 \leq \rho_m(t) \leq c_m t^m$ and $\int_0^1 \rho_m(t) dt = 1$. The same conclusion holds under the smooth-weight hypothesis of [theorem 1.3](#).

Proof. The distribution function of [equation \(6.5\)](#) is $1 - X_n(t)$. The uniform convergence of the forward paths therefore proves convergence of these distribution functions to $1 - F_m$. The derivative formula in [equation \(5.4\)](#), including its integrable bound at zero, shows that the limit is absolutely continuous and gives [equation \(6.6\)](#). For smooth weights, use the forward-profile part of [theorem 1.3](#), proved below. $\square$

No randomness has been introduced in this statement. The measure records what fraction of a fixed initial value is lost at each normalized stage.

[write] **Added 7 October 2026, batch 110.** Part III's [theorem 33.1](#) extends this corollary to every schedule of its class and marks each loss with the increment state of its step; for the law $\delta_{(m,1)}$ its time marginal is (6.6) ([remark 24.1](#)). In its example with active zero phases a fraction $2/\pi$ of the mass is lost at steps whose weight increments tend to zero.

6.3 Monotonicity and endpoint behavior of c_m

The explicit constant permits useful parameter calculations. Let γ be Euler's constant and ζ the Riemann zeta function.

Proposition 6.3 (Dependence on m). *The function $m \mapsto c_m$ is strictly increasing on $(0, \infty)$. Moreover,*

$$\log c_m = -m \log m + (1 - \gamma)m + O(m^2), \quad m \downarrow 0, \quad (6.7)$$

$$\log c_m = \log m + \gamma + \sum_{r=2}^{\infty} \frac{\zeta(r)}{r(m+1)^{r-1}}, \quad m > 0, \quad (6.8)$$

$$c_m = e^\gamma m \left(1 + \frac{\pi^2}{12(m+1)} + O(m^{-2}) \right), \quad m \rightarrow \infty. \quad (6.9)$$

In particular, $c_m \rightarrow 1$ as $m \downarrow 0$, and c_m/m decreases strictly to e^γ as $m \rightarrow \infty$.

Proof. The convergent expansion

$$\log \Gamma(1 - z) = \gamma z + \sum_{r=2}^{\infty} \frac{\zeta(r)}{r} z^r, \quad |z| < 1,$$

is a standard Gamma identity [10]. Substitute $z = 1/(m+1)$ to obtain equation (6.8). Its positive terms decrease with m , which proves the assertion about c_m/m and the expansion equation (6.9). For small m , put $a = m/(m+1)$ and use $\log \Gamma(a) = -\log a - \gamma a + O(a^2)$; substitution in equation (1.2) gives equation (6.7).

For strict monotonicity of c_m itself, logarithmic differentiation gives

$$\frac{d}{dm} \log c_m = \frac{1}{m} + \log \Gamma(a) + (1 - a)\psi(a), \quad a = \frac{m}{m+1},$$

where ψ is the digamma function. Using its functional equation, this is

$$g(a) = \log \Gamma(1 + a) - \log a + (1 - a)\psi(1 + a).$$

We have $g(1) = 0$, and the standard trigamma series [10] gives

$$g'(a) = -\frac{1}{a} + (1 - a)\psi'(1 + a), \quad \psi'(1 + a) = \sum_{r=1}^{\infty} \frac{1}{(r + a)^2} < \int_a^{\infty} \frac{du}{u^2} = \frac{1}{a}.$$

Thus $g'(a) < 0$ for $0 < a < 1$ and consequently $g(a) > g(1) = 0$. $\square$

The limit $m \downarrow 0$ is singular for the original process. At $m = 0$, every modulus is 1, so a positive integer initial value never changes and has no finite extinction time. The fixed- m theorem does not justify interchanging the limits $m \downarrow 0$ and $n \rightarrow \infty$.

[write] **Added 7 October 2026, batch 110.** Part II's section 21 prints the same series and small- m expansion, one further term of (6.9), and the exponentiated form $c_m = 1 + m(\log(1/m) + 1 - \gamma) + O(m^2 \log^2 m)$ as $m \downarrow 0$. Its remainder bounds are uniform only on compact m -sets, so it too leaves this interchange of limits open.

7 Smoothly varying moduli

This section proves theorem 1.3 in full. The useful information in equation (1.12) is the local relative increment of the modulus; no differentiability of an interpolation is required.

7.1 Elementary consequences of the ratio hypothesis

Lemma 7.1 (Regular variation and a summable tail). *Under [equation \(1.12\)](#), the weights tend to infinity and are eventually strictly increasing. For every compact interval $[\delta, M] \subset (0, \infty)$,*

$$\frac{w_{\lfloor Kt \rfloor}}{w_K} \longrightarrow t^m \quad \text{uniformly for } \delta \leq t \leq M. \quad (7.1)$$

For each $0 < \delta \leq 1$,

$$\frac{1}{Kw_K} \sum_{1 \leq j < \delta K} w_j \longrightarrow \frac{\delta^{m+1}}{m+1}. \quad (7.2)$$

Proof. The ratio hypothesis implies

$$\log(w_k/w_{k-1}) = \frac{m + o(1)}{k}. \quad (7.3)$$

In particular the ratios are eventually larger than one, and the divergence of the harmonic series implies $w_k \rightarrow \infty$. Summing [equation \(7.3\)](#) between K and $\lfloor Kt \rfloor$ proves [equation \(7.1\)](#). Uniformity follows because all relevant indices are at least a fixed positive multiple of K , while the sums of their reciprocals are uniformly bounded on $[\delta, M]$.

For the sum near zero, choose $0 < \varepsilon < m$. For all sufficiently large ℓ ,

$$\log(w_\ell/w_{\ell-1}) \geq \frac{m - \varepsilon}{\ell}.$$

It follows, for all sufficiently large $j \leq K$, that

$$\frac{w_j}{w_K} \leq \exp\left(- (m - \varepsilon) \sum_{\ell=j+1}^K \frac{1}{\ell}\right) \leq \left(\frac{j+1}{K+1}\right)^{m-\varepsilon}. \quad (7.4)$$

The finitely many omitted weights have contribution tending to zero after division by Kw_K . On $[\eta, \delta]$ for $\eta > 0$, the locally uniform limit in [equation \(7.1\)](#) gives the usual Riemann sum. The bound in [equation \(7.4\)](#) makes the normalized sum over $j < \eta K$ uniformly small as $\eta \downarrow 0$. Thus the full limit is $\int_0^\delta t^m dt$, proving [equation \(7.2\)](#). $\square$

The proof of this lemma is included to keep the main theorem independent of general regular-variation machinery. Its conclusion is the familiar regular variation of index m , obtained here from an explicit successive-ratio assumption.

7.2 The same backward profile

Define

$$q_K^{(K)} = 1, \quad q_{k-1}^{(K)} = \left\lceil \frac{w_k}{w_{k-1}} q_k^{(K)} \right\rceil, \quad T_w(K) = q_1^{(K)}. \quad (7.5)$$

The proof of [lemma 3.2](#) applies without change:

$$\tau_w(n) > K \iff n \geq T_w(K). \quad (7.6)$$

This remains valid even if finitely many early weights decrease. The normalization $w_1 = 1$ ensures that the integer initial values are exactly the admissible quotients at stage 1.

The exact error identity is now

$$0 \leq w_{k-1}q_{k-1} - w_k q_k < w_{k-1}. \quad (7.7)$$

It gives

$$w_k q_k \leq w_K + \sum_{j=k}^{K-1} w_j, \quad (7.8)$$

$$0 \leq T_w(K) - w_k q_k < \sum_{j=1}^{k-1} w_j \quad (k \geq 2). \quad (7.9)$$

For $k \geq \delta K$, eventual monotonicity bounds the right-hand side of [equation \(7.8\)](#) by $(K+1)w_K$. Together with [equation \(7.1\)](#), this gives a uniform bound on q_k/K .

Set $b_k = w_k/w_{k-1} - 1$. On these macroscopic intervals,

$$q_{k-1} - q_k = \lceil b_k q_k \rceil, \quad b_k = \frac{m + o(1)}{k} > 0.$$

Thus the increments are uniformly bounded and at least one. The compactness, positive-drift, and ceiling-discontinuity arguments of [section 4](#) apply verbatim. The limit equation is still $y' = -C(my/t)$, and its unique selected solution is y_m . Therefore

$$\frac{q_{\lfloor Kt \rfloor}^{(K)}}{K} \longrightarrow y_m(t) \quad \text{locally uniformly on } (0, 1]. \quad (7.10)$$

The finitely many weights before monotonicity begins are outside each fixed macroscopic interval once K is large.

7.3 Matching and inversion

Proof of [theorem 1.3](#). Set $k = \lfloor \delta K \rfloor$ in [equation \(7.9\)](#). By [equations \(7.1\)](#) and [\(7.10\)](#),

$$\frac{w_k q_k}{K w_K} \longrightarrow \delta^m y_m(\delta).$$

By [equation \(7.2\)](#), the normalized error sum tends to $\delta^{m+1}/(m+1)$. The same sandwich used for [theorem 1.1](#), followed by $\delta \downarrow 0$, gives $T_w(K)/(K w_K) \rightarrow L_m$.

Let $J = \tau_w(n)$. Since $w_k \rightarrow \infty$, extinction is finite; as $n \rightarrow \infty$, its index tends to infinity. Formula [equation \(7.6\)](#) gives

$$T_w(J-1) \leq n < T_w(J).$$

The ratio hypothesis implies $w_{J-1}/w_J \rightarrow 1$, so both bounds are asymptotic to $L_m J w_J$. This proves [equation \(1.13\)](#).

For the forward trajectory, put $Q_k = x_k/w_k$. Survival through $J-1$ but not J gives the exact sandwich

$$q_k^{(J-1)} \leq Q_k < q_k^{(J)} \quad (1 \leq k \leq J-1).$$

[Equation \(7.10\)](#) yields $Q_{\lfloor Jt \rfloor}/J \rightarrow y_m(t)$ locally uniformly in $(0, 1)$. Therefore

$$\frac{x_{\lfloor Jt \rfloor}}{n} = \frac{w_{\lfloor Jt \rfloor}}{w_J} \cdot \frac{Q_{\lfloor Jt \rfloor}/J}{n/(J w_J)} \longrightarrow \frac{t^m y_m(t)}{L_m} = F_m(t).$$

The endpoint values and the finite-mesh monotonicity argument from [theorem 1.2](#) again give uniform convergence on $[0, 1]$. $\square$

7.4 Examples and consequences

Corollary 7.2 (Logarithmic modifiers). *Let $m > 0$, $b \in \mathbb{R}$, and $A > 0$. Take $w_1 = 1$ and $w_k = Ak^m(\log k)^b$ for $k \geq 2$. Then*

$$\tau_w(n) \sim \left(\frac{c_m(m+1)^b}{A} \frac{n}{(\log n)^b} \right)^{1/(m+1)}. \quad (7.11)$$

The same leading result holds if the weights are changed at finitely many stages.

Proof. Logarithmic differences give

$$k \log(w_k/w_{k-1}) = m + \frac{b}{\log k} + o(1),$$

so [equation \(1.12\)](#) holds. The threshold theorem gives $AJ^{m+1}(\log J)^b \sim c_m n$. In particular $\log J \sim \log n/(m+1)$; substitution yields [equation \(7.11\)](#). A finite change leaves the ratio hypothesis and the asymptotic expression for w_K unchanged, so the same theorem applies. $\square$

One may similarly take $w_k = A(k + \sigma)^m$ for all sufficiently large k , where $k + \sigma > 0$. The stopping equivalent is $(c_m n/A)^{1/(m+1)}$. More generally, if $w_k = k^m \ell(k)$ satisfies [equation \(1.12\)](#), the clean inverse statement is

$$\tau_w(n)^{m+1} \ell(\tau_w(n)) \sim c_m n.$$

The profile theorem remains explicit even when this inverse has no elementary closed form.

Corollary 7.3 (Regular variation of the stopping index). *Under [equation \(1.12\)](#), for every $\lambda > 0$,*

$$\frac{\tau_w(\lfloor \lambda n \rfloor)}{\tau_w(n)} \longrightarrow \lambda^{1/(m+1)}.$$

Proof. The increasing function $K \mapsto L_m K w_K$ is eventually regularly varying with positive index $m+1$, by [equation \(7.1\)](#). For any constants $0 < a < \lambda^{1/(m+1)} < b$, its ratios at $\lfloor aK \rfloor$ and $\lfloor bK \rfloor$ tend respectively to $a^{m+1} < \lambda$ and $b^{m+1} > \lambda$. Apply these inequalities at $K = \tau_w(n)$ and use [equation \(1.13\)](#) and monotonicity of the stopping index. Let a and b approach $\lambda^{1/(m+1)}$. $\square$

8 Why the microscopic schedule matters

It is tempting to regard $w_k \sim Ak^m$ as all the information needed to determine extinction. The following exact construction shows why the local ratio condition is substantive.

Proposition 8.1 (Repeating each modulus). *Fix an integer $r \geq 2$ and put*

$$w_k = \lceil k/r \rceil^m. \quad (8.1)$$

For every integer $n > 0$ and terminal stage $K \geq 1$,

$$\tau_w(n) = r(\tau_m(n) - 1) + 1, \quad (8.2)$$

$$T_w(K) = T_m(\lceil K/r \rceil). \quad (8.3)$$

Although $w_k \sim (k/r)^m$ and the weights are regularly varying with index m ,

$$\frac{T_w(K)}{K w_K} \longrightarrow \frac{L_m}{r}. \quad (8.4)$$

Proof. The operation $R_a(x) = a \lfloor x/a \rfloor$ is idempotent: $R_a(R_a(x)) = R_a(x)$. Each modulus j^m occurs at the stages $(j-1)r+1, \dots, jr$. Only its first occurrence can change the current value. At the initial block the modulus is 1, so it leaves the integer n unchanged. If the unblocked process first vanishes when modulus J^m is used, the blocked process first vanishes at stage $(J-1)r+1$. This proves [equation \(8.2\)](#).

Survival through stage K of the blocked process is equivalent to survival through modulus $\lceil K/r \rceil^m$ of the unblocked process. Duality gives [equation \(8.3\)](#). Apply [equation \(1.4\)](#) and $w_K = \lceil K/r \rceil^m$ to obtain

$$\frac{T_w(K)}{Kw_K} \sim L_m \frac{\lceil K/r \rceil}{K} \longrightarrow \frac{L_m}{r}.$$

□

For $n = 0$, both stopping times are 1; the displayed positive- n formula need not be used. In the block example the quantity $k(w_k/w_{k-1} - 1)$ is zero inside each block and approaches rm at its jumps. It therefore has no limit equal to m .

For comparison, choose smooth weights $v_1 = 1$ and $v_k = (k/r)^m$ for $k \geq 2$. They have the same leading equivalent as [equation \(8.1\)](#), but satisfy the local ratio hypothesis. Their stopping equivalent is

$$\tau_v(n) \sim (c_m r^m n)^{1/(m+1)},$$

whereas [equation \(8.2\)](#) gives $\tau_w(n) \sim r(c_m n)^{1/(m+1)}$. The ratio tends to $r^{1/(m+1)}$. Thus even the leading stopping constant distinguishes the two schedules.

This example supplies a precise research boundary. A theorem for all regularly varying moduli must include additional information about how their growth is distributed among individual rounding steps.

[write] Added 7 October 2026, batch 110. Part III identifies that information for nondecreasing weights with bounded scaled increments: the joint limiting law of the scaled increment $k(w_k/w_{k-1} - 1)$ and of the flag recording whether the weight strictly increases ([theorem 27.2](#)). The block weights above are its case $a = 0$ of [corollary 30.2](#). Part III also shows that weights agreeing with the block weights to every inverse power of the index can have any constant in $[1/(2\pi), \pi/8]$ when $m = 1$, $r = 2$ ([theorem 31.2](#)), and that a schedule need not have a limiting constant at all ([theorem 32.1](#)).

9 Exact computation and numerical evidence

The accompanying program serves two purposes: it checks that independent finite implementations agree with the definitions, and it illustrates the asymptotic theorems. All floor and ceiling decisions in the reported checks use exact integers. Special-function evaluations and plots are separately identified as numerical diagnostics.

9.1 Integer and rational exponents

For integer $m \geq 1$, the backward recurrence has the purely integer form

$$q_{k-1} = \left\lfloor \frac{k^m q_k + (k-1)^m - 1}{(k-1)^m} \right\rfloor. \quad (9.1)$$

It computes $T_m(K)$ in $K-1$ arithmetic steps and uses only one current quotient if no profile is retained. These are counts of arithmetic operations, not a bit-complexity bound. For fixed m , all integer values involved have $O_m(\log K)$ bits by [equation \(3.9\)](#).

Exact computation is also possible for rational $m = p/q > 0$, written in lowest terms. Given the integer quotient $v = q_k$, the next backward quotient is the least integer u satisfying

$$u^q(k-1)^p \geq v^q k^p. \quad (9.2)$$

This is an integer-root problem. Let $N = v^q k^p$, $D = (k-1)^p$, and let r be the integer q th root rounded down of $\lfloor N/D \rfloor$. The desired ceiling is r if $r^q D = N$, and $r+1$ otherwise. The final comparison is exact, including the cases where the algebraic expression is an integer.

Independently, the forward quotients satisfy

$$Q_1 = n, \quad Q_k = \left\lfloor \left(\frac{k-1}{k} \right)^m Q_{k-1} \right\rfloor. \quad (9.3)$$

For $m = p/q$, this is the greatest integer u such that $u^q k^p \leq Q_{k-1}^q (k-1)^p$. Thus rational exponents can be checked without ever representing an irrational intermediate x_k .

9.2 Independent checks

The recorded verification performs the following mathematical checks:

- For $m = 1, 2, 3$ and every $0 \leq n \leq 2000$, direct simulation of the original x_k recurrence agrees with the quotient recurrence and with the exact backward-threshold interval for the stopping time.
- For $m = 1/2, 2/3, 3/2, 5/3$ and $0 \leq n \leq 200$, the algebraic integer-root implementation of the forward process agrees with the backward survival intervals.
- All displayed initial terms in the inspected OEIS entries agree: 79 terms for A073047 with offset 1, and 105 terms each for A082527 and A082528 with offset 0.
- For block length $r = 3$ and $m = 1, 2, 3$, the exact threshold identity is checked through $K = 200$, and the stopping identity through $n = 2000$. The zero input is checked separately.
- The finite-product constant enclosures are constructed as exact fractions for $m = 1, 2, 3, 5$ and $j = 10, 100, 1000$, and their stated relative width is checked exactly.

These checks all passed. Their bounds and results are recorded in `verification.json`. They audit the finite definitions and calculations; the infinite statements rest on the preceding proofs.

9.3 Threshold convergence

Table 1 shows both the correct scale and the irregular finite errors. For example, for $m = 2$ the normalized error at $K = 10000$ is larger than at $K = 1000$. Neither monotonic convergence nor a particular rate follows from the theorem or from these samples.

9.4 The predicted shapes

Figure 2 directly illustrates the compact convergence proved in [proposition 4.3](#). Figure 3 uses actual forward trajectories, which were not used to construct the limiting profile. For the cubic process, the exact stopping indices for initial values $10^3, 10^6, 10^9$ are respectively 9, 51, 287. The step curves converge to the continuous weighted profile F_3 .

m	K	Exact $T_m(K)$	$c_m T_m(K)/K^{m+1}$
$\frac{1}{2}$	10	17	1.1785908081
$\frac{1}{2}$	100	470	1.0304145588
$\frac{1}{2}$	1000	14440	1.0011088982
$\frac{1}{2}$	10000	456247	1.0002628749
1	10	34	1.0681415022
1	100	3234	1.0159910642
1	1000	318570	1.0008171717
1	10000	31833630	1.0000829815
2	10	244	1.2116837876
2	100	205688	1.0214295693
2	1000	201388916	1.0000806743
2	10000	201457639236	1.0004219482
3	10	1816	1.2284918534
3	100	15338896	1.0376491616
3	1000	148522784144	1.0047303434
3	10000	1478758856906056	1.0003541899
5	10	141440	1.4628989077
5	100	103280328768	1.0682174784
5	1000	97237370572647616	1.0057158031
5	10000	96759789128216456247328	1.0007762290

Table 1: Exact backward thresholds and numerical normalizations. The last column tends to 1 by [theorem 1.1](#); its displayed values use high-precision evaluation of c_m . The full data file also includes $K = 100000$. The exponent $m = 1/2$ uses exact integer square roots.

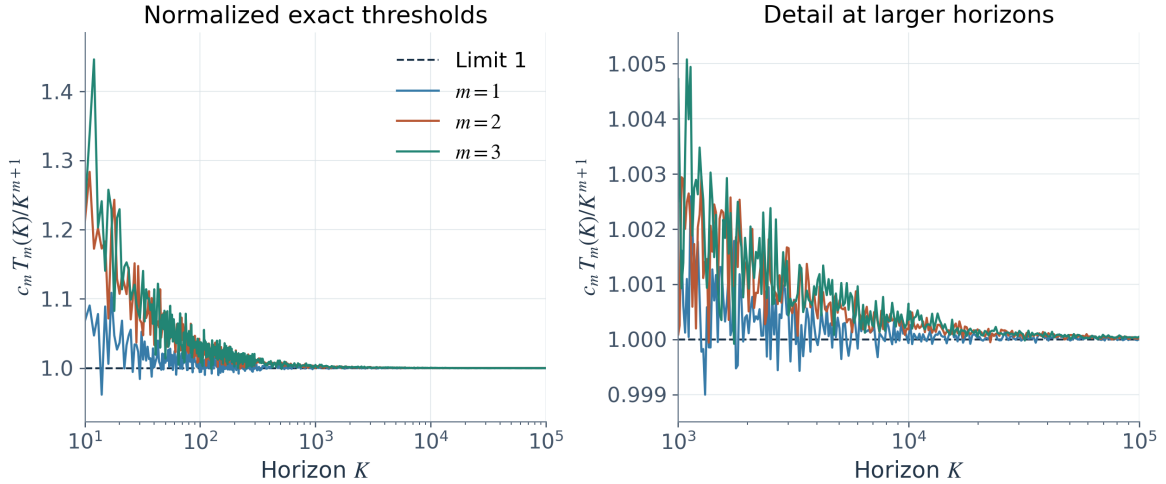


Figure 1: Normalized thresholds for $m = 1, 2, 3$, calculated by the exact backward recurrence. The Gamma constants are used only in the normalization. The irregular excursions illustrate why estimating a discrete error law requires more than a smooth fitted curve.

m	Lower endpoint	$m\Gamma(m/(m+1))^{m+1}$	Upper endpoint
1	3.1408077460	3.1415926536	3.1423781500
2	4.9642631432	4.9659171624	4.9675726521
3	6.7622883375	6.7648229810	6.7673600538
5	10.3385883285	10.3428938609	10.3472038189

Table 2: Finite-product enclosures from [proposition 6.1](#) at $j = 1000$. The outer decimal endpoints are rounded outward by exact integer arithmetic, so the displayed intervals retain the enclosure. The middle column is a numerical evaluation of the exact Gamma formula.

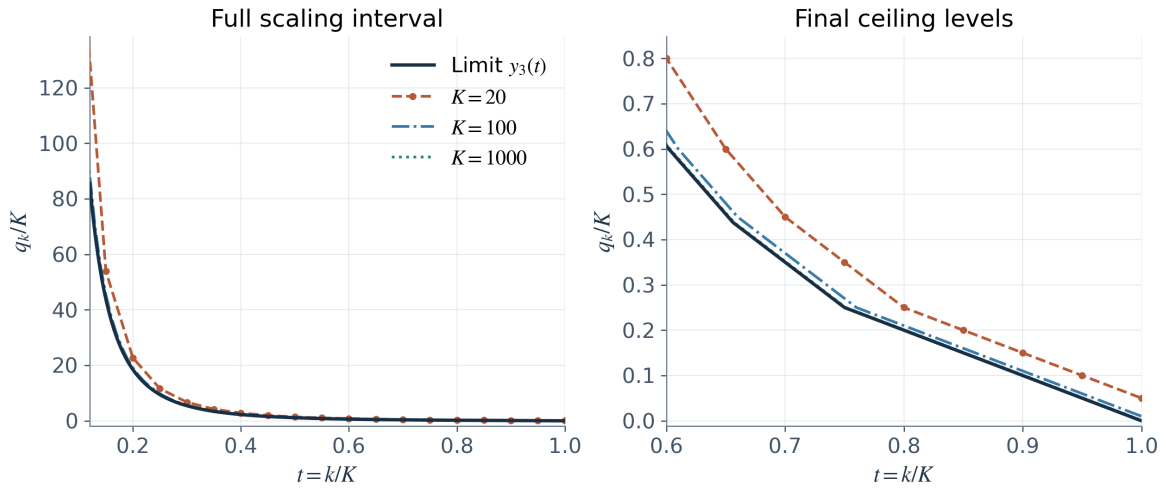


Figure 2: The cubic backward quotient profile. Discrete arrays, divided by their terminal stage K , approach the explicit piecewise linear function y_3 . The plot stays away from $t = 0$, where y_3 diverges; the proof handles that endpoint through weighted tail bounds.

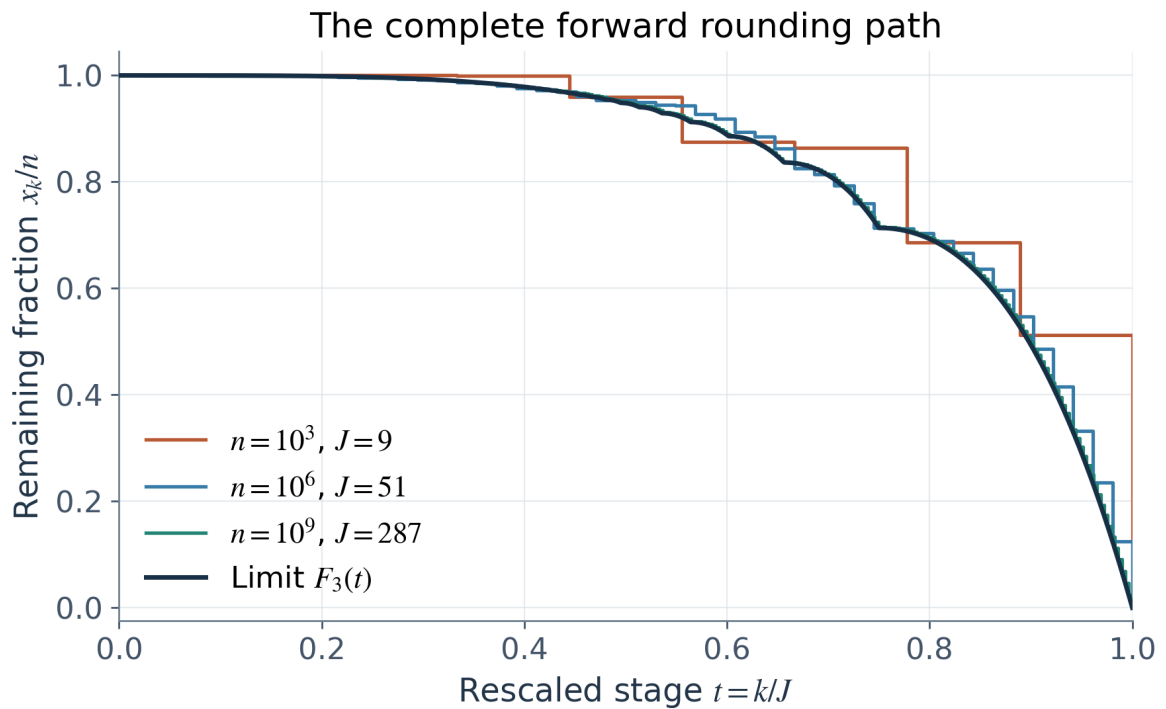


Figure 3: The fraction of the initial value remaining in three cubic forward trajectories, with time divided by the exact stopping index. The limiting curve is $F_3(t) = c_3 t^3 y_3(t)$. Its continuous value at $t = 0$ is 1, despite the divergence of the unweighted quotient profile.

9.5 Certified constant intervals

The exact rational numerators and denominators underlying Table 2 are included in

```
data/constant_certificates.json
```

The table’s enclosure is provided by the theorem and exact endpoint arithmetic. A numerical comparison of a Gamma evaluation with those intervals is an additional consistency check, not the basis of their certification.

9.6 Reproduction and formalization

The archive includes the LaTeX source, this PDF, the figures, the Python program, its dependency versions, and recorded data. From the archive’s root directory, the full calculation is reproduced by

```
python reproduce.py
```

The exact checks alone require only the Python standard library:

```
python reproduce.py --exact-only --out replay_exact
```

The full replay additionally uses `mpmath` and `matplotlib`; it requires no network access. Compile `article.tex` twice with `pdflatex` to rebuild the PDF. The README gives query examples, output descriptions, and precision conventions.

A formalization can be divided at natural mathematical interfaces. The finite recurrence and survival duality need only integer order, floor/ceiling inequalities, and finite induction. Rational exponents admit an exact integer-root specification. The analytic part requires compactness of uniformly Lipschitz functions on compact intervals, dominated convergence, the Gamma functional equation, and the beta integral. The proof isolates the delicate points as reusable lemmas: the positive drift that selects the terminal solution, the finite number of ceiling discontinuities on compact intervals, and the final weighted tail estimate.

No proof-assistant development accompanies this report. The executable checks are finite computations and are not described as a formal verification of the real-parameter theorem.

[write] Added 3 October 2026, batch 85. This section describes the delivered archive. In the repository the files keep their bytes but not all their places: the program is `code/reproduce.py`, `requirements.txt` and the recorded `verification.json` are in `data/`, the delivered `README.txt` is replaced by the report’s `README.md`, and the delivered 25-page PDF was not shipped (the repository’s `article.pdf` is a rebuild of this source with the [write] notes; the delivered one survives in the arrival commit). The program reads `data/oeis_prefixes.json` relative to its own directory and, without `--out`, writes into that directory, so it must not be run in place: run from `code/` it would not find its fixture file. The command above was printed in the delivered PDF with en dashes for its double hyphens (a typewriter-font ligature); the writing step set them as two hyphens, the only change to the delivered text outside the labels, the notes and the preamble. The report’s `README.md` rebuilds the delivered layout on a scratch copy and replays there. At intake (3 October 2026) the full replay on such a copy passed in 92.5s; the seven CSV files were byte-identical to the recorded ones, the two JSON and two T_EX data files differed only by CRLF line ends (Python text mode on Windows), `verification.json` only in its runtime and Python-version fields, and the three PNG figures only in rendering. The OEIS terms in `data/oeis_prefixes.json` are OEIS content, licensed by The OEIS Foundation under CC BY-SA 4.0; they are not covered by the repository’s MIT-0 licence. No Lean or Rocq development in ProveIt treats this subject.

10 Further research questions

The leading constant is only the first part of the rounding problem. The following questions separate plausible next steps from conclusions already proved. They are proposed research problems; except for the historical linear question, their open status refers to this report and the sources inspected, not to an exhaustive classification of the literature.

Research question 1 (Quantitative threshold errors). For fixed $m > 0$, find a usable rate in

$$T_m(K) = L_m K^{m+1} + E_m(K).$$

Which exponents $\delta_m > 0$ admit $E_m(K) = O_m(K^{m+1-\delta_m})$? Can an explicit bound be obtained uniformly for m in a compact subinterval of $(0, \infty)$?

The proof identifies the correct place to work: make the crossing times of the discrete quotient increments quantitative, then balance their accumulated errors against the omitted small-time tail. The classical $m = 1$ estimate in [equation \(2.1\)](#) supplies a benchmark. Compactness alone contains no numerical rate.

[write] **Added 7 October 2026, batch 110.** Answered by Part II (Report 188): $E_m(K) = O_m(K^{(m+1)^2/(m+2)})$, that is $\delta_m = (m+1)/(m+2)$, with implied constants uniform for m in any compact subinterval of $(0, \infty)$, and an explicit enclosure of $T_m(K)$ at every index ([theorems 12.1](#) and [17.1](#)). The proof makes the crossing errors quantitative through a monotone integral coordinate and balances them against the small-time tail, as suggested above. At $m = 1$ the exponent is the classical $4/3$. Whether δ_m can be enlarged is Part II's question [1](#), open.

Research question 2 (Bounded stopping-time error). For which $m > 0$, if any, does

$$\tau_m(n) = (c_m n)^{1/(m+1)} + O_m(1)$$

hold? In particular, resolve the stronger bounded-error claim for $m = 1$ recorded in [A073047](#).

For pure powers, this is equivalent to $T_m(K) = L_m K^{m+1} + O_m(K^m)$. One direction follows by inverting the two neighboring threshold bounds. For the other, evaluate the stopping-time formula at the first integer of each survival interval, where the stopping index is $K + 1$, and expand the resulting power. The two equivalent formulations indicate that an error term at the scale of a single threshold gap is required.

[write] **Added 7 October 2026, batch 110.** Open. Part II's remainder gives $\tau_m(n) = (c_m n)^{1/(m+1)} + O_m(n^{1/((m+1)(m+2))})$, not $O_m(1)$; this question reappears as Part II's question [2](#) and Part III's Research question [11](#).

Research question 3 (Signs and fluctuation scales). Does $T_m(K) - L_m K^{m+1}$ have an eventual sign? What are the natural scales of its upper and lower envelopes? Does any centered and scaled version have a limiting distribution along integer horizons?

The selected tabulated horizons have positive errors, while the denser plotted data include both signs. Neither observation establishes an eventual-sign theorem. The nonmonotone normalized errors show that a simple monotonicity argument would need additional structure. Any probabilistic terminology here would describe a distribution over sampled horizons, not randomness in the recurrence.

[write] **Added 7 October 2026, batch 110.** Open; it reappears as Part II's question [3](#). Part II bounds the error, $|T_m(K) - L_m K^{m+1}| = O_m(K^{(m+1)^2/(m+2)})$, but says nothing about its sign or its distribution.

Research question 4 (A theory of microscopic schedules). Classify the limits for regularly varying weights when $k(w_k/w_{k-1} - 1)$ oscillates, has repeated zero values, or follows a periodic pattern. Which information beyond the regular-variation index determines the extinction constant and the limiting profile?

The block construction gives a completely solved test family. A possible next class consists of a fixed periodic pattern of small relative increments. The ceiling acts before any averaging, so an effective drift would have to be derived from the individual phases of that pattern. Establishing such an averaging theorem is a substantive extension; it is not a consequence of [theorem 1.3](#).

[write] **Added 7 October 2026, batch 110.** Answered by Part III for nondecreasing weights whose scaled increments $\beta_k = k(w_k/w_{k-1} - 1)$ lie in a fixed interval $[0, B]$ and whose joint empirical law of β_k and of the flag $\sigma_k = \mathbf{1}_{\{w_k > w_{k-1}\}}$ converges ([definition 27.1](#)). The information beyond the index is that augmented law: the ceiling is applied to each type before averaging, the averaged drift ([27.6](#)) determines the profile, the constant and the forward trajectory ([theorem 27.2](#)), and the drift determines the law ([theorem 35.4](#)). Periodic patterns, repeated values and oscillation are all covered; a zero slope counts differently according as the weight stays equal or increases by $o(1/k)$, which moves the constant from $1/(2\pi)$ to $\pi/8$ in the example of [theorem 31.1](#). Without a limiting law the constant need not exist ([theorem 32.1](#)). Outside this class the question stays open, re-scoped as Part III's Research questions [14](#) (unbounded or negative increments), [15](#) (sparse large jumps) and [17](#) (schedules without a limiting law), and the rate of the averaged law is its Research question [10](#).

Research question 5 (Stability under sparse exceptional steps). How many exceptional ratios can be allowed without changing L_m ? For example, suppose the local ratio condition holds outside a set of indices of density zero. Which bounds on the exceptional increments suffice, and can a sparse set of very large jumps change the constant?

Finite modifications are already covered. Passing from finitely many exceptions to a sparse infinite set requires controlling their combined effect on the quotient increments. Density alone need not control the size of those effects.

[write] **Added 7 October 2026, batch 110.** Answered in its bounded form by Part III: for nondecreasing weights, changes of the successive ratios on a set of density zero, with the scaled increments there bounded, change neither the augmented law, nor the constant, nor the profiles ([corollary 34.2](#) and the note after it), each schedule being normalized by its own w_K . Whether a sparse set of very large jumps can change the constant is open: Part III's Research question [15](#).

Research question 6 (Moving exponents and singular regimes). Describe joint limits with $m = m(n) \downarrow 0$ or $m = m(n) \rightarrow \infty$. Determine ranges in which the fixed-parameter equivalent remains uniform and identify any transition scales.

The discontinuity at $m = 0$ makes the first question particularly interesting: positive integer values never vanish when every weight is exactly 1. The explicit expansions in [proposition 6.3](#) describe the constants, but they do not control the speed of convergence of the underlying discrete process.

[write] **Added 7 October 2026, batch 110.** Open. Part II's bounds are uniform only on compact m -sets ([theorem 12.1](#)); its [section 21](#) expands c_m at both ends but proves no joint regime, and its question [4](#) asks the same.

Research question 7 (Rates for the full trajectory and loss distribution). Obtain explicit bounds for

$$\sup_{0 \leq t \leq 1} |X_n(t) - F_m(t)|$$

and for convergence of the loss measures in [corollary 6.2](#). Can one derive asymptotics for individual moments or quantiles of the loss-time distribution with a controlled error?

The limiting density is explicit on each increment regime. It permits certified computation of limiting observables using finite sums and an integrable tail. Rates for the finite process again depend on discrete crossing errors rather than solely on the Gamma constant.

[write] **Added 7 October 2026, batch 110.** Partly advanced, still open. Part II gives a rate for the backward minimal path, uniformly over the whole path: $\max_{1 \leq j \leq k} |j^p q_j / k^{p+1} - f_m(j/k)| = O_m(k^{-(m+1)/(m+2)})$ for $p = m$ (theorem 18.1, remark 16.4). It gives no rate for the forward paths X_n or for the loss measures asked about here. Part III marks the loss measures by the increment state (theorem 33.1), without a rate.

Research question 8 (Sieve and game interpretations for higher powers). Is there a natural sieve or solitaire model whose survival thresholds are $T_m(K)$ for integer $m \geq 2$? Can such a model explain the Gamma product combinatorially or connect the threshold gaps to an existing family of integer sequences?

The ordinary case has the Tchoukaillon interpretation. A higher-power model could expose monotonicity, congruence, or conservation properties not apparent in the floor recurrence. Any proposed model should first be proved equivalent to the exact backward recurrence, including its initial conditions.

Research question 9 (Faster exact algorithms and formal verification). Can long consecutive ranges with a constant quotient increment be skipped in a provably correct exact algorithm? Can the general real-parameter theorem be formalized with a reusable library lemma for monotone discontinuous drift limits?

The present algorithm takes one arithmetic step per stage. The limiting piecewise linear structure suggests acceleration, but the exact locations of discrete transitions must be certified before steps can be skipped. For formalization, a useful first milestone is the exact survival duality and rational-exponent algorithm, followed by the compactness and endpoint-selection lemmas.

10.1 A practical order for continuing the work

The most direct continuation is a quantitative crossing-time analysis for the pure power family. In parallel, a periodic-ratio model can test how the constant changes when the local smoothness condition is relaxed. These two directions address the principal limitations of the present theorem: the absence of a finite error bound, and the dependence on the microscopic schedule.

[write] **Added 7 October 2026, batch 110.** Both directions have been taken up: the quantitative analysis for pure powers is Part II, and a periodic-ratio model, generalized to every convergent bounded increment law, is Part III.

Part II

Sequential power rounding: quantitative thresholds, inverses, and global mass profiles

11 Provenance, scope and notation of Part II

[Added 7 October 2026, batch 110.] Part II was added in the write of batch 110. It is built from Report 188 of a session bundle of research reports, which studies exactly the pure power

process of Part I and was written without sight of this report. Its leading law is [theorem 1.1](#), proved again by a different route; what it adds is a quantitative remainder, uniform on compact sets of exponents, which answers Research question 1. Everything of Report 188 is printed below: its abstract and scope paragraph in this section, its twelve sections as [section 12](#) to [section 23](#), with Report 188’s own letters ([section 11.3](#)). Statements that Part I already proves are printed as marked second routes, with remarks of the write that identify them ([table 3](#)). Bracketed notes beginning “[Added 7 October 2026, batch 110” and remarks marked [write] are the write’s; all other text of Part II is Report 188’s.

11.1 The manuscript

Report 188 is the archive `Sequential_Power_Rounding_Gamma_Constants_and_Inverses_Source.zip` (564,257 bytes; 20 files), which arrived in commit 60f54ea06 and was placed in this report by commit 8622ca7e5 (batch 110). Its article `Report188.tex` (709 lines; a 17-page PDF) is titled *Report188: Sequential Power Rounding*, with the subtitle *Exact Gamma constants, quantitative thresholds, inverses, and global mass profiles* printed in its author slot, and is dated 4 October 2026. Its PDF author field reads “Research report”; it names no person, tool or addressee. It has no pin: it names no ProveIt commit and cites neither the repository nor Part I, which was written on 3 October 2026 (commit 2b3b69b4b) and which it does not know. It quotes the OEIS records as A082527 revision 8 (11 December 2025), A082528 revision 6 (30 March 2012), A073047 revision 15 (7 June 2020) and A002491 revision 109 (20 August 2026), inspected on 4 October 2026; the write did not re-check those revision numbers. On 7 October 2026 the write read the live entries: A082528 still states the general- m law as “Conjecture”, and A002491 still carries D. E. Knuth’s comment of 8 June 2021 as Report 188 summarizes it in [section 20](#) (the summed error of Broline and Loeb’s argument is $O(n^{3/2})$; summing over part of the range recovers $O(n^{4/3})$).

[Added 7 October 2026, independent check of the batch-110 write: the figure 564,257 bytes above is the total size of the archive’s twenty members; the archive itself has 566,521 bytes. Part I’s provenance, like the other reports of the collection, gives archive sizes. The rest of this paragraph was confirmed: the line and page counts, the title, the author field, the four revision numbers and dates (equal to the live entries on 7 October 2026, as are the 342 fixture terms), A082528’s “Conjecture” and Knuth’s comment.]

Report 188’s abstract and its scope paragraph read as follows.

Start with an integer mass n and round it successively down to multiples of $2^p, 3^p, \dots$, where $p > 0$ is any fixed real number. We prove that its first zero occurs at

$$A_p(n) = (p\Gamma(p/(p+1))^{p+1}n)^{1/(p+1)} + \mathcal{O}_p\left(n^{1/((p+1)(p+2))}\right).$$

This identifies the constant in the all-real-power conjecture recorded in OEIS A082528, including the square-power sequence A082527. The proof uses an exact reverse ceiling recursion and a monotone integral coordinate. A dyadic telescoping argument controls the coordinate error without a logarithmic loss. We obtain explicit all-index threshold enclosures, an elementary piecewise mass profile with Gamma-product breakpoints, and a uniform positive-terminal-mass law with an explicit inverse curve. Exact integer algorithms cover every positive rational power. The accompanying offline source archive separates exact finite checks from optional floating-point diagnostics and includes a deterministic PDF builder.

Scope and attribution. The first-power threshold estimate is classical: Erdős and Jabotinsky proved the relevant $k^2/\pi + \mathcal{O}(k^{4/3})$ behavior in 1958. The Gamma-function identities, log-convexity bounds, and monotone inversion used here are standard. We do

not claim a sharper first-power remainder, a bounded stopping-index error, an all-orders expansion of the discrete thresholds, or worldwide priority. A final section expands the exact constant as a function of p ; this classical parameter expansion does not resolve the deterministic rounding fluctuations.

Status and non-claims of Report 188, kept. The first-power remainder is classical and credited to Erdős and Jabotinsky; the Gamma identities, log-convexity bounds and monotone inversion are standard. It claims no sharper first-power remainder, no bounded stopping-index error, no all-orders expansion of the discrete thresholds, no uniformity as $p \rightarrow 0$ or $p \rightarrow \infty$ (only on compact p -sets), and no worldwide priority; its targeted source check “is not a worldwide novelty certification”. Its decimal displays are not certified intervals, its exact code covers rational p only, the real- p theorem being analytic, and its floating-point diagnostics are labelled as such. Its finite checks “do not constitute a formal proof of the asymptotic theorems”. Nothing in Part II is formalized in Lean or Rocq.

Stale wording, corrected here. Report 188 was written without Part I, so two of its sentences overstate what is new relative to this report. Its abstract says that it “identifies the constant in the all-real-power conjecture recorded in OEIS A082528”; its [section 20](#) says that “targeted source checks found no directly matching square- or cube-power theorem among the inspected material”. Part I ([theorem 1.1](#), dated 3 October 2026) had already proved the leading law $T_m(K) \sim L_m K^{m+1}$ for every real $m > 0$ and identified the square and cube constants. What Report 188 contributes relative to Part I is listed in [table 3](#): chiefly the remainder, the explicit enclosure, the global profile with a rate and the positive-terminal law. Dated notes at both passages say so. Its novelty statement remains, as it says, bounded by a targeted search, and the write searched no further.

11.2 What Part II adds

Table 3: Report 188 against Part I. “Second route” means the same statement proved here by a different argument; “same” means the same statement by essentially the same argument.

Report 188 (Part II)	Part I	Status here
Propositions 13.1 and 13.2	lemma 3.2, corollary 3.3	same; Report 188 adds the terminal quotient h and the no-overshoot path (13.2)
Section 13.1	section 9, (9.2)	same integer-root method
Lemmas 14.1 and 15.1, theorem 15.2	—	new: drift sandwich, uniform ratio bounds, dyadic telescoping
Proposition 16.1, (16.7)	(1.8), (5.1), (1.9)	the same objects: $P_r = t_r$, $tU_p(t) = y_m(t)$, $H_p = f_m$ (remark 16.4)
Lemma 16.2	(5.2)	two-sided bounds where Part I proves the asymptotic
Proposition 16.3	(5.4) , proof of proposition 6.1	the integral of (5.4) , which Part I evaluates at $t = t_j$, at every t
Theorem 17.1	—	new: an explicit enclosure at every index
Theorem 12.1, leading terms	theorem 1.1	second route (an integral coordinate instead of compactness)
Theorem 12.1, remainders	Research question 1	new; answers that question

Report 188 (Part II)	Part I	Status here
Theorem 18.1	proposition 4.3 , (4.7)	new: a rate, uniform over the whole minimal path
Theorem 19.1 , (19.7)	—	new: positive terminal quotients and the inverse curve
(19.8)	theorem 1.2	the same limit object, reparametrized (remark 19.2)
(20.1)	(2.1)	classical (Erdős–Jabotinsky)
(21.1) , (21.3)	(6.8) , (6.7)	same (classical Gamma expansions)
(21.2) , (21.4)	(6.9)	one further term at $p \rightarrow \infty$; the exponentiated form at $p \downarrow 0$

Questions of Part I. Research question 1 asks for exponents $\delta_m > 0$ with $T_m(K) = L_m K^{m+1} + O_m(K^{m+1-\delta_m})$, and for a bound uniform on compact m -intervals. [Theorem 12.1](#) gives $\delta_m = (m+1)/(m+2)$, uniformly on compact sets, and [theorem 17.1](#) is an explicit all-index enclosure from which such a bound is read off; Report 188 prints no numerical value of the implied constant. The question is answered; whether δ_m can be enlarged is Report 188’s own question 1. Its global profile ([theorem 18.1](#)) gives a rate for the backward minimal path only, so the question “Rates for the full trajectory and loss distribution” (Research question 7), which concerns the forward path and the loss measures, stays open. Research questions 2 and 3 reappear as its questions 2 and 3, unanswered. Dated notes at those questions say so.

11.3 Notation of Part II

Part II keeps Report 188’s symbols; no symbol is renamed and no normalization is changed. Report 188 writes p for the exponent that Part I calls m , and it indexes stages by j up to a terminal level k , where Part I indexes stages by k up to K . Several of its letters mean something else in Parts I and III. [Table 4](#) fixes the reading; wherever Part II says p , read Part I’s m .

Table 4: Report 188’s symbols in Part II, their counterparts in Part I, and the false readings they would have elsewhere in this report.

Part II	Meaning	Part I	False reading elsewhere
p	the exponent	m	in section 9 , p is the numerator of $m = p/q$; in Part III, p_s are frequencies and p a period length
A, B in $p = A/B$	numerator and denominator (section 13.1)	p, q	$A_p(n)$, $B_p(k)$ below; Part III’s bound B
$s = p + 1$		$m + 1$	Part III: s indexes phase types
$a = 1/s$	Gamma exponent	$1 - a$	Part I: $a = m/(m+1)$; Part III: a counts active zero phases
$\beta = s/(s+1)$	cutoff exponent	—	Part III: β_k are scaled increments
K_p, c_p	threshold and extinction constants	L_m, c_m	K is Part I’s terminal stage
$k; j$	terminal level; stage index	$K; k$	in Parts I and III, k is a stage index
$A_p(n)$	first zero index	$\tau_m(n)$	—

Part II	Meaning	Part I	False reading elsewhere
$B_p(k); B_p(k, h)$	survival threshold; terminal quotient at least h	$T_m(K); —$	Part III: B bounds the increments
X_j	retained mass	x_k	Part I: $X_n(t)$ is the normalized forward path
$q_j = q_j(k, h);$ $u_j = q_j/j$ $c; m$ in propositions 13.1 and 13.2 and in the inverse proof of theorem 12.1	minimal reverse path; scaled quotient the ratio $((j+1)/j)^p$; an integer, resp. $\max\{k : B_p(k) \leq n\}$	$q_k^{(K)} (h = 1);$ z/m —	Part I: Q_k, Q are forward quotients c_m and the exponent m of Part I
$f_p(u) = u + [pu],$ $f_p(0) = 1$		$u + C(pu)$	Part I: $f_m = t^m y_m$
I_p	integral coordinate	—	Part III: equals H for the law $\delta_{(p,1)}$
$R_j; Q_p; E_p$	step ratio; ratio bound; dyadic constant	—	Part I: Q_k forward quotients, $E_m(K)$ the threshold error of Research question 1; Part III: $E(z)$ residual drift; R_k in (19.8)
e_j	rounding defects in $[0, 1)$	—	—
P_r	Gamma products	t_r	—
$U_p(t), H_p(t)$	inverse profile, mass profile	$y_m(t)/t, f_m(t)$	Part III: $H(z)$ is an integral in z ; $U_p(k, J)$ in section 17 is an envelope endpoint
$L_p(k, J),$ $U_p(k, J)$	envelope endpoints	—	L_m ; the profile $U_p(t)$
$\Phi_p, H_{p,v}$	terminal-quotient cost; its profile	—	—
$M_{k,j}$	normalized retained mass	—	Part III: $M(t) = t^m y(t)$
$t; r$	time in $(0, 1]$; cell index	$t; j$ in t_j	Part III: $t(z)$; r a period length
h, V, N_k, R_k	terminal quotient, its bound, initial masses, retained quotient	—	Part III: h_j perturbation sizes, h a block index

Intake checks. The dossier of the intake (6 October 2026) reran Report 188’s mandatory checker on a copy: its outputs under normal and optimized Python are byte-identical to each other and to the recorded `data/02-rates-generated-exact_checks.json`; the package’s replay script fails on Windows only because it compares text-mode output with CRLF line ends (section 22; the README gives a rerun route). An independent backward recursion written at intake reproduced the first ten thresholds for $p = 2, 3$ printed in section 13 and the four values of the table in section 22, and found $(B_p(k) - K_p k^s)/k^{s^2/(s+1)}$ between 0.003 and 0.18 on the sampled ranges for $p = 1, 2, 3$. The write recomputed the ten thresholds, $B_2(1000)$ and $B_3(1000)$ with exact integers, and c_2, c_3 to forty digits (the last digits of Report 188’s decimal displays are wrong; remark 12.2).

[Added 7 October 2026, independent check of the batch-110 write: the band $[0.003, 0.18]$ quoted above from the intake dossier could not be reproduced. The dossier, which named the sampled ranges, is lost, and on the check’s own sample $k = 100, 200, \dots, 2000, 5000, 10^4$ the normalized remainder $(B_p(k) - K_p k^s)/k^{s^2/(s+1)}$ runs from -0.0446 ($p = 1, k = 1200$) to 0.2216

($p = 3$, $k = 100$); for $p = 1$ it takes both signs. It stays bounded on these samples, as (12.5) requires, but the quoted band is not a property of the thresholds, and nothing rests on it. The check's thresholds equal the 53 terms of A002491, Part I's `data/threshold_table.csv`, the twenty thresholds of section 13 and the table of section 22.]

Independent check of the write (7 October 2026). An independent adversarial check of the batch-110 write re-derived each remark marked [write] and each dated note of Parts I–III. It recomputed every number they print (the two constants, the expansions of section 21, the Gamma products and certificate of Part III, the thresholds) and read the live OEIS entries. It reran Report 188's checker and Part III's program on copies (outputs equal to the records). It found no error in remarks 12.2, 16.4, 19.2 and 24.1. It corrected the archive sizes in sections 11.1 and 24.1, the band above, and one sentence of the note at the end of section 23, by dated notes there. It also recorded the numbering changes in section 24.1. One omission is noted here: the dated notes at section 1 and at Research question 4 describe Part III's class without two of its hypotheses, the positive mean slope $m = \int \beta d\mu > 0$ and $w_1 = 1$ (definition 27.1, (25.1)).

12 The model and the conclusions

Fix a real number $p > 0$. For an integer $n \geq 0$, define

$$X_1 = n, \quad X_j = j^p \left\lfloor \frac{X_{j-1}}{j^p} \right\rfloor \quad (j \geq 2), \quad A_p(n) = \min\{j \geq 1 : X_j = 0\}. \quad (12.1)$$

The retained masses need not be integers when p is nonintegral. What matters is that X_j/j^p is a nonnegative integer. Since $0 \leq X_j \leq n$, a zero occurs no later than an index with $j^p > n$. At $n = 0$ the convention is $A_p(0) = 1$.

The square- and cube-power cases are OEIS A082527 and A082528, respectively, introduced by Benoit Cloitre in 2003 [2, 3]. The latter record conjectures an equivalent $(c(p)n)^{1/(p+1)}$ for every real $p > 0$. The first-power sequence, beginning at $n = 1$, is A073047 [1]. Define the survival threshold and, more generally, a terminal-quotient threshold by

$$B_p(k) = \min\{n \in \mathbb{Z}_{\geq 0} : X_k > 0\}, \quad (12.2)$$

$$B_p(k, h) = \min\{n \in \mathbb{Z}_{\geq 0} : X_k \geq hk^p\}, \quad k, h \in \mathbb{N}. \quad (12.3)$$

Thus $B_p(k) = B_p(k, 1)$. Throughout, $\mathbb{N} = \{1, 2, \dots\}$ and a subscript on $\mathcal{O}$ specifies the allowed dependence of its constant.

Use the parameters

$$s = p + 1, \quad a = \frac{1}{s}, \quad \beta = \frac{s}{s + 1}, \quad K_p = \frac{1}{p\Gamma(1 - a)^s}, \quad c_p = \frac{1}{K_p}. \quad (12.4)$$

The exponent a governs Gamma products; β is the cutoff exponent in the error estimates. They are different quantities.

Theorem 12.1 (Thresholds and first-zero indices). *For every fixed real $p > 0$, as $k, n \rightarrow \infty$ through integers,*

$$B_p(k) = K_p k^{p+1} + \mathcal{O}_p\left(k^{(p+1)^2/(p+2)}\right), \quad (12.5)$$

$$A_p(n) = p^{1/(p+1)} \Gamma\left(\frac{p}{p+1}\right) n^{1/(p+1)} + \mathcal{O}_p\left(n^{1/((p+1)(p+2))}\right). \quad (12.6)$$

The implied constants can be chosen uniformly when p ranges over a fixed compact subset of $(0, \infty)$.

In particular, the constants inside the OEIS equivalents are

$$c_2 = 2\Gamma(2/3)^3, \quad A_2(n) = (c_2 n)^{1/3} + \mathcal{O}(n^{1/12}), \quad (12.7)$$

$$c_3 = 3\Gamma(3/4)^4, \quad A_3(n) = (c_3 n)^{1/4} + \mathcal{O}(n^{1/20}). \quad (12.8)$$

Their diagnostic decimal values are $c_2 \approx 4.965917162430456$ and $c_3 \approx 6.764822981008765$. The exact Gamma expressions are the results; these decimal displays are not certified intervals.

Remark 12.2 ([write] The two decimal displays are wrong in their last digits). [Added 7 October 2026, batch 110.] The displays above are printed as delivered, and both are wrong. Evaluated to forty digits (mpmath),

$$\begin{aligned} c_2 &= 2\Gamma(2/3)^3 = 4.96591716243045547322\dots, \\ c_3 &= 3\Gamma(3/4)^4 = 6.76482298100876023357\dots, \end{aligned}$$

in agreement with Part I's (1.7). Rounded to fifteen decimals they are 4.965917162430455 and 6.764822981008760; Report 188's ...456 and ...765 are neither these roundings nor truncations. The error is confined to the two diagnostic displays, which Report 188 itself labels non-certified; no statement of Part II uses them.

The proof provides more than the equivalent. Define

$$Q_p = \max\{2^{p+1} + p + 2, 2^p(p + 3) + 1\}, \quad E_p = 2Q_p \log Q_p + 2. \quad (12.9)$$

For every pair of integers $1 \leq J \leq k$, we will prove the effective enclosure

$$K_p k^s e^{-sE_p/J} - \frac{J^s}{s} \leq B_p(k) \leq K_p k^s e^{sE_p/J} + \frac{J^s}{s}. \quad (12.10)$$

The constants are deliberately conservative. This is a rigorous real-valued inequality, not a claim that evaluating its endpoints with ordinary floating point produces a rigorous numerical enclosure.

A globally uniform profile for all intermediate masses is proved in Section 18. Section 19 gives positive-terminal-mass thresholds and their limiting inverse, including the important distinction between $h \geq 1$ and $h = 0$.

13 Exact reverse thresholds

The backward formulation removes the unknown stopping time from the recurrence.

Proposition 13.1 (Minimal reverse path). *For $k, h \in \mathbb{N}$, define integers $q_j = q_j(k, h)$ by*

$$q_k = h, \quad q_j = \left\lceil \left(\frac{j+1}{j} \right)^p q_{j+1} \right\rceil \quad (j = k-1, k-2, \dots, 1). \quad (13.1)$$

Then $B_p(k, h) = q_1$. Starting the forward recurrence at q_1 realizes the entire reverse path:

$$X_j = j^p q_j \quad (1 \leq j \leq k). \quad (13.2)$$

Proof. To reach a prescribed mass $(j+1)^p q_{j+1}$ after the next floor operation, the preceding mass must be at least that value. Among the allowed masses at stage j , namely the nonnegative integer multiples of j^p , the least such mass is $j^p q_j$ from (13.1). Backward induction proves both necessity and attainability at the initial level $1^p = 1$.

There is no overshoot in the subsequent forward process. With $c = ((j+1)/j)^p > 1$ and an integer $m \geq 1$,

$$m \leq \frac{\lceil cm \rceil}{c} < m + \frac{1}{c} < m + 1.$$

Consequently the forward floor quotient of q_j at level $j+1$ is exactly q_{j+1} . This proves (13.2), including its terminal value $X_k = hk^p$. $\square$

Proposition 13.2 (Exact discrete inverse). *The sequence $B_p(k)$ is strictly increasing, $B_p(1) = 1$, and for $n \geq 1$,*

$$A_p(n) = 1 + \max\{k \geq 1 : B_p(k) \leq n\}. \quad (13.3)$$

In particular, for every $k \geq 1$,

$$A_p(B_p(k) - 1) = k, \quad A_p(B_p(k)) = k + 1. \quad (13.4)$$

Proof. A reverse path ending at $k+1$ with quotient 1 has quotient at least 2 at level k . For $c > 1$, the map $m \mapsto \lceil cm \rceil$ preserves an integral difference of at least one. Reversing the remaining levels shows $B_p(k+1) > B_p(k)$. Monotonicity of the forward floor maps shows that survival through level k is equivalent to $n \geq B_p(k)$. Since zero is absorbing, (13.3) follows. At the two adjacent values it gives (13.4); the case $k=1$ uses $A_p(0) = 1$. $\square$

For reference, the first ten survival thresholds are

$$\begin{aligned} p = 2 : & \quad 1, 4, 12, 20, 36, 72, 84, 136, 216, 244, \\ p = 3 : & \quad 1, 8, 32, 88, 136, 272, 520, 920, 1168, 1816. \end{aligned}$$

These are thresholds indexed by k , rather than the OEIS first-zero sequences indexed by n .

[Added 7 October 2026, batch 110: [proposition 13.1](#) at $h=1$ and [proposition 13.2](#) are Part I's survival duality ([lemma 3.2](#)) and exact inverse ([corollary 3.3](#)), proved by the same backward induction; Report 188 adds a general terminal quotient h and the observation (13.2) that the minimal input realizes the whole reverse path. The thresholds above are those of Part I's example ($T_3(4) = 88$); the write recomputed all twenty with exact integers.]

13.1 An integer algorithm for rational powers

Let $p = A/B > 0$ be rational, with positive coprime integers A, B . If the next reverse quotient is q , the preceding quotient is the least integer r satisfying

$$j^A r^B \geq (j+1)^A q^B. \quad (13.5)$$

No algebraic or floating approximation to j^p is needed. For the forward step from level j to $j+1$, the quotient is the greatest nonnegative integer r satisfying

$$(j+1)^A r^B \leq j^A q^B. \quad (13.6)$$

Both are integer-root comparisons. To compute $\lfloor (U/V)^{1/B} \rfloor$, first set $w = \lfloor U/V \rfloor$ and find the largest r with $r^B \leq w$ by integer binary search. The ceiling equals this r precisely when $Vr^B = U$, and otherwise equals $r+1$. The equality test is essential at perfect powers.

This algorithm proves exact computability for rational p ; it does not restrict the real- p theorem. An arbitrary real parameter has no implied finite exact input representation. The software accepts exact positive rational parameters and validates them before calculation.

[Added 7 October 2026, batch 110: this is the integer-root method of Part I, (9.2) and (9.3), with the letters A/B for Part I's p/q .]

14 A monotone coordinate for the reverse path

For $u > 0$ define

$$f_p(u) = u + \lceil pu \rceil, \quad f_p(0) = 1, \quad I_p(u) = \int_0^u \frac{dv}{f_p(v)}. \quad (14.1)$$

The value at zero is the right-hand limit. The function f_p is positive and nondecreasing. The function I_p is continuous, strictly increasing, locally absolutely continuous, and unbounded. Indeed,

$$\max\{1, su\} \leq f_p(u) \leq su + 1 \quad (u \geq 0), \quad (14.2)$$

where the upper inequality is strict for $u > 0$. Thus

$$\frac{1}{s} \log(1 + su) \leq I_p(u) \leq u. \quad (14.3)$$

In particular the inverse I_p^{-1} exists on $[0, \infty)$.

Set $u_j = q_j/j$ for a reverse trajectory. These scaled quotients increase as j decreases. The crucial fact is an exact drift sandwich, rather than an assumed average rounding error.

Lemma 14.1 (Drift sandwich). *For every $p > 0$ and $1 \leq j < k$,*

$$f_p(u_{j+1}) \leq j(u_j - u_{j+1}) \leq f_p(u_j). \quad (14.4)$$

Proof. Write $u = u_{j+1}$, $x = 1 + 1/j$, and $d = (j+1)(x^p - 1)$. Integrality of $q_{j+1} = (j+1)u$ gives

$$j(u_j - u) = u + \lceil du \rceil. \quad (14.5)$$

By convexity of $x \mapsto x^{p+1}$ on $(0, \infty)$,

$$d = j(x^{p+1} - x) \geq p.$$

The reverse ceiling also gives $u_j \geq x^{p+1}u$. Convexity of $x \mapsto x^{-p}$ gives $x^{-p} \geq 1 - p/j$, hence

$$\frac{d}{x^{p+1}} = j(1 - x^{-p}) \leq p.$$

Therefore $pu \leq du \leq pu_j$. Applying the monotonicity of the ceiling in (14.5), and then using $u \leq u_j$, proves (14.4). The argument does not require convexity of x^p and remains valid when $0 < p < 1$. $\square$

Let

$$R_j = \frac{f_p(u_j)}{f_p(u_{j+1})} \geq 1. \quad (14.6)$$

Monotonicity of f_p , followed by (14.4), yields

$$\frac{1}{jR_j} \leq I_p(u_j) - I_p(u_{j+1}) \leq \frac{R_j}{j}.$$

Consequently

$$\left| I_p(u_j) - I_p(u_{j+1}) - \frac{1}{j} \right| \leq \frac{R_j - 1}{j}. \quad (14.7)$$

The jumps of f_p cause no difficulty: monotone endpoint bounds control the integral, and values at the finitely many cell endpoints in a bounded interval do not change that integral.

15 Dyadic control without a logarithmic loss

A uniform bound on R_j alone would only give a harmonic error when (14.7) is summed. The logarithms of these ratios telescope, and a dyadic decomposition uses this additional structure.

Define the exact rounding defects

$$e_j = q_j - \left(\frac{j+1}{j}\right)^p q_{j+1}, \quad 0 \leq e_j < 1. \quad (15.1)$$

Multiplication by j^p gives the mass identity

$$j^p q_j = (j+1)^p q_{j+1} + j^p e_j. \quad (15.2)$$

For any $1 \leq L \leq H \leq k$, it telescopes to

$$L^p q_L = H^p q_H + \sum_{j=L}^{H-1} j^p e_j. \quad (15.3)$$

Lemma 15.1 (Uniform ratio bounds). *With Q_p as in (12.9), one has $1 \leq R_j \leq Q_p$. If $1 \leq L \leq H \leq \min(2L, k)$, then*

$$\frac{f_p(u_L)}{f_p(u_H)} \leq Q_p. \quad (15.4)$$

These bounds do not depend on the terminal quotient $h \geq 1$.

Proof. The ceiling recursion implies

$$u_j \leq (1 + 1/j)^s u_{j+1} + 1/j \leq 2^s u_{j+1} + 1.$$

Using (14.2) separately on each numerator term gives

$$R_j \leq 2^s + s + 1 = 2^{p+1} + p + 2.$$

For the block estimate, divide (15.3) by L^s and bound the sum by $L(2L)^p$. Since $H/L \leq 2$,

$$u_L \leq 2^s u_H + 2^p.$$

It follows that

$$\frac{f_p(u_L)}{f_p(u_H)} \leq 2^s + s2^p + 1 = 2^p(p+3) + 1.$$

Only positivity of u_H and the defect bound were used, so the constants are independent of h . $\square$

Theorem 15.2 (Quantified integral invariant). *For every real $p > 0$, integers $k, h \geq 1$, and $1 \leq J \leq k$, the reverse path (13.1) satisfies*

$$|I_p(u_J) - I_p(h/k) - \log(k/J)| \leq \frac{E_p}{J}. \quad (15.5)$$

For $h = 1$ one also has

$$|I_p(u_J) - \log(k/J)| \leq \frac{E_p}{J}. \quad (15.6)$$

Proof. For $1 \leq R \leq Q_p$,

$$R - 1 = \int_1^R dv \leq Q_p \int_1^R \frac{dv}{v} = Q_p \log R.$$

On a block $[L, H]$ with $H \leq 2L$, the ratio product telescopes, so

$$\begin{aligned} \sum_{j=L}^{H-1} \frac{R_j - 1}{j} &\leq \frac{Q_p}{L} \sum_{j=L}^{H-1} \log R_j \\ &= \frac{Q_p}{L} \log \frac{f_p(u_L)}{f_p(u_H)} \leq \frac{Q_p \log Q_p}{L}. \end{aligned}$$

Partition $[J, k]$ at $J, 2J, 4J, \dots$, with its last endpoint truncated to k . The geometric sum of $1/L$ is at most $2/J$, giving

$$\sum_{j=J}^{k-1} \frac{R_j - 1}{j} \leq \frac{2Q_p \log Q_p}{J}. \quad (15.7)$$

This is the step that removes a potential logarithmic loss.

Sum (14.7) and use $u_k = h/k$. The elementary integral comparison

$$0 \leq \sum_{j=J}^{k-1} \frac{1}{j} - \log(k/J) \leq \frac{1}{J} - \frac{1}{k}$$

proves (15.5). When $h = 1$, use $0 \leq I_p(1/k) \leq 1/k$ from (14.3). Its contribution together with the harmonic comparison is at most $1/J$, proving (15.6). The $+2$ in E_p is a safe common allowance; these estimates would permit a slightly smaller constant. Empty sums when $J = k$ are covered by the same argument. $\square$

All estimates so far are deterministic. In particular, the proof has imposed no randomness, independence, or equidistribution assumption on the defects e_j .

16 Gamma products and the continuous profile

The coordinate I_p can be evaluated explicitly. Set

$$P_0 = 1, \quad P_r = \prod_{m=1}^r \left(1 - \frac{a}{m}\right) = \frac{\Gamma(r+1-a)}{\Gamma(1-a)\Gamma(r+1)} \quad (r \geq 1). \quad (16.1)$$

The Gamma identity follows repeatedly from $\Gamma(z+1) = z\Gamma(z)$.

Proposition 16.1 (Exact cell formula). *For every integer $r \geq 1$ and $(r-1)/p \leq u \leq r/p$,*

$$e^{-I_p(u)} = P_{r-1} \frac{r + (r-1)/p}{r + u}. \quad (16.2)$$

In particular, $e^{-I_p(r/p)} = P_r$.

Proof. Inside this cell, $f_p(u) = u + r$, except possibly at its lower endpoint. Its integral contributes a logarithm. Starting from $I_p(0) = 0$, the factor across a complete cell is

$$\frac{r + (r-1)/p}{r + r/p} = \frac{sr - 1}{sr} = 1 - \frac{a}{r}.$$

Multiplying these factors over the preceding cells and then integrating the last partial cell gives (16.2). Adjacent formulas agree at common endpoints by this same factor identity. $\square$

Lemma 16.2 (Gamma bounds and the constant). *For $r \geq 1$,*

$$\frac{(r+1-a)^{-a}}{\Gamma(1-a)} \leq P_r \leq \frac{r^{-a}}{\Gamma(1-a)}. \quad (16.3)$$

Consequently, as $u \rightarrow \infty$,

$$e^{I_p(u)} = \Gamma(1-a)(pu)^a(1 + \mathcal{O}_p(u^{-1})). \quad (16.4)$$

Proof. Here are the needed classical log-convexity estimates, included to specify every inequality. For $x > 0$ and $0 < t < 1$, interpolation between x and $x+1$ gives

$$\Gamma(x+t) \leq x^t \Gamma(x).$$

Interpolation between $x+t$ and $x+t+1$, at the point $x+1$, gives

$$\Gamma(x+1) \leq (x+t)^{1-t} \Gamma(x+t).$$

Together,

$$\left(\frac{x}{x+t}\right)^{1-t} \leq \frac{\Gamma(x+t)}{x^t \Gamma(x)} \leq 1.$$

Taking $x = r$ and $t = 1-a$ gives (16.3). These are Wendel-type bounds derived directly from log-convexity; standard Gamma inequalities are discussed in [10, §5.6].

The bounds imply $P_r = r^{-a} \Gamma(1-a)^{-1} (1 + \mathcal{O}_p(r^{-1}))$. On the r th cell, $r = pu + \mathcal{O}(1)$ and the partial-cell factor in (16.2) is $1 + \mathcal{O}_p(r^{-1})$. Taking reciprocals proves (16.4). $\square$

Define for $0 < t \leq 1$

$$U_p(t) = I_p^{-1}(-\log t), \quad H_p(t) = t^s U_p(t). \quad (16.5)$$

Equation (16.4) gives the finite limit

$$H_p(0) := \lim_{t \downarrow 0} H_p(t) = \frac{1}{p \Gamma(1-a)^s} = K_p. \quad (16.6)$$

The profile is explicit at every positive t . On the interval $P_r \leq t \leq P_{r-1}$,

$$U_p(t) = \frac{(sr-1)P_{r-1}}{pt} - r, \quad H_p(t) = \frac{(sr-1)P_{r-1}}{p} t^p - rt^s. \quad (16.7)$$

Thus the profile is a sequence of elementary arcs joined at Gamma-product breakpoints. In particular $U_p(1) = H_p(1) = 0$. The limit at zero was proved independently of any numerical interpolation.

Proposition 16.3 (A signed global profile bound). *The function H_p is continuous and nonincreasing on $[0, 1]$, and*

$$0 \leq K_p - H_p(t) \leq \frac{t^s}{s} \quad (0 \leq t \leq 1). \quad (16.8)$$

Equivalently, for every $u \geq 0$,

$$0 \leq K_p e^{s I_p(u)} - u \leq \frac{1}{s}. \quad (16.9)$$

Proof. On every closed subinterval of $(0, 1]$, the inverse coordinate and the profile are absolutely continuous. At almost every t ,

$$tU_p'(t) = -f_p(U_p(t)),$$

and hence

$$H_p'(t) = t^p [pU_p(t) - \lceil pU_p(t) \rceil] \in [-t^p, 0]. \quad (16.10)$$

Integrate between δ and t , then let $\delta \downarrow 0$ and use (16.6). The derivative bound is integrable at zero, so the result is (16.8); it also proves absolute continuity on $[0, 1]$. Substituting $t = e^{-I_p(u)}$ and dividing by t^s gives (16.9), including $u = 0$ by continuity. $\square$

The sign and the $1/s$ constant in (16.9) are useful. Mere asymptotic equivalence from Gamma ratios would not by itself provide this all- u interpolation bound.

Remark 16.4 ([write] The profile of this section is Part I's). [Added 7 October 2026, batch 110.] Take $p = m$, so that $s = m + 1$ and $a = 1/(m + 1)$. The factor $1 - a/r$ in (16.1) is then $1 - 1/((m + 1)r)$, so $P_r = t_r$ for every $r \geq 0$: the products of this section are the crossing times (1.8) of Part I, and (16.1) is (5.1), because $1 - a$ here is Part I's $a = m/(m + 1)$. On $P_r \leq t \leq P_{r-1}$, since $sr - 1 = (m + 1)r - 1$, formula (16.7) gives

$$tU_p(t) = \frac{(m + 1)r - 1}{m} t_{r-1} - rt,$$

which is $y_m(t)$ on $t_r \leq t \leq t_{r-1}$ by (1.9). Hence $tU_p(t) = y_m(t)$ on $(0, 1]$ and $H_p(t) = t^m \cdot tU_p(t) = t^m y_m(t) = f_m(t)$, the function of lemma 5.1; (16.6) is (5.3), and $K_p = L_m$. With $z = my_m(t)/t = pU_p(t)$ as in the proof of lemma 5.1, (16.10) is (5.4) in the form $f_m'(t) = t^m(z - C(z))$, for $pU_p(t) > 0$, that is for $t < 1$. Integrating it from 0 to t is (16.8); Part I performs the same integration at the crossing times in the proof of proposition 6.1, and at $u = j/p$, where $e^{-I_p(u)} = P_j = t_j$, (16.9) multiplied by t_j^{m+1} is exactly Part I's bracket (6.1). Proposition 16.3 is thus Part I's estimate at every t rather than at the crossing times, by the same argument. In Part III the coordinate I_p reappears as the integral H of the one-type law $\delta_{(p,1)}$ (remark 24.1).

17 Effective thresholds and their inverse

We now prove the all-index enclosure and Theorem 12.1.

Theorem 17.1 (All-index threshold enclosure). *For every real $p > 0$ and integers $1 \leq J \leq k$, (12.10) holds.*

Proof. Use the reverse trajectory with $q_k = 1$. The invariant (15.6) implies

$$K_p(k/J)^s e^{-sE_p/J} \leq K_p e^{sI_p(u_J)} \leq K_p(k/J)^s e^{sE_p/J}.$$

By (16.9), u_J lies between the middle quantity minus $1/s$ and the middle quantity. Multiply by J^s to bound $J^p q_J$.

The exact tail of (15.3) gives

$$0 \leq B_p(k) - J^p q_J = \sum_{j=1}^{J-1} j^p e_j \leq \sum_{j=1}^{J-1} j^p \leq \frac{J^s}{s}. \quad (17.1)$$

The last bound follows by comparison with $\int_0^J x^p dx$, since x^p is increasing. Combining the two inequalities gives the lower endpoint with $-J^s/s$ and the upper endpoint with $+J^s/s$. $\square$

Proof of the threshold assertion in Theorem 12.1. Take $J = \max\{1, \lfloor k^\beta \rfloor\}$. As $k \rightarrow \infty$, J is comparable to k^β , and

$$e^{\pm s E_p/J} = 1 + \mathcal{O}_p(J^{-1}).$$

Theorem 17.1 therefore gives

$$B_p(k) = K_p k^s + \mathcal{O}_p(k^s/J + J^s).$$

The choice $\beta = s/(s+1)$ balances these two errors, because

$$s - \beta = s\beta = \frac{s^2}{s+1}.$$

This is (12.5). Bounded k can be absorbed into the implied constant. The formulas for Q_p, E_p, K_p are bounded on each compact positive p -interval, with K_p bounded away from zero; all elementary estimates used above are uniform there. This proves the stated local uniformity. $\square$

Proof of the inverse assertion in Theorem 12.1. Let $m = \max\{k : B_p(k) \leq n\}$. Then

$$B_p(m) \leq n < B_p(m+1).$$

The threshold equivalent first gives $m \asymp_p n^{1/s}$. Apply (12.5) at m and $m+1$. The change in the smooth main term is $\mathcal{O}_p(m^{s-1})$, smaller than $m^{s-\beta}$ because $\beta < 1$. It follows that

$$n = K_p m^s + \mathcal{O}_p(m^{s-\beta}).$$

Dividing by $K_p m^s$ and taking the $1/s$ power yields

$$m = K_p^{-1/s} n^{1/s} + \mathcal{O}_p(n^{(1-\beta)/s}).$$

Finally $A_p(n) = m+1$, and 1 is absorbed because $(1-\beta)/s = 1/(s(s+1)) > 0$. Since $K_p^{-1/s} = p^{1/s} \Gamma(p/s)$, this proves (12.6). Only the explicit smooth power $K_p x^s$ was expanded; no unknown discrete remainder was differentiated. $\square$

[Added 7 October 2026, batch 110: the leading terms of theorem 12.1 are Part I's theorem 1.1, here by a second route. Part I obtains them by compactness of the rescaled backward arrays (lemmas 4.1 and 4.2, proposition 4.3), which yields no rate; Report 188 obtains them through the coordinate I_p and the explicit enclosure of theorem 17.1. The remainders are new. They answer Part I's Research question 1: with $m = p$, (12.5) is $E_m(K) = O_m(K^{m+1-\delta_m})$ with $\delta_m = (m+1)/(m+2)$, since $(m+1) - (m+1)^2/(m+2) = (m+1)/(m+2)$, and the implied constants are uniform on compact m -intervals, as that question asks. At $p = 1$ the exponent $(p+1)^2/(p+2)$ is $4/3$, the classical Erdős–Jabotinsky bound (2.1), which is not improved (section 20).]

17.1 Using the enclosure for finite inverse bounds

The all-index formula also gives rigorous inverse tests without an asymptotic onset. For $1 \leq J \leq k$ write

$$L_p(k, J) = K_p k^s e^{-s E_p/J} - J^s/s, \quad U_p(k, J) = K_p k^s e^{s E_p/J} + J^s/s.$$

If $U_p(k, J) \leq n$, then $B_p(k) \leq n$ and $A_p(n) \geq k+1$. If $L_p(k, J) > n$, then $B_p(k) > n$ and $A_p(n) \leq k$. One may optimize over the finite set of cutoffs J to improve these sufficient tests. Exact rational-power thresholds from Section 13.1 give exact inverse answers. Ordinary floating evaluations of L_p, U_p are useful diagnostics but do not certify either comparison.

18 The global minimal mass profile

For the reverse threshold path with $q_k = 1$, normalize its actual retained mass as

$$M_{k,j} = \frac{j^p q_j}{k^s} = \left(\frac{j}{k}\right)^s u_j. \quad (18.1)$$

A local inverse-function argument alone would only control j away from zero. The exact mass tail extends the approximation across the entire trajectory.

Theorem 18.1 (Uniform trajectory profile). *For every fixed $p > 0$,*

$$\max_{1 \leq j \leq k} \left| \frac{j^p q_j}{k^s} - H_p(j/k) \right| = \mathcal{O}_p(k^{-\beta}). \quad (18.2)$$

For every fixed $\varepsilon > 0$, uniformly for $\varepsilon k \leq j \leq k$,

$$\frac{q_j}{k} = \frac{j}{k} U_p(j/k) + \mathcal{O}_{p,\varepsilon}(k^{-1}). \quad (18.3)$$

Proof. First we derive a location-dependent bound. For nonnegative v, w , (14.2) and integration imply

$$\left| \log \frac{sv + 1}{sw + 1} \right| \leq s |I_p(v) - I_p(w)|. \quad (18.4)$$

Let $t = j/k$ and $w = U_p(t)$. By (15.6), the right side with $v = u_j$ is at most sE_p/j . Also (14.3) gives

$$t^s (sU_p(t) + 1) \leq 1. \quad (18.5)$$

Consequently

$$\begin{aligned} |t^s u_j - H_p(t)| &\leq \frac{1}{s} (e^{sE_p/j} - 1) \\ &\leq \frac{e^{sE_p} - 1}{sj}. \end{aligned} \quad (18.6)$$

For the last inequality use convexity of $x \mapsto e^{sE_p x}$ on $[0, 1]$, at $x = 1/j$. This proof also works at $j = k$, where $U_p(1) = 0$, and does not assume differentiability at that endpoint.

Take $J = \max\{1, \lfloor k^\beta \rfloor\}$. For $j \geq J$, (18.6) is $\mathcal{O}_p(J^{-1})$. For $j \leq J$, (15.3) gives

$$0 \leq M_{k,j} - M_{k,J} \leq \frac{J^s}{sk^s}.$$

The monotone profile bound, or direct integration of (16.10), similarly gives

$$0 \leq H_p(j/k) - H_p(J/k) \leq \frac{J^s}{sk^s}.$$

Compare the two masses at J using (18.6); the triangle inequality then gives a uniform error

$$\mathcal{O}_p(J^{-1} + (J/k)^s) = \mathcal{O}_p(k^{-\beta}),$$

since $s(1 - \beta) = \beta$. This proves (18.2).

If $j \geq \varepsilon k$, the invariant bounds $I_p(u_j)$ uniformly, so u_j and $U_p(j/k)$ belong to a fixed compact interval depending only on p, ε . On such an interval I_p^{-1} is Lipschitz, because its almost-everywhere derivative is the bounded function f_p . Their I_p coordinates differ by $\mathcal{O}_p(1/j) = \mathcal{O}_{p,\varepsilon}(1/k)$, hence the quotients themselves differ by that amount. Multiply by j/k to obtain (18.3). $\square$

The function H_p decreases from K_p to zero. There is no conflict between $H_p(1) = 0$ and survival at the final discrete level: the surviving terminal mass is k^p , which becomes $1/k$ after division by k^s . The theorem concerns the minimal surviving trajectory. An arbitrary initial condition should not be identified with this particular profile without specifying its terminal quotient.

[Added 7 October 2026, batch 110: by [remark 16.4](#), $H_p = f_m$ and $U_p(t) = y_m(t)/t$, so (18.3) is the convergence (4.7) of Part I with a rate $O(1/k)$ on $[\varepsilon, 1]$, and (18.2) extends it, after multiplication by t^m , to the whole interval with rate $O(k^{-\beta})$. Both concern the backward minimal path. They do not give a rate for the forward trajectories of [theorem 1.2](#) or for the loss measures of [corollary 6.2](#), so Part I's Research question 7 stays open.]

19 Positive terminal mass and its inverse curve

Define

$$\Phi_p(v) = K_p e^{sI_p(v)} \quad (v \geq 0). \quad (19.1)$$

This is continuous and strictly increasing, with $\Phi_p(0) = K_p$ and $\Phi_p(v) \rightarrow \infty$. Proposition 16.3 gives the useful global bounds

$$v \leq \Phi_p(v) \leq v + \frac{1}{s}. \quad (19.2)$$

On $(r-1)/p \leq v \leq r/p$, its exact formula is

$$\Phi_p(v) = K_p \left(\frac{r+v}{P_{r-1}[r+(r-1)/p]} \right)^s. \quad (19.3)$$

Theorem 19.1 (Uniform terminal-mass law). *For every real $p > 0$, integers $k, h \geq 1$, and $1 \leq J \leq k$,*

$$\begin{aligned} k^s \Phi_p(h/k) e^{-sE_p/J} - \frac{J^s}{s} &\leq B_p(k, h) \\ &\leq k^s \Phi_p(h/k) e^{sE_p/J} + \frac{J^s}{s}. \end{aligned} \quad (19.4)$$

For each fixed $V > 0$, uniformly over integers $1 \leq h \leq Vk$,

$$B_p(k, h) = k^s \Phi_p(h/k) + \mathcal{O}_{p,V}(k^{s-\beta}). \quad (19.5)$$

Proof. Use (15.5) rather than (15.6). It places $K_p e^{sI_p(u_J)}$ between

$$(k/J)^s \Phi_p(h/k) e^{-sE_p/J} \quad \text{and} \quad (k/J)^s \Phi_p(h/k) e^{sE_p/J}.$$

Now apply (16.9), multiply by J^s , and add the same mass tail (17.1). The defects still lie in $[0, 1)$ and the invariant constants are independent of h , proving (19.4) for every positive h . When $h/k \leq V$, (19.2) bounds the main coefficient by $V + 1/s$. Choosing $J = \max\{1, \lfloor k^\beta \rfloor\}$ proves (19.5). $\square$

The zero-terminal boundary is excluded. If one extends the definition to $h = 0$, then $B_p(k, 0) = 0$ exactly: no initial mass is required to retain at least zero. In contrast, $\Phi_p(0) = K_p > 0$ describes the limiting cost of a *positive* terminal quotient with $h/k \rightarrow 0$. Thus neither (19.4) nor (19.5) is asserted for $h = 0$. This is a genuine boundary discontinuity of the finite threshold problem. It cannot be removed by substituting $h = 0$ into a formula proved only for $h \geq 1$.

19.1 The limiting inverse

The inverse of Φ_p has domain $[K_p, \infty)$. If

$$\frac{K_p}{P_{r-1}^s} \leq \lambda \leq \frac{K_p}{P_r^s}, \quad (19.6)$$

then direct inversion of (19.3) gives

$$\Phi_p^{-1}(\lambda) = P_{r-1} \left[r + \frac{r-1}{p} \right] \left(\frac{\lambda}{K_p} \right)^{1/s} - r. \quad (19.7)$$

At adjacent endpoints these formulas coincide; at $\lambda = K_p$ they give zero. From (19.2),

$$\max\{0, \lambda - 1/s\} \leq \Phi_p^{-1}(\lambda) \leq \lambda \quad (\lambda \geq K_p).$$

The formula in (19.7) is an exact expression for the *limiting curve*. It is not an exact finite- k quotient obtained by a single floor or ceiling.

For clarity, this curve has the following direct retention interpretation. Let N_k be nonnegative integers with $N_k/k^s \rightarrow \lambda \geq 0$, and let $R_k = X_k/k^p$ be the quotient retained from initial mass N_k . Then

$$\frac{R_k}{k} \rightarrow \begin{cases} 0, & 0 \leq \lambda \leq K_p, \\ \Phi_p^{-1}(\lambda), & \lambda > K_p. \end{cases} \quad (19.8)$$

Indeed, $R_k \geq h$ is equivalent to $N_k \geq B_p(k, h)$ for each integer $h \geq 1$. Fix any $v > 0$ and take $h = \lceil vk \rceil$. By (19.5), $B_p(k, h)/k^s \rightarrow \Phi_p(v)$. Comparing λ with $\Phi_p(v)$ bounds the upper and lower limits of R_k/k . If $\lambda > K_p$, take v successively below and above the unique inverse value. If $\lambda \leq K_p$, every positive v has $\Phi_p(v) > \lambda$, forcing the limit to be zero. At $\lambda = K_p$ this proves only a zero *normalized* quotient; the unnormalized quotient need not vanish.

Remark 19.2 ([write] The retention limit is Part I's forward profile). [Added 7 October 2026, batch 110.] Take $p = m$, so $K_p = L_m$, and let $\lambda > L_m$. Put $t = (L_m/\lambda)^{1/(m+1)} \in (0, 1)$. Then $I_p(\Phi_p^{-1}(\lambda)) = \frac{1}{s} \log(\lambda/K_p) = -\log t$, so $\Phi_p^{-1}(\lambda) = U_p(t) = y_m(t)/t$ by [remark 16.4](#), and since $\lambda = L_m t^{-(m+1)}$ and $F_m = c_m t^m y_m$,

$$\Phi_p^{-1}(\lambda) = \lambda F_m(t).$$

Thus (19.8) says that the mass $X_k = k^p R_k$ retained at stage k is the fraction $F_m(t)$ of N_k in the limit, where $t = \lim k/\tau_m(N_k)$. This also follows from Part I. Indeed $N_k \rightarrow \infty$, and [theorem 1.1](#) gives $\tau_m(N_k) \sim (c_m \lambda)^{1/(m+1)} k$, so $k/\tau_m(N_k) \rightarrow t$. With $J = \tau_m(N_k)$ one has $X_k/N_k = X_{N_k}(k/J)$ in the notation of [theorem 1.2](#), and

$$\left| \frac{X_k}{N_k} - F_m(t) \right| \leq \sup_{0 \leq u \leq 1} |X_{N_k}(u) - F_m(u)| + |F_m(k/J) - F_m(t)| \rightarrow 0$$

by (1.11) and the continuity of F_m ; this is the case $\lambda > K_p$ of (19.8), because $R_k/k = (X_k/N_k)(N_k/k^s)$. If $0 < \lambda < L_m$, then $k/\tau_m(N_k) \rightarrow t > 1$, so eventually $k \geq \tau_m(N_k)$ and $R_k = 0$. If $\lambda = L_m$, then for $k < J$ the same estimate gives $X_k/N_k \rightarrow F_m(1) = 0$, and $R_k = 0$ for $k \geq J$. If $\lambda = 0$, then $R_k/k \leq N_k/k^s \rightarrow 0$. So the whole of (19.8) is a consequence of [theorems 1.1](#) and [1.2](#), reparametrized; Report 188's proof through (19.5) is a second route. The positive-terminal law (19.5) itself, with its rate and its uniformity in $h \leq V_k$, is new.

19.2 The associated terminal-mass profile

The same proof also controls the whole minimal trajectory for a prescribed positive terminal quotient. For $v \geq 0$ put

$$H_{p,v}(t) = t^s I_p^{-1}(I_p(v) - \log t) \quad (0 < t \leq 1), \quad H_{p,v}(0) = \Phi_p(v). \quad (19.9)$$

It has terminal value $H_{p,v}(1) = v$. The Gamma asymptotic proves continuity at zero, and the derivative calculation in (16.10) gives

$$0 \leq \Phi_p(v) - H_{p,v}(t) \leq t^s/s. \quad (19.10)$$

For each fixed $V > 0$, the reverse path with $q_k = h$ satisfies, uniformly over $1 \leq h \leq Vk$,

$$\max_{1 \leq j \leq k} \left| \frac{j^p q_j}{k^s} - H_{p,h/k}(j/k) \right| = \mathcal{O}_{p,V}(k^{-\beta}). \quad (19.11)$$

Here is the full reduction to the preceding proof. With $w = I_p^{-1}(I_p(v) - \log t)$, (15.5) replaces (15.6). The logarithmic comparison (18.4) is unchanged, while (14.3) gives

$$t^s(sw + 1) \leq e^{sI_p(v)} \leq e^{sI_p(V)}.$$

Thus (18.6) acquires only a factor bounded in terms of p, V . The exact mass tail is unchanged, and (19.10) supplies its continuous counterpart. Splitting at $J = \lfloor k^\beta \rfloor$ proves (19.11), with J replaced by 1 when necessary.

20 The classical first power and related models

For $p = 1$, the constant simplifies to $K_1 = 1/\pi$, and our estimates become

$$B_1(k) = \frac{k^2}{\pi} + \mathcal{O}(k^{4/3}), \quad A_1(n) = \sqrt{\pi n} + \mathcal{O}(n^{1/6}). \quad (20.1)$$

This threshold leading term and error order are classical, not a new first-power result. The connection to sieving and Tchoukaillon is described by Erdős and Jabotinsky [5], Broline and Loeb [6], and the threshold record A002491 [4].

It is important not to import a stronger historical remainder without its qualifications. The current A002491 record contains a June 2021 comment by Donald Knuth explaining a problem with the $\mathcal{O}(k)$ remainder claimed by Broline and Loeb. The comment identifies an accumulation of local area errors that costs $\mathcal{O}(k^{3/2})$ in that argument, and suggests that splitting the range recovers $\mathcal{O}(k^{4/3})$. We rely on neither the disputed $\mathcal{O}(k)$ assertion nor the conjectural bounded remainder $A_1(n) - \sqrt{\pi n} = \mathcal{O}(1)$ recorded in A073047. The present proof supplies the safe estimate (20.1) independently.

Several related generalizations deserve distinction at the level of their defining recurrence.

- Erdős and Jabotinsky's general sieve uses integer parameters in ratios of the form $b_j/(b_j - 1)$. Its first-power specialization is historical groundwork for nested-ceiling analysis; the inspected theorem does not state the arbitrary power-grid result (12.6).
- A 2023 generalized rounding problem [8] changes the selected multiple on each *linear* grid. Its Gamma-ratio constant and piecewise-parabola argument concern a different parameterized model with quadratic growth. Gamma products and piecewise profile reasoning are established ideas, not inventions claimed here.

- Li's 2026 paper [9] studies a two-coordinate first-power Tchoukaillon array and proves an interior uniform estimate. Its two coordinates do not change the grid from j to j^p . We do not claim to improve or correct that independent array theorem.

The result addressed here is the particular sequential j^p grid for every fixed real $p > 0$, together with its explicit constant, deterministic error control, and minimal-mass profiles. Targeted source checks found no directly matching square- or cube-power theorem among the inspected material. That bounded check is not a worldwide novelty certification.

[Added 7 October 2026, batch 110: this sentence is stale relative to the repository. Part I of this report, dated 3 October 2026 and written into the repository that day (commit [2b3b69b4b](#)), proves the leading law for the j^p grid for every real $p > 0$, with the constant c_p and the square and cube cases ([theorem 1.1](#), [\(1.5\)](#), [\(1.6\)](#)), and the minimal-mass profile without a rate ([proposition 4.3](#)). Report 188, dated a day later, did not know it. Relative to Part I, what is new here is the deterministic quantitative remainder of [theorem 12.1](#), the enclosure of [theorem 17.1](#), the rates of [theorem 18.1](#) and the positive-terminal law of [section 19](#) ([table 3](#)). The display [\(20.1\)](#) is the classical bound [\(2.1\)](#) and its consequence printed in [section 2](#).]

21 Dependence of the exact constant on the power

This section concerns the smooth parameter function $c_p = p\Gamma(p/(p+1))^{p+1}$. It does not assert additional asymptotic orders in the integer threshold index k .

Let $z = 1/(p+1)$, so $0 < z < 1$. The classical convergent log-Gamma expansion [[10](#), Equation 5.7.3] gives

$$\log \Gamma(1-z) = \gamma z + \sum_{r=2}^{\infty} \frac{\zeta(r)}{r} z^r.$$

Consequently

$$\log \frac{c_p}{p} = \gamma + \sum_{r=2}^{\infty} \frac{\zeta(r)}{r(p+1)^{r-1}}. \quad (21.1)$$

The series converges for each $p > 0$. It supplies every fixed parameter order as $p \rightarrow \infty$; for example,

$$\begin{aligned} c_p = e^{\gamma} p \left[1 + \frac{\zeta(2)}{2(p+1)} \right. \\ \left. + \frac{\zeta(3)/3 + \zeta(2)^2/8}{(p+1)^2} + \mathcal{O}((p+1)^{-3}) \right]. \end{aligned} \quad (21.2)$$

In particular $c_p \sim e^{\gamma} p$. At the other endpoint, the expansion of $\log \Gamma(x)$ at $x = 0^+$ yields

$$\log c_p = -p \log p + (1-\gamma)p + \mathcal{O}(p^2), \quad (21.3)$$

$$c_p = 1 + p(\log(1/p) + 1 - \gamma) + \mathcal{O}(p^2 \log^2 p), \quad p \downarrow 0. \quad (21.4)$$

To see [\(21.3\)](#) explicitly, use

$$\log \Gamma\left(\frac{p}{1+p}\right) = -\log p + \log(1+p) - \gamma \frac{p}{1+p} + \mathcal{O}(p^2)$$

in $\log c_p = \log p + (1+p) \log \Gamma(p/(1+p))$. Thus $c_p \rightarrow 1$ as $p \downarrow 0$.

These are attributed consequences of classical Gamma expansions once the exact constant is known. The proof of [Theorem 12.1](#) is uniform only on compact positive p -intervals; it

supplies no joint k, p regime approaching either endpoint. Expanding c_p or the smooth arcs of H_p cannot eliminate or characterize the remaining deterministic rounding defects.

[Added 7 October 2026, batch 110: (21.1) and (21.3) are Part I's (6.8) and (6.7) (proposition 6.3), from the same Gamma expansion. (21.2) adds the term of order $(p+1)^{-2}$ to Part I's (6.9), and (21.4) is the exponentiated form of (6.7), where Part I states only $c_m \rightarrow 1$. The write checked both numerically to forty digits: at $p = 10$ and $p = 100$ the remainder of (21.2), divided by $e^\gamma p(p+1)^{-3}$, is 0.76 and 0.70; at $p = 10^{-3}$ and $p = 10^{-5}$ the remainder of (21.4), divided by $(p \log p)^2$, is 0.59 and 0.55.]

22 Exact checks and reproducibility

The source archive contains original Python standard-library code, the L^AT_EX source, bounded attributed integer fixtures, and an offline deterministic builder. Its tests are finite mathematical checks, not a replacement for the proofs.

22.1 What is checked exactly

The mandatory checker uses only integers and exact rational arithmetic. It exercises the reverse recursion, independent forward threshold search, forward/reverse path agreement, adjacent threshold boundaries, the drift sandwich, integral-power mass-tail bounds, and Gamma-product interpolation identities at rational breakpoints. In particular, the quantity $e^{I_p(u)}$ is rational when p, u are rational: formula (16.2) evaluates it using only rational products. Testing that formula does not require approximating Γ .

The bounded fixtures contain a total of 342 terms from the official records A002491, A073047, A082527, and A082528, with their original offsets and source attribution. The first is tested as a survival-threshold sequence, while the other three are tested as first-zero sequences. This distinction catches a possible off-by-one convention error. Fixture integrity and malformed-input cases are checked explicitly. The archive contains no full OEIS exports or downloaded third-party papers.

The package also checks the positive-terminal boundary $h \geq 1$ and rejects invalid exact algorithm inputs. Its error checks use explicit exceptions rather than Python `assert` statements, so they remain active under optimized execution. The builder runs the mandatory checker both normally and with `-O`, requiring byte-identical deterministic JSON results. Independent algorithms reduce the risk of merely testing one implementation against itself; they do not prove statements for untested parameters.

22.2 Diagnostics are separate

Optional floating-point diagnostics evaluate exact thresholds against the smooth main term, using ordinary library Gamma values. These outputs are labeled as diagnostics. They do not certify real intervals, prove error bounds, or establish a threshold's exact integer value by rounding a Gamma approximation. For illustration, integer-only computations give

p	$B_p(1000)$	$B_p(10000)$
2	201388916	201457639236
3	148522784144	1478758856906056

The proofs apply to irrational real powers as well; the exact finite software tests intentionally use rational parameters.

22.3 Building and verifying the artifact

The top-level `README.md` and `README_REPRODUCIBILITY.md` specify the precise offline commands and prerequisites. The builder uses an explicit closed source inventory, compiles in isolated temporary storage with fixed metadata, rejects $\text{\TeX}$ warnings and bad boxes, and refuses to replace existing output. It runs package integrity and adversarial guard checks, including optimized execution. The output manifest describes every archive member by size and SHA-256; an external receipt records final output hashes without a self-referential ZIP hash.

[Added 7 October 2026, batch 110: this section describes the delivered archive. In this report its programs are `code/02-rates-*.py`, its fixtures and recorded outputs `data/02-rates-*.json`, and its `README_REPRODUCIBILITY.md`, `DATA_SOURCES.md`, `code/README.md` and `data/README.md` are shipped as `02-rates-*.md`; the top-level `README.md`, `Report188.tex`, the delivered PDF and the checksum manifest `SHA256SUMS.json` are not shipped and survive in the arrival commit. The replay, test and build scripts import one another by their delivered names and check the closed delivered inventory, including `Report188.tex` and `SHA256SUMS.json`, so they run only in a re-extracted archive; the report's `README.md` gives a route that runs the exact checker on a copy. The release receipt mentioned above is written by the builder; none was delivered.]

Reproduction requires a compatible Python and $\text{\TeX}$ toolchain already installed. No network access or downloaded code is needed for the build. Byte-for-byte reproduction is a claim about the same documented toolchain, not about arbitrary $\text{\TeX}$ versions or font packages. The source tree is not modified during a build, and the archive can be extracted and rebuilt into a fresh destination.

23 Further questions

The proof isolates several precise remaining problems.

1. **Sharper deterministic remainders.** Can the exponent $s^2/(s+1)$ in $B_p(k) - K_p k^s$ be reduced for a fixed p ? The present cutoff balances an invariant error against a worst-case positive mass tail. Any improvement must exploit additional structure rather than assume cancellation.
2. **Bounded inverse errors.** Does $A_p(n) - (c_p n)^{1/s}$ remain bounded for particular powers? This would require substantially finer threshold control. The first-power conjecture is not settled here.
3. **Fluctuations and genuine discrete expansions.** Do normalized threshold errors have limiting distributions, deterministic oscillations, or large subsequential excursions? A formal expansion of the Gamma constant does not answer these questions. No all-orders discrete- k expansion is asserted.
4. **Varying powers and perturbed grids.** Which joint regimes in p and k admit uniform estimates? Which perturbations of j^p , including regularly varying grids, preserve an explicit coordinate or profile? The exact drift sandwich uses the power grid and is not a theorem for arbitrary perturbations.
5. **Sharper finite certificates.** The all-index enclosures are explicit but conservative. Better dyadic constants and independently certified Gamma/exponential intervals could make the finite inverse tests practically useful while retaining rigorous endpoints.

The established conclusion is the all-real-positive-power leading law, with a deterministic quantitative remainder and explicit positive-mass profiles. The unresolved questions concern finer discrete fluctuations, additional uniform regimes, and extensions of the model.

[Added 7 October 2026, batch 110: the “established conclusion” above is, for its leading law, Part I’s [theorem 1.1](#); the remainder and the profiles with rates are Report 188’s ([table 3](#)). On the five questions. Question 1 is open; it is the sharpening of Part I’s Research question 1, which [theorem 12.1](#) answers. Question 2 is Part I’s Research question 2, and question 3 is Part I’s Research question 3; both are open. For question 4, Part III answers the second half at leading order for one large class: for nondecreasing weights whose scaled increments are bounded and whose augmented empirical increment law converges, the coordinate H of (27.7) and the profile (27.9) are explicit, and $H = I_p$ for the one-type law $\delta_{(p,1)}$ ([theorem 27.2](#), [remark 24.1](#)). Part III proves no rate and no uniformity in the parameter, and it shows that perturbations far smaller than every power of the index can change the constant ([theorem 31.2](#)); the joint regimes in p and k are Part I’s Research question 6, open, and rates for perturbed grids are Part III’s question 10, open. Question 5 is open.]

[Added 7 October 2026, independent check of the batch-110 write: the clause on [theorem 31.2](#) above concerns perturbations of the repeated weights $\lceil k/2 \rceil$, not of the power grid j^p that question 4 asks about. The mechanism there is that the perturbation makes zero increments active. The grid j^p has no zero increments, and perturbations of it smaller than every inverse power cannot change the constant. Indeed, let $w_1 = 1$ and $w_k = k^p + e_k > 0$ with $e_k = o(k^{-N})$ for every N , and put $\Delta_k = e_k(k-1)^p - e_{k-1}k^p$. Then

$$k\left(\frac{w_k}{w_{k-1}} - 1\right) = k\left(\left(\frac{k}{k-1}\right)^p - 1\right) + \frac{k \Delta_k}{(k-1)^p w_{k-1}} \longrightarrow p,$$

because $|\Delta_k| \leq (|e_k| + |e_{k-1}|)k^p = o(k^{p-1})$ and $(k-1)^p w_{k-1} \asymp k^{2p}$. So Part I’s [theorem 1.3](#) applies (no monotonicity is needed) and gives $T_w(K) \sim L_p K w_K \sim L_p K^{p+1}$; for nondecreasing w the augmented law is $\delta_{(p,1)}$. The same holds whenever the relative perturbation of the successive ratios is $o(1/k)$. Which larger perturbations of j^p keep the constant, and with what rate, remains open, as stated.]

Part III

Microscopic schedules in rounding extinction: averaged ceiling laws, exact Gamma constants, discontinuous perturbations, and a sharp spectrum of limits

24 Provenance, scope and notation of Part III

[Added 7 October 2026, batch 110.] Part III was added in the write of batch 110. It is built from the manuscript *Microscopic Schedules in Rounding Extinction*, which starts from Part I’s Research question 4 and answers it for a large class of weight schedules. Everything of that manuscript is printed below: its abstract and scope paragraph in this section, its sections as [section 25](#) to [section 37](#), and its two appendices, “Proof dependencies and source audit” and

“Notation reference”, as the closing [sections 38 and 39](#) of Part III (this report’s own appendices follow Part III). Its symbols are kept ([section 24.3](#)). Bracketed notes beginning “[Added 7 October 2026, batch 110]” and remarks marked [write] are the write’s; all other text of Part III is the manuscript’s.

24.1 The manuscript

The manuscript is the archive `Periodic_Rounding_Extinction.zip` (1,681,098 bytes; 23 files in an inner directory `periodic_rounding_extinction/`), which arrived in commit `e4d5dcf9e`, was held from batch 114 so that this report would receive one write, and was placed in this report by commit `8622ca7e5` (batch 110). Its article `periodic_rounding_extinction.tex` (1,901 lines; a 29-page PDF) is titled *Microscopic Schedules in Rounding Extinction*, subtitled *Averaged ceiling laws, exact Gamma constants, discontinuous perturbations, and a sharp spectrum of limits*, and dated 5 October 2026. Its author line and PDF author field read “Research article prepared for Vladimir Reshetnikov with OpenAI”, as Part I’s do; it names OpenAI as the tool and no human author. Its pin is ProveIt commit `52d8ca404b076c38eb7c513569592ebd8c479115` (5 October 2026), quoted in [section 25](#) and in its bibliography entry for this report. Its source audit, `03-schedules-SOURCE_AUDIT.txt`, records the Git blob identities of Part I’s `body.tex`, `extensions.tex`, `questions.tex`, `bibliography.tex` and `README.md` at the pin; the write confirmed that all five equal the blobs at the pin and at the parent of this write. The manuscript’s statements about the repository (Part I has the backward-threshold method, the power-weight constant, smooth-ratio universality and the repeated-weight example; its open problem is Research question 4, label `rex:q:schedules`) are true. It does not know Part II, which reached the repository later.

The manuscript cited this report twice, as its bibliography entries `repo-report` (the report, with the pin and directory) and `repo-questions` (its Section 10 and the label `rex:q:schedules`). In Part III the one sentence citing them points to Part I instead ([section 25](#)); the other citations go to this report’s bibliography, the two papers of Erdős and Jabotinsky to its single entry for both, and DLMF to its Chapter 5 entry with the section the manuscript names. Figure files carry the prefix `03-schedules-`. The uncaptioned notation table of the manuscript’s second appendix received a caption ([table 10](#)), so that its PDF destination is unique, and the `latexmk` command line of [section 36](#) is set in a smaller font so that it fits this report’s narrower A4 text width. Labels carry the prefix `rex:sch:`, and the ten questions of [section 37](#) received labels. No other text was changed.

[Added 7 October 2026, independent check of the batch-110 write: the figure 1,681,098 bytes above is the total size of the archive’s 23 members; the archive itself has 1,129,393 bytes. Part I’s author line reads “Research report prepared for Vladimir Reshetnikov with OpenAI”; “as Part I’s do” above means the same form, with “article” in place of “report”. The numbering differs from the delivered PDF, although no label was lost. The manuscript’s Sections 1–13 and Appendices A and B are Sections 25–37 and 38–39 here, its Figures 1–4 are Figures 4–7, its Tables 1–3 are Tables 7–9 (its uncaptioned notation table is Table 10), and its Questions 1–10 are Research questions 10–19, which continue Part I’s count. Part II likewise numbers Report 188’s equations (1)–(63) within its Sections 12–21, as (12.1)–(21.4). Neither text cites a number literally, so every reference follows. The five blob identities, the pin and its date, and the manuscript’s statements about the repository were confirmed.]

The manuscript’s abstract and its scope paragraph read as follows.

Starting from an integer n , repeatedly round down to a multiple of a positive weight w_k . The power weights $w_k = k^m$ lead to OEIS A073047, A082527, and A082528. A recent ProveIt report determines their leading extinction constants and asks how oscillating

microscopic weight changes alter the answer. We prove a general averaging theorem for bounded local increments whose empirical distribution converges. The necessary state records both $\beta_k = k(w_k/w_{k-1} - 1)$ and whether the increment is strictly positive. An active increment tending to zero contributes one integer unit to the backward dynamics; an exact repetition contributes none. The limiting drift is an average of ceilings, and a single integral determines the complete trajectory and extinction constant. Commensurable positive slopes give finite Gamma products. We construct weights that differ by less than every inverse power of their index but have different leading constants, and weights with periodic limiting slopes for which the constant does not exist. Bounded modifications on a set of density zero preserve the law. Finally, at each fixed growth index $m > 0$, every constant in the sharp interval $(0, 1/(m+1))$ is realized by a finite-type schedule. A finite Möbius inversion recovers the entire bounded increment law from its drift. The proofs include finite enclosures for the constants, an explanation of the losses caused by almost invisible weight changes, and reproducible exact-arithmetic checks.

Scope and status. This is a self-contained research manuscript with proofs and accompanying computations. The periodic-schedule and bounded sparse-exception questions of the cited repository report are resolved under the hypotheses stated here. The classical linear asymptotic and the previously reported power-weight Gamma constant are credited. We do not claim to settle the classical bounded-error conjecture or to establish exhaustive historical priority. The manuscript has not been peer reviewed or formally verified.

Status and non-claims of the manuscript, kept. It credits the classical linear asymptotic, the backward-threshold method and the power-weight Gamma constant (Erdős–Jabotinsky; Part I), credits successive increment regimes to the sieve literature and discontinuous averaging to the literature on piecewise-smooth averaging, and does “not claim to settle the classical bounded-error conjecture or to establish exhaustive historical priority”; its targeted search “does not certify global novelty”. Its main theorem has no rate in K . The spectrum theorem concerns positive real weights, allows the increment bound to depend on the constant, and uses dormant phases; it is not claimed for integer weights or strictly increasing schedules. The recovery theorem concerns exact data, not stable recovery from noisy observations. Its decimal displays are rounded, and its exact rational certificates are the stored fractions. The manuscript “has not been peer reviewed or formally verified”; nothing in Part III is formalized in Lean or Rocq. Neither the manuscript nor the write submitted anything to OEIS.

24.2 What Part III adds

Table 5: The manuscript of Part III against Parts I and II.

Part III	Parts I and II	Status here
Lemma 26.1, (26.4)–(26.6)	(7.6), (7.7)–(7.9)	same, without strict increase of the thresholds
Lemma 28.1	lemma 7.1	the same conclusions from convergent empirical means instead of convergent ratios
Theorem 27.2	theorem 1.3; proposition 16.3	new; the law $\delta_{(m,1)}$ gives Part I’s theorem for nondecreasing weights, and its certificate (27.16) is (16.9) (remark 24.1)
Theorem 30.1, (30.5), corollary 30.2	(1.2), proposition 8.1	new; one type gives L_m , and $a = 0$ gives the block constant L_m/r
Theorems 31.1, 31.2 and 32.1, proposition 31.3	section 8	new: instability below every power of the index; a schedule with no limit

Part III	Parts I and II	Status here
Theorem 33.1	corollary 6.2	new: the loss measure marked by the increment state; its time marginal for $\delta_{(m,1)}$ is (6.6)
Theorem 34.1 , corollary 34.2	question “Stability under sparse exceptional steps”	new; answers its bounded form
Proposition 35.1 , theorem 35.2	—	new: the constants at fixed m are exactly $(0, 1/(m+1))$
Proposition 35.3 , theorem 35.4	—	new: the drift determines the law

Remark 24.1 ([write] The one-type law is Parts I and II). [Added 7 October 2026, batch 110.] Let $\mu = \delta_{(m,1)}$, the law of nondecreasing weights with $k(w_k/w_{k-1} - 1) \rightarrow m$. Then $\eta = 0$, $\rho = 1$, and $D(z) = C(mz)$. Since $C(mu) = \lceil mu \rceil$ for $u > 0$ and $C(0) = 1$, the integrand $u + D(u)$ is Part II’s $f_p(u)$ with $p = m$, so $H = I_m$ and $t(z) = e^{-I_m(z)}$. Hence $z = U_m(t)$ along the profile, and (27.9) gives $y(t) = tU_m(t) = y_m(t)$ by [remark 16.4](#). The constant is $L(\mu) = \lim_{z \rightarrow \infty} z e^{-(m+1)I_m(z)} = L_m$ by (16.4) and (16.6), F is F_m , and the certificate (27.16), which reads $z t(z)^{m+1} \leq L \leq (z + 1/(m+1)) t(z)^{m+1}$, is (16.9) with $u = z$; at $z = j/m$, where $t(z) = t_j$, it is Part I’s bracket (6.1). [Theorem 27.2](#) thus contains Part I’s [theorem 1.3](#) for nondecreasing weights; Part I’s hypothesis also admits finitely many early decreases, which [theorem 27.2](#) excludes. The two Parts use different normalizations of one letter: Part I’s $z(t)$ is $my(t)/t$, Part III’s $z(t)$ in (28.5) is $y(t)/t$. With that, the time marginal of [theorem 33.1](#) for $\delta_{(m,1)}$ has density $t^m(C(mz) - mz)/L_m$ with Part III’s z , which is Part I’s (6.6).

Questions of Part I. Research question 4 asks for the limits when $k(w_k/w_{k-1} - 1)$ oscillates, has repeated zero values or follows a periodic pattern, and which information beyond the index determines the constant and the profile. [Theorem 27.2](#) answers it for nondecreasing weights with uniformly bounded scaled increments whose augmented empirical law converges: the additional information is that law, the joint frequency of the scaled increment and of the activity flag, through the averaged drift D ; the order of the steps does not enter, and D determines the law ([theorem 35.4](#)). Outside that class the question stays open, re-scoped as Part III’s questions on unbounded or signed increments (Research question 14) and on schedules without a limiting law (Research question 17); without a limiting law the constant need not exist ([theorem 32.1](#)). The question “Stability under sparse exceptional steps” (Research question 5) is answered in its bounded form by [corollary 34.2](#): bounded changes on a set of density zero leave the law, the constant and the profiles unchanged. Whether a sparse set of very large jumps can change the constant stays open (Research question 15). Dated notes at Part I’s questions say so.

24.3 Notation of Part III

Part III keeps the manuscript’s symbols; no symbol is renamed and no normalization is changed. It shares most of Part I’s letters with the same meanings: w_k , x_k , $\tau_w(n)$, $T_w(K)$, $q_k^{(K)}$, b_k , the positive ceiling C , and m , L and F in the power case ([remark 24.1](#)). [Table 6](#) lists the letters that mean something else in Parts I or II.

Table 6: Letters of Part III that have other meanings in Parts I and II.

Part III	Meaning	Meaning elsewhere
m	mean slope $\int \beta d\mu$	Part I: the exponent; equal for power weights
$L = L(\mu)$	limit of $T_w(K)/(Kw_K)$	Part I: $L_m = L(\delta_{(m,1)})$; Part II: K_p
$z(t) = y(t)/t$		Part I: $z(t) = my(t)/t$
β_k, σ_k	scaled increment, activity flag	Part II: $\beta = s/(s+1)$ a cutoff exponent
$\mu; \mu_N; \nu_n$	limiting law; empirical laws; marked loss measures	Part I: μ_n are the loss measures (6.5)
ρ, η	activity, active-zero mass	Part I: ρ_m is the loss density
$D(z), E(z)$	averaged drift, residual $D(z) - mz$	Part I: $E_m(K)$ in Research question 1; Part II: the constant E_p
$H(z), t(z)$	integral and time of the profile	Part II: $H_p(t)$ is a mass profile; Part I: t_j are crossing times (here $t(j/m) = t_j$ for $\delta_{(m,1)}$)
$M(t) = t^m y(t)$		Part I: f_m ; Part II: $M_{k,j}$
$y(t)$	backward profile	Part I: y_m , the case $\delta_{(m,1)}$
$Q; Q_k$	cell length $1/\lambda$; forward quotients	Part I: Q_k forward quotients; Part II: Q_p a ratio bound
A_j, B_j	Gamma arguments	Part II: $A_p(n), B_p(k)$
$B; \mathcal{E}_B$	increment bound; state space	Part II: B_p ; the denominator B of $p = A/B$
a	number (or frequency a/r) of active zero phases	Part I: $a = m/(m+1)$; Part II: $a = 1/s$
r	period length	Part I: the block length r (the same role, proposition 8.1)
$s; p_s; \alpha_s; h_s$	type index; frequency; slope; integer multiple	Part II: $s = p+1$, p the exponent
p in section 36	length of a periodic tuple	Part II: the exponent
$h_j; h$ in theorem 32.1	perturbation sizes; block index	Part II: terminal quotient
$A \subseteq \mathbb{N}, \theta$	set of perturbed indices, its density	Part II: $A_p(n)$
c	thinning factor	Part I: c_m
R	cutoff in proofs	Part II: R_j, R_k
$f(x), G(x), \mathbf{m}$	tail of the slope law, its sum, Möbius function	Part I: $f_m = t^m y_m$

Intake checks. The dossier of the intake (6 October 2026) reran the manuscript’s program on a copy with `--no-figures`: it exited normally in 41 s; the four CSV files were byte-identical to the recorded ones, the three JSON files equal apart from line ends, and the console output equal to the recorded log apart from the line reporting the figures, which that run did not regenerate. It recomputed $T/(Kw_K) = 0.1591535$ for $\varepsilon = 0$ and 0.3927036 for $\varepsilon = 10^{-4}$ at $K = 10^5$ ([table 7](#)), and found the two-phase constant (30.5) inside its own finite certificate $[0.33185619, 0.33193915]$ at $z = 2000$. The write evaluated the Gamma products to forty digits: the two-phase constant is $0.33189767362899860\dots$, the three-phase constant of [table 8](#) is $0.32927650881086035\dots$, and the expansion in (35.8) holds, the remainder times r^2 being 0.0625 for $m = 1$ and 0.0278 for $m = 2$ at $r = 1000$.

25 The OEIS problem and the new questions

For positive weights $w_1 = 1, w_2, w_3, \dots$, define

$$x_1 = n, \quad x_k = w_k \left\lfloor \frac{x_{k-1}}{w_k} \right\rfloor \quad (k \geq 2), \quad \tau_w(n) = \min\{k \geq 1 : x_k = 0\}. \quad (25.1)$$

Here n is a nonnegative integer, and $\tau_w(0) = 1$. Intermediate values may be real, although x_k/w_k is always an integer. Every step removes a nonnegative amount. If $w_k \rightarrow \infty$, extinction occurs in finite time.

The choices $w_k = k, k^2, k^3$ are respectively [A073047](#), [A082527](#), and [A082528](#). The cubic entry records Cloitre's question about a leading equivalent for general real $m > 0$ [[1](#), [2](#), [3](#)]. The associated least initial value surviving through stage K is

$$T_w(K) = \min\{n \in \mathbb{N} : \tau_w(n) > K\}. \quad (25.2)$$

For $w_k = k$, this is the classical Tchoukaillon threshold sequence [A002491](#). Erdős and Jabotinsky proved $T_w(K) = K^2/\pi + O(K^{4/3})$ [[5](#)].

The ProveIt report *Gamma Constants and Scaling Limits for Rounding Extinction* establishes the fixed- m power-weight formula

$$L_m = \frac{1}{m\Gamma(m/(m+1))^{m+1}}, \quad T_{k^m}(K) \sim L_m K^{m+1}, \quad (25.3)$$

and extends it to weights with $k(w_k/w_{k-1} - 1) \rightarrow m$. Its microscopic schedule question explicitly asks what happens when those local ratios oscillate, vanish repeatedly, or have a periodic pattern (Part I of this report: [theorem 1.3](#) and Research question 4). We take that question as the starting point, and recover [\(25.3\)](#) as a specialization of the theory below.

25.1 What the article establishes

The new results concern the dependence on a schedule, rather than a new proof of the classical constant in isolation.

1. **An empirical-distribution theorem.** A bounded local increment law, augmented by a binary activity flag, determines a unique backward profile, the leading threshold constant, the stopping-time equivalent, and the full normalized forward trajectory. Periodic patterns are a finite-support case; arbitrary deterministic schedules with convergent empirical distributions are included.
2. **Exact constants.** Every finite pattern of commensurable positive slopes has a finite Gamma-product expression. Weakly active zero slopes contribute an explicit additive term to the drift.
3. **A sharp instability.** For $w_{2j-1} = j$ and $w_{2j} = j + \varepsilon/j$, the normalized threshold constant is $1/(2\pi)$ at $\varepsilon = 0$, and $\pi/8$ at every $0 < \varepsilon < 1$. The relative factor is $\pi^2/4$. The perturbation can instead be smaller than every inverse power.
4. **Failure of a limit.** Even the same periodic limiting slopes and the same leading weights $w_k \sim k/2$ can coexist with two distinct subsequential threshold constants.
5. **Stability and a complete range of constants.** Bounded changes on a density-zero set do not alter the normalized law. For fixed $m > 0$, the set of constants over the stated class of schedules is exactly $(0, 1/(m+1))$.

6. **Exact inverse recovery.** The limiting drift determines the entire bounded augmented increment law by a finite Möbius inversion. The normalized forward shape determines it up to an explicitly characterized drift scaling.

25.2 Attribution and the inspected repository

The source audit is pinned to the repository revision

52d8ca404b076c38eb7c513569592ebd8c479115.

The inspected directory is

```
SetTheory/Cardinals/docs/reports/
generating-functions-and-asymptotics/
oeis-sequence-asymptotics/a082528-rounding-extinction/.
```

The source files `body.tex`, `extensions.tex`, and `questions.tex` were read, and their Git blob identities were checked against that revision. The existing report already contains the backward-threshold method, the smooth power law, and repeated-weight examples. The distributional averaging theorem, the general finite Gamma product, and the instability and spectrum results below are extensions developed here. The present proofs do not require accepting an unproved claim in the repository.

Successive integer-increment regimes and product limits go back to the classical sieve literature [5]. Discontinuous averaging is also an established subject [11]. Our assertion of contribution is specific to the theorem, hypotheses, constants, and counterexamples for (25.1); it is not a claim to have invented averaging. A targeted search did not identify the same schedule theorem in the primary sources examined. That search does not certify global novelty.

26 Exact survival duality and rounding-error accounting

The forward process repeatedly discards information. Reversing the survival condition produces an exact integer recurrence with a useful one-sided error estimate.

Lemma 26.1 (Backward thresholds). *For a fixed horizon K , define*

$$q_K^{(K)} = 1, \quad q_{k-1}^{(K)} = \left\lceil \frac{w_k}{w_{k-1}} q_k^{(K)} \right\rceil \quad (k = K, K-1, \dots, 2). \quad (26.1)$$

Then $T_w(K) = q_1^{(K)}$. More generally, an integer quotient Q at stage k survives through K if and only if $Q \geq q_k^{(K)}$. Consequently,

$$\tau_w(n) > K \iff n \geq T_w(K), \quad \tau_w(n) = J \iff T_w(J-1) \leq n < T_w(J). \quad (26.2)$$

The second equivalence is stated for $n > 0$ and $J \geq 2$.

Proof. At the terminal stage, survival means quotient at least one. If the least surviving quotient at stage k is q_k , then an input $w_{k-1}Q$ survives precisely when

$$\left\lceil \frac{w_{k-1}Q}{w_k} \right\rceil \geq q_k, \quad \text{equivalently} \quad Q \geq \frac{w_k q_k}{w_{k-1}}.$$

Taking the least integer Q gives (26.1). Backward induction reaches stage one, where $w_1 = 1$ and the initial values are integers. The inverse statements follow directly. $\square$

The thresholds are nondecreasing in K . They need not be strictly increasing when some weights are repeated; no strictness will be used. If $w_K \rightarrow \infty$, they are unbounded because a surviving initial value is at least w_K .

Suppress the superscript K , and write $b_k = w_k/w_{k-1} - 1$. The integrality of q_k gives

$$q_{k-1} - q_k = \lceil b_k q_k \rceil. \quad (26.3)$$

The original ceiling also gives the exact accounting inequality

$$0 \leq w_{k-1}q_{k-1} - w_k q_k < w_{k-1}. \quad (26.4)$$

Summing it yields both of the estimates used throughout the paper:

$$w_k q_k \leq w_K + \sum_{j=k}^{K-1} w_j, \quad (26.5)$$

$$0 \leq T_w(K) - w_k q_k < \sum_{j=1}^{k-1} w_j \quad (k \geq 2). \quad (26.6)$$

At $k = 1$ the latter difference is zero. Thus the entire initial segment of the reversed process has a controlled weighted cost. No hypothesis about equidistribution of fractional parts is involved.

27 The local increment law and the main theorem

Assume from now on that the weights are nondecreasing. Define

$$\beta_k = k \left(\frac{w_k}{w_{k-1}} - 1 \right), \quad \sigma_k = \mathbf{1}_{\{w_k > w_{k-1}\}} \quad (k \geq 2). \quad (27.1)$$

The number β_k measures the local relative change on its natural $1/k$ scale. The flag σ_k records whether a change occurs at all.

For some finite B , suppose $0 \leq \beta_k \leq B$. The compact state space relevant to limits is

$$\mathcal{E}_B = \{(0, 0)\} \cup ([0, B] \times \{1\}). \quad (27.2)$$

Although a finite-stage pair cannot equal $(0, 1)$, that point can be a limit of positive increments tending to zero. It is deliberately distinct from $(0, 0)$.

Definition 27.1 (Convergent increment law). The schedule has limiting law μ if the empirical probability measures satisfy

$$\mu_N = \frac{1}{N-1} \sum_{k=2}^N \delta_{(\beta_k, \sigma_k)} \Rightarrow \mu \quad (N \rightarrow \infty) \quad (27.3)$$

weakly on $\mathcal{E}_B$. Set

$$m = \int \beta \, d\mu, \quad \rho = \int \sigma \, d\mu, \quad \eta = \mu(\{(0, 1)\}), \quad (27.4)$$

and assume $m > 0$. Here ρ is the asymptotic fraction of active steps, while η is the mass of active steps whose scaled increments vanish in the limiting law.

Weak convergence means convergence of integrals of continuous functions. For a finite set of limiting phase types it reduces to convergence of their frequencies. No randomness is required.

Use the positive ceiling

$$C(u) = \max\{1, \lceil u \rceil\} \quad (u \geq 0), \quad (27.5)$$

and define

$$D(z) = \int_{\mathcal{E}_B} \sigma C(\beta z) d\mu(\beta, \sigma), \quad (27.6)$$

$$H(z) = \int_0^z \frac{du}{u + D(u)}, \quad t(z) = e^{-H(z)}. \quad (27.7)$$

These are ordinary Lebesgue integrals. Changing the values assigned at positive-integer ceiling jumps may change D at a countable set of z -values, but leaves H , the profiles, and their integral formulas unchanged. The convention $C(0) = 1$ remains fixed. Notice the order in (27.6): apply the ceiling to each local type, then average.

Theorem 27.2 (Averaged extinction law). *Under definition 27.1, the weights are regularly varying with index m :*

$$\frac{w_{\lfloor Kt \rfloor}}{w_K} \longrightarrow t^m \quad \text{locally uniformly for } t > 0. \quad (27.8)$$

The function $t(z)$ decreases continuously from 1 to 0. It defines a unique profile by

$$y(t(z)) = z t(z), \quad y(1) = 0. \quad (27.9)$$

The limit

$$L = L(\mu) = \lim_{z \rightarrow \infty} z t(z)^{m+1} \quad (27.10)$$

exists, is positive, and has the strict bounds

$$\frac{\eta}{m+1} < L < \frac{\rho}{m+1} \leq \frac{1}{m+1}. \quad (27.11)$$

The following conclusions hold:

$$\frac{q_{\lfloor Kt \rfloor}^{(K)}}{K} \longrightarrow y(t) \quad \text{locally uniformly on } (0, 1], \quad (27.12)$$

$$T_w(K) \sim LKw_K, \quad (27.13)$$

$$n \sim L\tau_w(n)w_{\tau_w(n)}. \quad (27.14)$$

For $J = \tau_w(n)$, the entire forward trajectory satisfies

$$\sup_{0 \leq t \leq 1} \left| \frac{x_{\max(1, \lfloor Jt \rfloor)}}{n} - F(t) \right| \longrightarrow 0, \quad F(t) = \frac{t^m y(t)}{L}, \quad F(0) = 1. \quad (27.15)$$

Finally, every finite $z \geq 0$ gives the certified enclosure

$$\left(z + \frac{\eta}{m+1} \right) t(z)^{m+1} \leq L \leq \left(z + \frac{\rho}{m+1} \right) t(z)^{m+1}. \quad (27.16)$$

The interpolated version of (27.12) and its floor-indexed version differ by $O_\delta(1/K)$ on every $[\delta, 1]$. We use either without changing the limiting assertion.

The theorem identifies the leading term and the limiting shape. The mere convergence in (27.3) can be arbitrarily slow; no numerical rate in K is part of its conclusion.

Corollary 27.3 (Finite phase patterns). *Let s_k take values in a finite set S , with $\#\{k \leq N : s_k = s\}/N \rightarrow p_s$. For each type fix $\alpha_s \geq 0$ and an activity flag $e_s \in \{0, 1\}$, with $\alpha_s > 0$ implying $e_s = 1$. Suppose*

$$\beta_k - \alpha_{s_k} \longrightarrow 0, \quad \sigma_k = e_{s_k} \quad \text{eventually.}$$

Then [theorem 27.2](#) applies with

$$\mu = \sum_s p_s \delta_{(\alpha_s, e_s)}, \quad m = \sum_s p_s \alpha_s, \quad D(z) = \sum_s p_s e_s C(\alpha_s z), \quad (27.17)$$

provided $m > 0$. In particular, a periodic pattern of length r has $p_s = 1/r$ when its phases are listed separately.

For a zero slope there are two possibilities: exact plateau phases have $e_s = 0$, and phases with positive $o(1/k)$ relative growth have $e_s = 1$. The numerical slopes alone do not specify the theorem's input.

28 Proof of the averaged profile theorem

We give the averaging argument in detail because ordinary pointwise convergence of the discrete slopes is false for a periodic schedule. The averaging must also cross the ceiling discontinuities and select the nonzero solution at the terminal endpoint.

28.1 Regular variation from empirical means

Lemma 28.1 (Weight ratios and small-time sums). *Under the hypotheses of [theorem 27.2](#), $w_K \rightarrow \infty$ and [\(27.8\)](#) holds. For every $0 < \delta \leq 1$,*

$$\frac{1}{K w_K} \sum_{1 \leq j < \delta K} w_j \longrightarrow \frac{\delta^{m+1}}{m+1}. \quad (28.1)$$

Proof. The function $(\beta, \sigma) \mapsto \beta$ is continuous and bounded, so $A(N) = \sum_{k=2}^N (\beta_k - m) = o(N)$. Also, uniformly in k ,

$$\log \frac{w_k}{w_{k-1}} = \log(1 + \beta_k/k) = \frac{\beta_k}{k} + O_B(k^{-2}).$$

Abel summation of $(\beta_k - m)/k$ over $a < k \leq b$ expresses its sum as endpoint terms $A(b)/b - A(a)/a$ plus a sum bounded by a constant times $\sum_{k=a}^{b-1} |A(k)|/k^2$. Therefore that sum is $o(1)$ uniformly when $\delta K \leq a \leq b \leq MK$, for fixed positive δ, M . Adding the harmonic sum of m/k proves the locally uniform ratio limit.

For completeness the same argument controls the singular endpoint. Given $0 < \varepsilon < m$, choose a sufficiently small $\varepsilon' > 0$ so that $|A(j)| \leq \varepsilon' j$ for large j . Abel summation then bounds the centered harmonic sum over any $j < k \leq K$ by $2\varepsilon' + \varepsilon' \log(K/j)$, up to a fixed initial constant. The $O(k^{-2})$ remainder is summable. Enlarging a constant if necessary gives

$$\frac{w_j}{w_K} \leq C_\varepsilon \left(\frac{j}{K} \right)^{m-\varepsilon} \quad (1 \leq j \leq K). \quad (28.2)$$

The same harmonic estimates show $w_K \rightarrow \infty$. On $[a, \delta]$ with $a > 0$, the ratio limit yields the Riemann-sum limit $\int_a^\delta t^m dt$. Bound [\(28.2\)](#) makes the omitted sum below aK uniformly tend to zero as $a \downarrow 0$. This proves [\(28.1\)](#). $\square$

28.2 Compactness and terminal selection

Interpolate the points $y_K(k/K) = q_k^{(K)}/K$ linearly. On a fixed $[\delta, 1]$, (26.5), monotonicity of the weights, and lemma 28.1 give $q_k/K = O_\delta(1)$. Its discrete increments obey

$$0 \leq q_{k-1} - q_k = \left\lceil \frac{\beta_k q_k}{k} \right\rceil \leq \frac{Bq_k}{k} + 1 = O_\delta(1). \quad (28.3)$$

The interpolants are consequently uniformly bounded and uniformly Lipschitz there. Arzelà–Ascoli and a diagonal extraction yield a locally uniform subsequential limit y on $(0, 1]$, with $y(1) = 0$.

At every active step, the argument of the ceiling is strictly positive, so the increment is at least one. Hence

$$q_{k-1} - q_k \geq \sigma_k.$$

Since the empirical activity frequency tends to ρ , summing over an interval of indices and passing to the limit gives

$$y(s) - y(t) \geq \rho(t - s) \quad (0 < s < t \leq 1). \quad (28.4)$$

In particular, $\rho > 0$, $y(t) \geq \rho(1 - t)$, and

$$z(t) = \frac{y(t)}{t} \quad (28.5)$$

is continuous and strictly decreasing on $(0, 1]$. Indeed,

$$\frac{y(s)}{s} - \frac{y(t)}{t} = \frac{y(s) - y(t)}{s} + y(t) \left(\frac{1}{s} - \frac{1}{t} \right) > 0.$$

It runs from infinity to zero. The lower bound in (28.4) prevents an artificial interval on which the terminal solution remains identically zero.

28.3 Why the ceilings can be averaged

Lemma 28.2 (Joint empirical averaging). *For each $0 < \delta < 1$, the finite measures*

$$\frac{1}{K} \sum_{\delta K \leq k \leq K} \delta_{(k/K, \beta_k, \sigma_k)} \Rightarrow \mathbf{1}_{[\delta, 1]}(t) \, dt \, \mu(d\beta, d\sigma). \quad (28.6)$$

Along a subsequence for which $y_K \rightarrow y$ locally uniformly, the corresponding integrated ceiling increments converge to $\int D(y(t)/t) \, dt$ on every compact time interval.

Proof. For a continuous function g on $\mathcal{E}_B$, convergence of its prefix averages implies

$$\frac{1}{K} \sum_{aK < k \leq bK} g(\beta_k, \sigma_k) \longrightarrow (b - a) \int g \, d\mu$$

for fixed $0 \leq a < b \leq 1$. Approximating a continuous function of t by interval-step functions proves (28.6), first for products and then for arbitrary continuous functions by uniform approximation on the compact product space.

The ceiling integrand is $h(t, \beta, \sigma) = \sigma C(\beta z(t))$. It is bounded on the compact time interval. Its discontinuities can occur only when $\sigma = 1$, $\beta > 0$, and $\beta z(t)$ is a positive integer. For every fixed $\beta > 0$, the strict monotonicity of z makes this a finite set of times on a compact interval. For $\beta = 0$, $C(0) = 1$ is continuous from the right, and the two values of σ belong to separate

components of the state space. Fubini's theorem thus shows that the discontinuity set of h has zero measure for the product limit in (28.6). The usual bounded almost-everywhere-continuous form of weak convergence therefore applies to h .

There is also a moving argument: the discrete increment uses $z_K(k/K) = q_k/k$, not $z(k/K)$. Uniform convergence of z_K on the compact interval resolves it as follows. Outside a small closed neighborhood of the finitely many integer levels in the bounded range of βz , the ceilings are eventually identical. The limiting product measure of these neighborhoods tends to zero as their width decreases, by the preceding zero-measure assertion. Weak convergence bounds their empirical masses by arbitrarily small numbers after choosing a slightly larger closed neighborhood. On the omitted set the ceiling is uniformly bounded. Thus replacing z_K by z changes the average by $o(1)$.

Equivalently, one can squeeze the moving ceiling between the ceilings at $z(t) - \varepsilon$ and $z(t) + \varepsilon$ on intervals bounded away from $t = 1$, then let $\varepsilon \downarrow 0$. The final short interval near 1 has uniformly bounded slopes and can be removed first. Both arguments yield the stated integrated limit. $\square$

The preceding proof uses a zero-measure set in the joint state space. It does not presume that the individual periodic slopes converge at each fixed time.

By (26.3) and the definition of C , the slope of y_K on its k th interpolation interval is $-\sigma_k C(\beta_k q_k/k)$. Integrating and applying lemma 28.2 gives

$$y(t) = \int_t^1 D(y(s)/s) ds, \quad y'(t) = -D(y(t)/t) \quad \text{a.e.} \quad (28.7)$$

28.4 Solving the limiting equation

The drift is nondecreasing and locally bounded. More precisely,

$$D(z) \geq \rho, \quad \eta \leq E(z) := D(z) - mz \leq \rho \quad (z \geq 0). \quad (28.8)$$

For $z > 0$ the upper inequality is strict, since positive slopes have $\lceil \beta z \rceil - \beta z < 1$ and positive total mass. The change of variable (28.5) in (28.7) gives

$$\frac{dz}{dt} = -\frac{z + D(z)}{t}. \quad (28.9)$$

Its right-hand side is strictly negative. On compact intervals the inverse change of variable is absolutely continuous; the possible jump values of the monotone function D form a countable set and do not affect the chain rule. Separation gives

$$\int_0^{z(t)} \frac{du}{u + D(u)} = -\log t.$$

Thus every subsequential limit is exactly (27.9). Conversely, (27.7) defines that locally absolutely continuous solution. The denominator is positive, and $D(z) = mz + O(1)$, so H increases continuously from zero to infinity. This proves existence, uniqueness, and convergence of the whole sequence of interpolants.

29 Matching, inversion, and the complete trajectory

29.1 Existence and finite enclosures of the constant

For $z > 0$, differentiation almost everywhere yields

$$\frac{d}{dz} \log(zt(z)^{m+1}) = \frac{D(z) - mz}{z(z + D(z))} \geq 0. \quad (29.1)$$

For $z \geq 1$ the right-hand side is at most $\rho/((m+1)z^2)$. Its integral is finite. Therefore $zt(z)^{m+1}$ increases to a finite positive limit L .

Set $M(t) = t^m y(t)$. Equation (28.7) gives

$$M'(t) = -t^m E(z(t)) \quad \text{a.e.}, \quad M(0+) = L, \quad M(1) = 0. \quad (29.2)$$

Integrating and applying (28.8) proves

$$L = \int_0^1 t^m E(z(t)) dt, \quad (29.3)$$

$$M(t) + \frac{\eta t^{m+1}}{m+1} \leq L \leq M(t) + \frac{\rho t^{m+1}}{m+1}. \quad (29.4)$$

The latter is (27.16). The upper universal bound is strict because $E(z) < \rho$ for all $z > 0$. The lower one is strict because, for z sufficiently small and positive, the positive-slope part of $E(z)$ has positive mass. This proves (27.11).

The lower expression in (27.16) is nondecreasing in z , and the upper expression is nonincreasing. Their relative gap, measured against the lower expression, is

$$\frac{\rho - \eta}{(m+1)z + \eta} \quad (29.5)$$

when that lower expression is positive. This is a finite, explicit certificate rather than an asymptotic estimate with an unspecified constant.

29.2 Threshold matching

Take $k = \lfloor \delta K \rfloor$ in (26.6). The profile and weight-ratio limits give

$$\frac{w_k q_k}{K w_K} \longrightarrow \delta^m y(\delta).$$

The normalized upper error in (26.6) tends to $\delta^{m+1}/(m+1)$ by lemma 28.1. Consequently,

$$\delta^m y(\delta) \leq \liminf_K \frac{T_w(K)}{K w_K} \leq \limsup_K \frac{T_w(K)}{K w_K} \leq \delta^m y(\delta) + \frac{\delta^{m+1}}{m+1}.$$

Letting $\delta \downarrow 0$ proves (27.13).

29.3 Stopping-time inversion

Extinction is finite by lemma 28.1. Its index tends to infinity with n , because each fixed threshold is finite. If $J = \tau_w(n)$, (26.2) gives $T_w(J-1) \leq n < T_w(J)$. Since $w_J/w_{J-1} = 1 + O(1/J)$, both bounds are asymptotic to LJw_J . This proves (27.14), including schedules whose threshold sequence has repeated values.

If $w_K \sim AK^m$ with $A > 0$, it takes the familiar explicit form

$$\tau_w(n) \sim \left(\frac{n}{AL} \right)^{1/(m+1)}. \quad (29.6)$$

For more general regularly varying weights, (27.14) is the intrinsic inverse statement; no artificial constant is assigned to a slowly varying factor.

29.4 Forward paths from two backward barriers

At a stage $k \leq J - 1$, write $Q_k = x_k/w_k$. This value survives to $J - 1$ but not to J . The stagewise part of [lemma 26.1](#) gives

$$q_k^{(J-1)} \leq Q_k < q_k^{(J)}. \quad (29.7)$$

On every compact subinterval of $(0, 1)$, use the two profile limits, the ratio limit for w_k/w_J , and $n \sim LJw_J$ to obtain the forward limit $F(t) = t^m y(t)/L$, uniformly there. Near $t = 1$, both the limiting profile and the last discrete values tend to zero. Near $t = 0$, the forward path is between its value at any fixed $\delta > 0$ and its initial value 1 after normalization. Since $F(\delta) \rightarrow 1$, this proves uniform convergence on all of $[0, 1]$. The endpoint definitions agree with [\(29.2\)](#), so F is continuous and nonincreasing. This completes the proof of [theorem 27.2](#).

30 Exact finite products and Gamma evaluations

For a finite-type schedule, put

$$D(z) = \eta + \sum_{s=1}^r p_s C(\alpha_s z), \quad \alpha_s > 0, \quad p_s > 0, \quad \eta + \sum_s p_s \leq 1. \quad (30.1)$$

Dormant phases do not appear. Let $0 = u_0 < u_1 < \dots < u_N = z$ contain all breakpoints of this drift below a chosen finite z , including z itself. If d_j is the constant value on (u_{j-1}, u_j) , then integration of [\(27.7\)](#) gives the exact formula

$$t(z) = \prod_{j=1}^N \frac{u_{j-1} + d_j}{u_j + d_j}. \quad (30.2)$$

If the slopes, frequencies, and z are rational, this is a rational number. Together with [\(27.16\)](#) it gives exact rational bounds when $m + 1$ is an integer, and exact algebraic bounds when m is rational. Floating-point rounding is not part of those certificates.

Theorem 30.1 (Finite Gamma product). *Suppose the positive slopes in [\(30.1\)](#) are commensurable: $\alpha_s = \lambda h_s$ for some $\lambda > 0$ and positive integers h_s . Put $Q = 1/\lambda$. List the distinct breakpoints in one cell as*

$$0 = u_0 < u_1 < \dots < u_R = Q,$$

and let $d_j = D(u)$ for $u \in (u_{j-1}, u_j)$. Define

$$A_j = \frac{u_{j-1} + d_j}{(m+1)Q}, \quad B_j = \frac{u_j + d_j}{(m+1)Q}. \quad (30.3)$$

Then all these arguments are positive and

$$L = Q \left(\prod_{j=1}^R \frac{\Gamma(B_j)}{\Gamma(A_j)} \right)^{m+1}. \quad (30.4)$$

The frequencies need not be rational.

Proof. Away from cell endpoints, $C(\alpha_s(z + Q)) = C(\alpha_s z) + h_s$. Thus $D(z + Q) = D(z) + mQ$ almost everywhere. Formula (30.2), applied to n complete cells, gives

$$\begin{aligned} t(nQ) &= \prod_{\ell=0}^{n-1} \prod_{j=1}^R \frac{(m+1)\ell Q + u_{j-1} + d_j}{(m+1)\ell Q + u_j + d_j} \\ &= \prod_{j=1}^R \frac{\Gamma(B_j)\Gamma(n + A_j)}{\Gamma(A_j)\Gamma(n + B_j)}. \end{aligned}$$

The Gamma functional equation proves the second equality exactly. Also,

$$\sum_{j=1}^R (B_j - A_j) = \frac{1}{m+1}.$$

The standard Gamma-ratio asymptotic $\Gamma(n+a)/\Gamma(n+b) \sim n^{a-b}$ [10, §5.11(iii)] therefore gives

$$t(nQ) \sim n^{-1/(m+1)} \prod_{j=1}^R \frac{\Gamma(B_j)}{\Gamma(A_j)}.$$

Substitute in $L = \lim_n nQ t(nQ)^{m+1}$. □

30.1 The smooth power constant is recovered

If every limiting slope is m , then $D(z) = C(mz)$, $Q = 1/m$, and there is one cell. Its arguments are $A = m/(m+1)$ and $B = 1$. Formula (30.4) is exactly (25.3). In particular $L_1 = 1/\pi$. The appearance of the classical Gamma constant is thus a one-type case of the schedule theorem, not an additional claim of historical novelty.

[Added 7 October 2026, batch 110: for the whole one-type case, profile, certificate and loss density included, see [remark 24.1](#).]

30.2 Two strictly positive unequal phases

Take two phases with slopes $1/2$ and $3/2$, each of frequency $1/2$. Their mean is $m = 1$, but their drift is

$$D(z) = \frac{1}{2}C(z/2) + \frac{1}{2}C(3z/2),$$

which differs from $C(z)$. Choose $Q = 2$. The three cell intervals end at $2/3, 4/3, 2$, and their drifts are $1, 3/2, 2$. Theorem 30.1 gives

$$L = 2 \left[\frac{\Gamma(5/12)\Gamma(17/24)}{\Gamma(1/4)\Gamma(13/24)\Gamma(5/6)} \right]^2 = 0.3318976736 \dots \quad (30.5)$$

This differs from $1/\pi = 0.3183098861 \dots$ even though both mean slopes equal one. Replacing the drift by the ceiling of the mean would give the wrong constant without involving any zero phase.

30.3 One positive slope and active zero phases

Corollary 30.2 (An explicit activation family). *For a period of length r , suppose one phase has slope rm , a phases have zero slope but remain active, and the other phases are dormant. Here $a \in \{0, 1, \dots, r-1\}$ and $m > 0$. Then*

$$D(z) = \frac{C(rmz) + a}{r}, \quad L_{r,m,a} = \frac{1}{rm} \left[\frac{\Gamma(1 + ma/(m+1))}{\Gamma(m(a+1)/(m+1))} \right]^{m+1}. \quad (30.6)$$

The same formula holds for real $a \in [0, r - 1]$ if active zero-slope steps have frequency a/r , and the positive phase has frequency $1/r$.

Proof. The cell length is $Q = 1/(rm)$ and its constant drift is $(1+a)/r$. The Gamma arguments are $A = m(1+a)/(m+1)$ and $B = 1 + ma/(m+1)$. Apply [theorem 30.1](#). Frequencies, rather than literal phase counts, are all that enters the calculation. $\square$

At $a = 0$ this recovers the repeated-modulus constant L_m/r . At $a > 0$, the Gamma factor changes even though those additional phases contribute zero to the mean slope m .

[Added 7 October 2026, batch 110: the case $a = 0$ is Part I's [proposition 8.1](#). Its weights $w_k = \lceil k/r \rceil^m$ are constant inside each block of r stages and increase at the first stage $k = (j-1)r + 1$ of block $j \geq 2$, where $\beta_k = k((j/(j-1))^m - 1) \rightarrow rm$; their augmented law is $\frac{1}{r}\delta_{(rm,1)} + (1 - \frac{1}{r})\delta_{(0,0)}$, the law of the corollary with $a = 0$, and [\(30.6\)](#) gives L_m/r , which is [\(8.4\)](#).]

31 Discontinuous perturbations and a continuum of constants

31.1 An arbitrarily small perturbation with a fixed effect

Theorem 31.1 (A discontinuity at repeated weights). *For $0 \leq \varepsilon < 1$, define*

$$w_{2j-1}^{(\varepsilon)} = j, \quad w_{2j}^{(\varepsilon)} = j + \frac{\varepsilon}{j} \quad (j \geq 1). \quad (31.1)$$

All these schedules satisfy $w_k^{(\varepsilon)} \sim k/2$ and have the same limiting slopes: $\beta_{2j} \rightarrow 0$, $\beta_{2j+1} \rightarrow 2$. Nevertheless,

$$L(0) = \frac{1}{2\pi}, \quad L(\varepsilon) = \frac{\pi}{8} \quad (0 < \varepsilon < 1). \quad (31.2)$$

Equivalently,

$$T_{w^{(0)}}(K) \sim \frac{K^2}{4\pi}, \quad \tau_{w^{(0)}}(n) \sim 2\sqrt{\pi n}, \quad (31.3)$$

$$T_{w^{(\varepsilon)}}(K) \sim \frac{\pi K^2}{16}, \quad \tau_{w^{(\varepsilon)}}(n) \sim 4\sqrt{n/\pi} \quad (\varepsilon > 0). \quad (31.4)$$

The uniform distance between the weight schedules is exactly $\sup_k |w_k^{(\varepsilon)} - w_k^{(0)}| = \varepsilon$.

Proof. The restriction $\varepsilon < 1$ ensures $j \leq j + \varepsilon/j < j + 1$, so the weights are nondecreasing. Their exact successive ratios are

$$\frac{w_{2j}}{w_{2j-1}} = 1 + \frac{\varepsilon}{j^2}, \quad \frac{w_{2j+1}}{w_{2j}} = \frac{j+1}{j + \varepsilon/j}.$$

These give the stated limiting slopes. At $\varepsilon = 0$, the limiting law is $\frac{1}{2}\delta_{(2,1)} + \frac{1}{2}\delta_{(0,0)}$; at positive ε , it is $\frac{1}{2}\delta_{(2,1)} + \frac{1}{2}\delta_{(0,1)}$. Both have mean $m = 1$. Corollary [30.2](#), with $(r, m, a) = (2, 1, 0)$ and $(2, 1, 1)$, gives

$$\frac{1}{2\Gamma(1/2)^2} = \frac{1}{2\pi}, \quad \frac{1}{2} \left(\frac{\Gamma(3/2)}{\Gamma(1)} \right)^2 = \frac{\pi}{8}.$$

Finally $w_K \sim K/2$, so [\(27.13\)](#) and [\(29.6\)](#) give [\(31.3\)](#) and [\(31.4\)](#). The maximal perturbation occurs at $j = 1$. $\square$

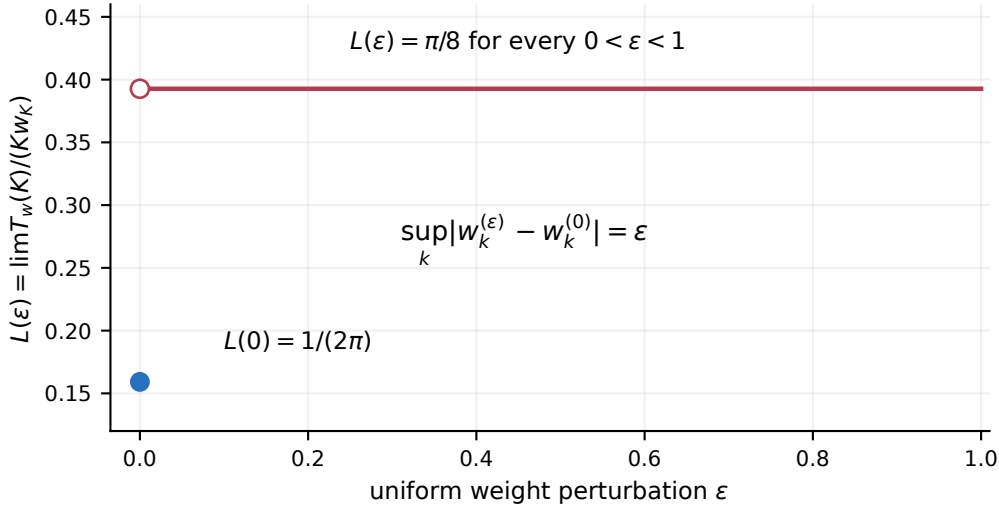


Figure 4: The proved discontinuity in the leading normalized threshold constant for (31.1). The open endpoint on the upper branch excludes $\varepsilon = 0$; its filled value lies on the lower branch. This is a plot of the exact formulas, not an extrapolation of finite-horizon data.

The factor $1/2$ in the leading weights matters. The constants L normalize $T_w(K)$ by Kw_K , not by K^2 . Keeping that distinction produces the stopping-time factor

$$\frac{T_{w^{(\varepsilon)}}(K)}{T_{w^{(0)}}(K)} \rightarrow \frac{\pi^2}{4}, \quad \frac{\tau_{w^{(\varepsilon)}}(n)}{\tau_{w^{(0)}}(n)} \rightarrow \frac{2}{\pi} \quad (\varepsilon > 0). \quad (31.5)$$

31.2 Perturbations smaller than every inverse power

Theorem 31.2 (Identical algebraic expansions, different constants). *Let $A \subseteq \mathbb{N}$ have natural density $\theta \in [0, 1]$, and let $0 < h_j < 1$ with $h_j \rightarrow 0$. Define*

$$w_{2j-1} = j, \quad w_{2j} = j + h_j \mathbf{1}_A(j). \quad (31.6)$$

Then $w_k \sim k/2$ and

$$L_\theta = \frac{1}{2} \left[\frac{\Gamma(1 + \theta/2)}{\Gamma((1 + \theta)/2)} \right]^2, \quad \tau_w(n) \sim 2 \frac{\Gamma((1 + \theta)/2)}{\Gamma(1 + \theta/2)} \sqrt{n}. \quad (31.7)$$

In particular, choose $h_j = \varepsilon 2^{-j^2}$ with $0 < \varepsilon < 1$. Every schedule in this family satisfies

$$w_k - \lceil k/2 \rceil = o(k^{-N}) \quad \text{for every } N > 0, \quad (31.8)$$

while its constant ranges continuously and strictly increasingly over $[1/(2\pi), \pi/8]$ as θ ranges over $[0, 1]$.

Proof. At even stages, $\beta_{2j} = 2h_j \mathbf{1}_A(j) \rightarrow 0$, and activity is exactly $\mathbf{1}_A(j)$. At odd stages,

$$\beta_{2j+1} = (2j+1) \frac{1 - h_j \mathbf{1}_A(j)}{j + h_j \mathbf{1}_A(j)} \rightarrow 2.$$

The joint empirical law is therefore

$$\mu_\theta = \frac{1}{2} \delta_{(2,1)} + \frac{\theta}{2} \delta_{(0,1)} + \frac{1-\theta}{2} \delta_{(0,0)}.$$

Its drift is $D_\theta(z) = (C(2z) + \theta)/2$. Apply [corollary 30.2](#) with $r = 2, m = 1, a = \theta$, and then invert with $w_K \sim K/2$. The flatness assertion follows directly from the chosen h_j . Strict monotonicity follows from the next proposition, and continuity follows either from the Gamma formula or from [theorem 34.1](#) below. $\square$

An explicit set of every desired density is

$$A_\theta = \{j \geq 1 : \lfloor (j+1)\theta \rfloor - \lfloor j\theta \rfloor = 1\}. \quad (31.9)$$

The count telescopes to $\theta N + O(1)$. Thus the family requires no nonconstructive choice of a set of density θ .

Equation (31.8) gives a stronger obstruction than failure of ordinary regular-variation universality. Even knowing the full algebraic asymptotic expansions of the weights along both parity classes leaves the leading extinction constant undetermined. The missing datum is the occurrence of strictly positive perturbations at the repeated weights.

Proposition 31.3 (Quantitative dependence on activation). *Keep the positive-slope part of the drift fixed and write $D_\eta = D_0 + \eta$, where $D_0(0) > 0$ and the available dormant mass permits the stated values of η . Then*

$$\log \frac{L(\eta_1)}{L(\eta_0)} = (m+1) \int_0^\infty \frac{\eta_1 - \eta_0}{(z + D_0(z) + \eta_0)(z + D_0(z) + \eta_1)} dz, \quad (31.10)$$

$$\frac{d}{d\eta} \log L(\eta) = (m+1) \int_0^\infty \frac{dz}{(z + D_0(z) + \eta)^2} > 0, \quad (31.11)$$

$$\frac{d^2}{d\eta^2} \log L(\eta) = -2(m+1) \int_0^\infty \frac{dz}{(z + D_0(z) + \eta)^3} < 0. \quad (31.12)$$

Thus activation strictly increases L , with strictly concave logarithm.

Proof. Use $\log L = \lim_{R \rightarrow \infty} [\log R - (m+1)H(R)]$ for the two drifts and subtract. The difference integrand is integrable: the denominator is bounded away from zero near the origin and grows quadratically at infinity. This gives (31.10). The same bounds, locally uniform in η , permit differentiation under the integral and prove the last two formulas. $\square$

32 Periodic limiting slopes without a limiting constant

The activity variable is necessary for existence as well as for the value of a limit. We now remove the density assumption in [theorem 31.2](#) in a controlled way.

Theorem 32.1 (A fixed schedule with two subsequential constants). *There is a nondecreasing weight schedule with $w_k \sim k/2$, $\beta_{2j} \rightarrow 0$, and $\beta_{2j+1} \rightarrow 2$, for which $T_w(K)/(Kw_K)$ does not converge. Two of its subsequential limits are $1/(2\pi)$ and $\pi/8$.*

Proof. Let $N_h = 2^{2^h}$, for integers $h \geq 1$. Define a set A that is alternately empty and full on the integer blocks $[N_h, N_{h+1})$; its values before N_1 are arbitrary. Use (31.6) with $h_j = 1/(2j)$, and take horizons

$$K_h = 2(N_{h+1} - 1).$$

The weights and slopes have the stated asymptotics regardless of A , by the direct calculations in the previous proof.

Fix $\delta > 0$. Since $N_h/N_{h+1} \rightarrow 0$, all pairs whose stages lie in $[\delta K_h, K_h]$ eventually belong to the last block, apart from at most a bounded number of endpoints. Along the subsequence of

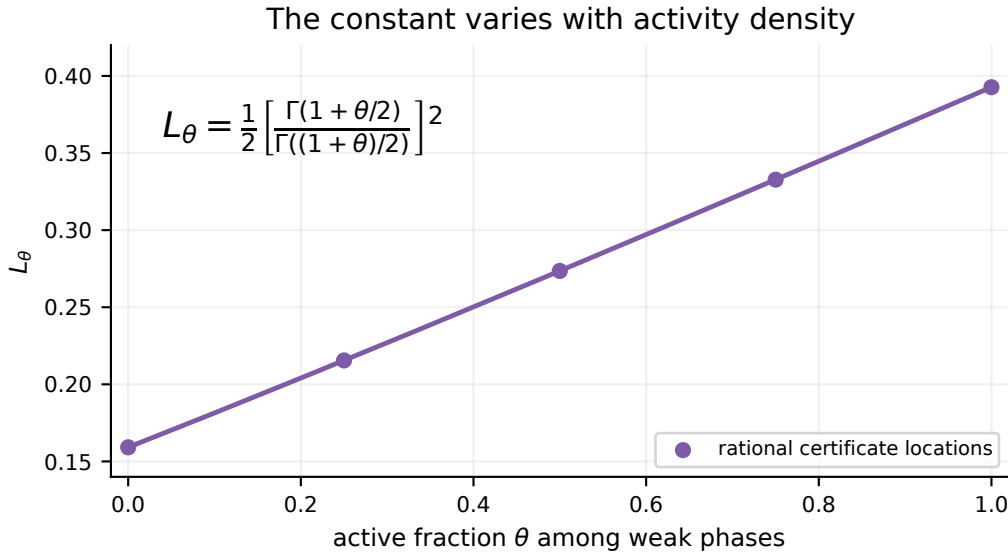


Figure 5: The exact Gamma family (31.7). The marked density values are also covered by finite rational certificates in the accompanying data. The curve comes from the theorem; the displayed numerical evaluations are rounded.

empty blocks, the joint empirical law on every macroscopic time interval is the plateau law $\mu_0 = \frac{1}{2}\delta_{(2,1)} + \frac{1}{2}\delta_{(0,0)}$. Along full blocks, it is the active-zero law $\mu_1 = \frac{1}{2}\delta_{(2,1)} + \frac{1}{2}\delta_{(0,1)}$.

The compactness and integrated-ceiling proof in section 28 uses only empirical averages within such macroscopic intervals. It therefore applies along each of these two horizon subsequences. Weight regular variation is already available directly from $w_k \sim k/2$. The initial-tail estimate (26.6) has, on either subsequence, normalized upper limit $\delta^2/2$. Sending $\delta \downarrow 0$ exactly as in the threshold matching proof yields

$$\frac{T_w(K_h)}{K_h w_{K_h}} \longrightarrow \begin{cases} 1/(2\pi), & \text{along empty terminal blocks,} \\ \pi/8, & \text{along full terminal blocks.} \end{cases}$$

The constants differ. Earlier blocks disappear from the fixed macroscopic interval and are controlled by the separate tail bound; they have not been silently replaced by a global density assumption. $\square$

The same proof works with $h_j = \varepsilon 2^{-j^2}$. Thus even weights agreeing with the repeated schedule to every algebraic order can lack a normalized threshold limit.

33 Where the lost mass goes

The average residual $E(z) = D(z) - mz$ already gives

$$-F'(t) = \frac{t^m}{L} E(z(t)) \quad \text{a.e. on } (0, 1). \quad (33.1)$$

It describes the limiting distribution of the time at which an infinitesimal fraction of the initial mass is removed. Keeping the local increment state reveals which kinds of steps cause that loss.

Theorem 33.1 (Joint loss time and increment state). *Under [theorem 27.2](#), let $J = \tau_w(n)$. The probability measures*

$$\nu_n = \sum_{k=2}^J \frac{x_{k-1} - x_k}{n} \delta_{(k/J, \beta_k, \sigma_k)} \quad (33.2)$$

converge weakly on $[0, 1] \times \mathcal{E}_B$ to

$$\boxed{\frac{t^m}{L} (\sigma C(\beta z(t)) - \beta z(t)) dt \mu(d\beta, d\sigma)}. \quad (33.3)$$

The limiting loss mass carried by state $(0, 1)$ equals

$$\frac{\eta}{(m+1)L}. \quad (33.4)$$

Proof. On a compact interval $[\delta, 1 - \delta]$, the positive forward quotients satisfy $Q_k/J \rightarrow y(k/J)$ uniformly. The integer $d'_k = Q_{k-1} - Q_k$ obeys, directly from the floor recurrence,

$$b_k Q_k \leq d'_k < b_k Q_k + 1 + b_k. \quad (33.5)$$

When $b_k = 0$, this forces $d'_k = 0$. When $b_k > 0$ and $b_k Q_k$ tends to zero, it forces $d'_k = 1$ for sufficiently large J . Away from small neighborhoods of positive integers in $\beta_k z(k/J)$, the only integer allowed by (33.5) is $\sigma_k C(\beta_k z(k/J))$. The joint averaging argument of [lemma 28.2](#) shows that the exceptional neighborhoods have vanishing limiting empirical mass as their widths tend to zero. All terms are uniformly bounded on the chosen compact interval.

The exact loss at a step is

$$x_{k-1} - x_k = w_{k-1} \left(d'_k - \beta_k \frac{Q_k}{k} \right).$$

Use $w_{k-1}/w_J \rightarrow (k/J)^m$ and $n/(Jw_J) \rightarrow L$, then apply joint averaging against any continuous test function. This gives (33.3) on the compact time interval. The total omitted early loss has limiting upper bound $1 - F(\delta)$, and the late loss has upper bound $F(1 - \delta)$, by uniform forward convergence. Both tend to zero when $\delta \downarrow 0$. Hence the full weak convergence follows.

The density is nonnegative and its total integral is one by (29.3). At state $(0, 1)$ its residual factor is exactly one; integrating $\eta t^m/L$ over time proves (33.4). $\square$

For a general empirical law, (33.4) refers to the atom in the *limiting* measure. It can be measured by first including active states with $0 < \beta_k \leq a$, taking the large- n limit at continuity values of a , and then sending $a \downarrow 0$. It does not assert that a finite-stage pair can equal $(0, 1)$. In finite-type examples with a separated positive slope, it is simply the fraction lost at the designated zero-slope active phases.

For (31.1) with $\varepsilon > 0$, $\eta = 1/2$, $m = 1$, and $L = \pi/8$. Therefore the even steps remove an asymptotic fraction

$$\frac{2}{\pi} = 0.6366197723 \dots \quad (33.6)$$

of the initial mass. Their increments in the weights tend to zero, yet their rounding losses remain a macroscopic portion of the total. At the relevant stages the input is close to a multiple of the previous weight; increasing that weight by any positive amount can force the integer quotient down by one. The theorem makes that mechanism quantitative without a random-fractional-part model.

The product measure used in [lemma 28.2](#) describes one stage sampled at a macroscopic time. It does not say that consecutive phases of a periodic or deterministic schedule are independent.

34 Continuity and bounded sparse exceptions

The instability above has a complementary positive result. The constant is continuous when the activity information is retained.

Theorem 34.1 (Continuity in the augmented increment law). *Let $\mu_j \Rightarrow \mu$ on a common compact state space $\mathcal{E}_B$, with $m = \int \beta d\mu > 0$. Write m_j, D_j, H_j, L_j, F_j for the corresponding quantities, restricting to the sufficiently large j for which $m_j > 0$. Then*

$$m_j \rightarrow m, \quad L_j \rightarrow L, \quad \sup_{0 \leq t \leq 1} |F_j(t) - F(t)| \rightarrow 0.$$

The backward profiles converge locally uniformly on $(0, 1]$.

Proof. The mean and activity are continuous functions of the law, so $m_j \rightarrow m > 0$ and $\rho_j \rightarrow \rho > 0$. For all but countably many $z > 0$, the bounded function $(\beta, \sigma) \mapsto \sigma C(\beta z)$ is μ -almost everywhere continuous. The exceptional z occur when a positive atom β of μ satisfies $\beta z \in \mathbb{N}$. Thus $D_j(z) \rightarrow D(z)$ for almost every z . Since $D_j(z) \geq \rho_j$, dominated convergence on bounded z -intervals yields locally uniform convergence $H_j \rightarrow H$. Strict monotonicity and continuity of the integrals give convergence of their inverses, and hence of y_j , on compact time intervals.

To pass from a fixed R to L_j , use the weaker form of (27.16) with lower endpoint $Re^{-(m_j+1)H_j(R)}$ and upper-to-lower ratio at most $1 + 1/((m_j + 1)R)$. The lower endpoint converges for fixed R . Let $R \rightarrow \infty$ after $j \rightarrow \infty$; this proves $L_j \rightarrow L$. Local convergence of F_j follows. Finally,

$$0 \leq 1 - F_j(t) \leq \frac{\rho_j}{(m_j + 1)L_j} t^{m_j+1},$$

which controls the initial endpoint uniformly. Continuity and monotonicity of the profiles control the terminal endpoint. $\square$

In particular, a positive slope tending to zero approaches the state $(0, 1)$. Its limiting drift contribution is one times its frequency. Assigning that limit to $(0, 0)$ changes the law, which explains the discontinuity of the unaugmented description.

Corollary 34.2 (Bounded changes on a density-zero set). *Suppose two nondecreasing schedules have bounded scaled increments and agree in their successive ratios outside a set $E \subseteq \mathbb{N}$ with $\#(E \cap [1, N]) = o(N)$. If the first augmented empirical law converges with positive mean, then the second has the same limiting law. Their constants L , backward profiles, and normalized forward profiles coincide.*

Proof. For a bounded continuous test function, the two empirical averages differ in absolute value by at most twice its supremum norm times $\#(E \cap [1, N])/(N - 1)$, which tends to zero. Apply theorem 27.2 to both schedules. $\square$

The normalization in this conclusion uses each schedule's own w_K . Sparse changes in ratios can accumulate into a nontrivial factor in the weights themselves. Thus the corollary asserts $T_w(K)/(Kw_K) \rightarrow L$ and $T_{\tilde{w}}(K)/(K\tilde{w}_K) \rightarrow L$; it does not assert $T_{\tilde{w}}(K)/T_w(K) \rightarrow 1$ without a corresponding comparison of the weights. Unbounded exceptional increments are outside the corollary's assumptions.

[Added 7 October 2026, batch 110: this answers the bounded form of Part I's Research question 5. If nondecreasing weights satisfy $k(w_k/w_{k-1} - 1) \rightarrow m$ outside a set of density zero and their scaled increments are bounded on that set, they agree in their successive ratios,

off that set, with nondecreasing weights whose ratios satisfy the hypothesis everywhere (put $w_k/w_{k-1} = 1 + m/k$ on the exceptional set), so their augmented law is $\delta_{(m,1)}$ and their constant is L_m . Part I's question whether a sparse set of very large jumps can change the constant is Research question 15 below, open.]

35 The sharp spectrum and equivalence of normalized shapes

The previous examples vary a constant while preserving the mean slope. The full possible range can also be determined.

Proposition 35.1 (Activity bound and scaling). *Every law in [theorem 27.2](#) satisfies*

$$L \geq \frac{\rho m^m}{(m+1)^{m+1}}. \quad (35.1)$$

For a finite-type drift D and any $0 < c \leq 1$, multiply every active frequency by c , every positive slope by $1/c$, and put the remaining mass in a dormant type. The resulting drift and constant satisfy

$$D_c(z) = cD(z/c), \quad t_c(z) = t(z/c), \quad y_c(t) = cy(t), \quad L_c = cL. \quad (35.2)$$

The mean slope and normalized forward shape remain unchanged.

Proof. The terminal selection inequality gives $y(t) \geq \rho(1-t)$. Since $L \geq t^m y(t)$, maximize $\rho t^m(1-t)$ at $t = m/(m+1)$ to obtain (35.1). The drift identity in (35.2) follows directly from the phase construction, including its active zero types. In the integral defining H_c , substitute $u = cv$ to get $H_c(z) = H(z/c)$. The remaining identities and invariance of $F = t^m y/L$ follow. $\square$

Theorem 35.2 (Every interior constant is attainable). *Fix $m > 0$. Among the nondecreasing positive real weight schedules with a convergent bounded augmented increment law of mean m , the set of normalized constants is exactly*

$$\{L\} = \left(0, \frac{1}{m+1}\right). \quad (35.3)$$

Every value can be realized by a finite-type schedule. It can also be realized with the additional normalization $w_K \sim K^m$, so that $T_w(K) \sim LK^{m+1}$. The finite bound on the increments may depend on the desired constant; a single bound prescribed for all schedules is not assumed.

Proof. The strict inclusion in the displayed interval is already proved in (27.11). To approach its upper endpoint, take the finite law with positive slope rm of frequency $1/r$ and active-zero mass $1 - 1/r$, for an integer $r \geq 1$. Its drift is

$$D_r(z) = 1 - \frac{1}{r} + \frac{1}{r}C(rmz),$$

and its residual lies between $1 - 1/r$ and 1. Therefore

$$\frac{1 - 1/r}{m+1} \leq L_r < \frac{1}{m+1}, \quad L_r \longrightarrow \frac{1}{m+1}. \quad (35.4)$$

Given any $0 < \ell < 1/(m+1)$, choose r with $L_r > \ell$ and set $c = \ell/L_r$. Apply [proposition 35.1](#). The new finite law has mean m and constant $cL_r = \ell$.

It remains to realize possibly irrational frequencies by actual weights with a controlled leading asymptotic. For a finite alphabet with frequencies p_s , let block h have length h . Assign each active symbol s exactly $\lfloor p_s h \rfloor$ positions in that block, and assign the remainder to the dormant symbol. Arrange the block in any fixed order. The counts satisfy

$$N_s(N) = p_s N + O(\sqrt{N}). \quad (35.5)$$

Indeed, the cumulative rounding error through block h is $O(h)$, and an incomplete final block has length $O(h)$, while the total length is $h(h+1)/2$.

At position $k \geq 2$, choose the ratio

$$\frac{w_k}{w_{k-1}} = \begin{cases} 1 + \alpha_s/k, & \text{a positive-slope type } s, \\ 1 + k^{-2}, & \text{an active zero-slope type,} \\ 1, & \text{a dormant type,} \end{cases} \quad w_1 = 1. \quad (35.6)$$

This realizes the required augmented law. Set $a_k = \alpha_s$ on positive types and $a_k = 0$ otherwise. By (35.5), $\sum_{k \leq N} (a_k - m) = O(\sqrt{N})$. Abel summation shows that $\sum_{k \leq N} (a_k - m)/k$ converges, with error $O(N^{-1/2})$. Also $\log(w_k/w_{k-1}) = a_k/k + O(k^{-2})$. Hence

$$w_N = AN^m(1 + O(N^{-1/2})) \quad (35.7)$$

for some $A > 0$.

To obtain $w_N \sim N^m$, choose N_0 so large that $A^{-1}w_{N_0} \geq 1$. Set the first $N_0 - 1$ weights equal to one and replace the tail by $A^{-1}w_k$ for $k \geq N_0$. The new sequence is nondecreasing, starts at one, and has the same successive ratios after one transition. Its augmented law and constant are unchanged. The one exceptional finite ratio has a finite scaled increment, so the boundedness hypothesis remains valid after increasing its bound. $\square$

This theorem concerns positive real weights, as does the real-power version of the original OEIS question. It does not assert the same spectrum under an additional requirement that every weight be an integer. The construction also uses dormant phases; it is not a claim about the smaller class in which every transition must be strictly increasing.

The approximating constants have an explicit form, useful for constructing the desired law:

$$L_r = \frac{1}{rm} \left[\frac{\Gamma((rm+1)/(m+1))}{\Gamma(rm/(m+1))} \right]^{m+1} = \frac{1}{m+1} \left(1 - \frac{1}{2r} + O_m(r^{-2}) \right). \quad (35.8)$$

The exact identity is [corollary 30.2](#) with $a = r - 1$. The expansion follows from the first correction in the standard Gamma-ratio expansion [10, §5.11(iii)]. The bounds (35.4) alone suffice for the spectrum proof.

Proposition 35.3 (When two normalized trajectories agree). *Consider two laws with the same mean $m > 0$. Their normalized forward profiles agree if and only if*

$$D_2(z) = cD_1(z/c) \quad \text{for almost every } z > 0, \quad c = L_2/L_1. \quad (35.9)$$

Proof. The drift relation gives the integral change of variables used in [proposition 35.1](#), so it implies equality of the normalized profiles. Conversely, equality of F_1 and F_2 gives $y_2 = cy_1$. Differentiating their locally absolutely continuous representatives and using the differential equation yields

$$D_2(cz_1(t)) = cD_1(z_1(t)) \quad \text{for a.e. } t.$$

The map $t \mapsto z_1(t)$ and its inverse are locally Lipschitz away from the endpoints, by (28.9) and the local drift bounds. They preserve null sets locally, so the identity transfers to almost every $z > 0$. The integral formulas are insensitive to changes at individual jump points. With the fixed ceiling convention used here, those values are determined by left continuity. $\square$

Theorem 35.4 (Exact recovery of every bounded law). *The drift D determines the entire augmented increment law μ on $\mathcal{E}_B$. In particular, let $\mathbf{m}(n)$ be the arithmetic Möbius function, characterized by $\sum_{d|n} \mathbf{m}(d) = \mathbf{1}_{\{n=1\}}$, and put $f(x) = \mu\{(\beta, \sigma) : \beta > x\}$ for $x > 0$. Then*

$$f(x) = \sum_{1 \leq n \leq B/x} \mathbf{m}(n) \left[D\left(\frac{1}{nx}\right) - \rho \right], \quad \rho = \lim_{z \downarrow 0} D(z). \quad (35.10)$$

Every sum is finite. The remaining masses are

$$\mu\{(0, 1)\} = \rho - \lim_{x \downarrow 0} f(x), \quad \mu\{(0, 0)\} = 1 - \rho. \quad (35.11)$$

Knowing D almost everywhere suffices, since its canonical representative is left-continuous on $(0, \infty)$.

Proof. For $u \geq 0$, the chosen convention gives the exact identity

$$C(u) = 1 + \sum_{j \geq 1} \mathbf{1}_{\{u > j\}}.$$

All positive slopes have activity flag one. Integrating and substituting $z = 1/x$ therefore gives

$$G(x) := D(1/x) - \rho = \sum_{j \geq 1} f(jx).$$

Because the slopes are bounded by B , terms with $jx \geq B$ vanish. The proposed inverse can consequently be rearranged as a finite sum:

$$\sum_{n \geq 1} \mathbf{m}(n) G(nx) = \sum_{n, j \geq 1} \mathbf{m}(n) f(jnx) = \sum_{k \geq 1} f(kx) \sum_{n|k} \mathbf{m}(n) = f(x).$$

This proves (35.10). The tail function f determines the positive-slope measure, including all its atoms. Its total mass is $\lim_{x \downarrow 0} f(x)$. The activity and total masses then give (35.11).

For each fixed $\beta > 0$, $C(\beta z)$ is left-continuous in $z > 0$, and for $\beta = 0$ it is constant. Bounded convergence on compact z -intervals proves left continuity of D . Two left-continuous functions agreeing almost everywhere agree everywhere on $(0, \infty)$, which proves the final assertion. $\square$

Thus an exact backward profile determines the law: its differential equation gives $D(z(t)) = -y'(t)$ almost everywhere, and the coordinate change used in [proposition 35.3](#) transfers that information to z -space. The schedule's ordering is absent from the law and is not recoverable. If only the normalized forward shape is known at a fixed m , [proposition 35.3](#) describes the remaining scaling ambiguity. Formula (35.10) addresses exact data; it is not an assertion of stable recovery from noisy observations.

36 Exact computations and numerical illustrations

The accompanying program checks the discrete definitions separately from the asymptotic calculations. It uses integer ceiling and floor division for all threshold and survival decisions. The integer recurrence is short enough to describe completely.

ε	$T_w(100000)$	$T_w(K)/(Kw_K)$	Predicted L
0	795767254	0.159153450800	0.159154943092
1/2	1962394026	0.392478805122	0.392699081699
1/100	1963448713	0.392689742598	0.392699081699
1/10000	1963518231	0.392703646200	0.392699081699

Table 7: Exact thresholds for (31.1) at $K = 100000$. Decimal ratios and constants are rounded numerical displays. The convergence for $\varepsilon = 1/2$ is visibly slower than for the smaller perturbations; a rate is not inferred.

36.1 Exact algorithms and conventions

If a successive ratio is the positive rational a_k/b_k , the backward update is

$$q \leftarrow \left\lfloor \frac{a_k q + b_k - 1}{b_k} \right\rfloor. \quad (36.1)$$

The forward quotient update is $Q \leftarrow \lfloor b_k Q / a_k \rfloor$. Both are exact integer operations. A single threshold calculation uses $K - 1$ updates; this operation count does not treat arbitrary-size integer arithmetic as unit cost.

For a periodic tuple $(\alpha_0, \dots, \alpha_{p-1})$, the implemented transition $k - 1 \rightarrow k$ uses $\alpha_{k \bmod p}$, for $k \geq 2$. Positive phases use ratio $1 + \alpha_{k \bmod p}/k$; an active zero phase uses $1 + k^{-2}$; a dormant phase uses one. Thus the tuple (1) in the numerical tables has $w_k = (k + 1)/2$, not $w_k = k$. It has the same limiting drift and $L = 1/\pi$, but its finite thresholds differ from A002491. No newly computed table is presented as an existing OEIS b-file.

For (31.1), if $\varepsilon = a/b$ in lowest terms, the exact ratios are

$$r_{2j} = \frac{bj^2 + a}{bj^2}, \quad r_{2j+1} = \frac{bj(j+1)}{bj^2 + a}. \quad (36.2)$$

The weights at the selected horizons are available directly, so no numerical product is needed to define that model.

36.2 Recorded verification scope

The completed run records the following results in `data/verification.json`:

- 3,200 exact boundary checks across ten models, for every horizon $K \leq 40$ and terminal target quotient 1, 2, 3, 4. The computed backward threshold reaches the terminal target; decreasing the initial integer by one does not.
- 2,847 exhaustive small-survival comparisons at $K \leq 15$, covering initial integers through slightly above each threshold.
- 310 exact integer thresholds, including the horizons 100, 1000, 10000, 100000.
- 22 finite rational certificates, including five activation densities and endpoints $z = 100$ and $z = 1000$.

All recorded checks passed. They verify implementations and finite identities on their stated ranges; they are not a substitute for the asymptotic proofs.

Positive slope tuple	$T_w(100000)$	Normalized value	Predicted L
(1)	1591627536	0.318322323977	0.318309886184
(1/2, 3/2)	1432132519	0.331907175562	0.331897673629
(1/2, 1, 3/2)	1708313875	0.329295202144	0.329276508811
(1/2, 3/2, 1)	1520900013	0.329280395971	0.329276508811

Table 8: Positive periodic phases. The last two rows have the same empirical law and therefore the same normalized limit, despite their different finite weights and thresholds. All displayed decimals are rounded.

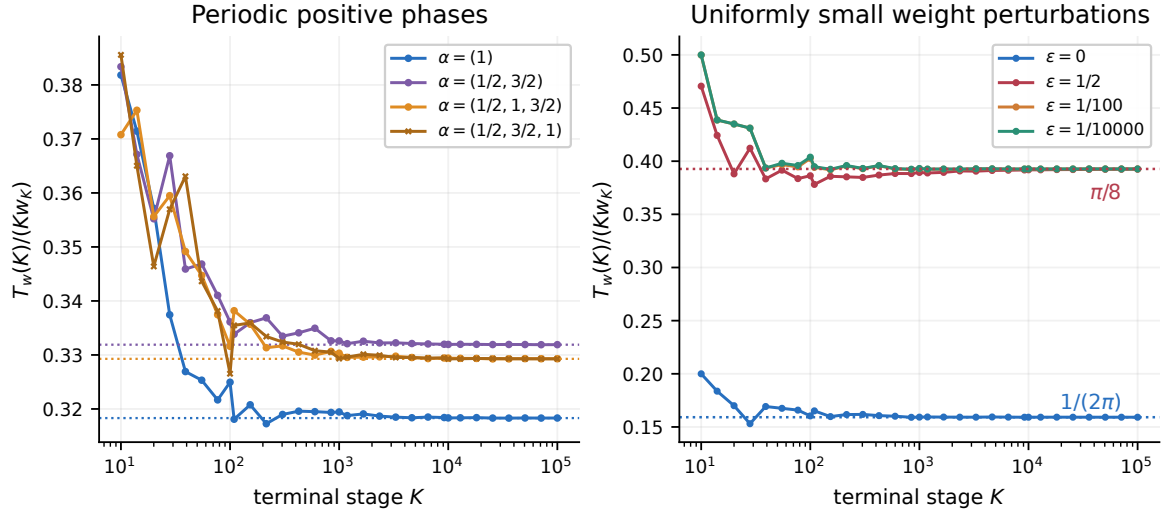


Figure 6: Exact thresholds divided by numerical evaluations of Kw_K . Horizontal reference values are proved constants. The finite-horizon curves are diagnostics of convergence, with no fitted error law asserted.

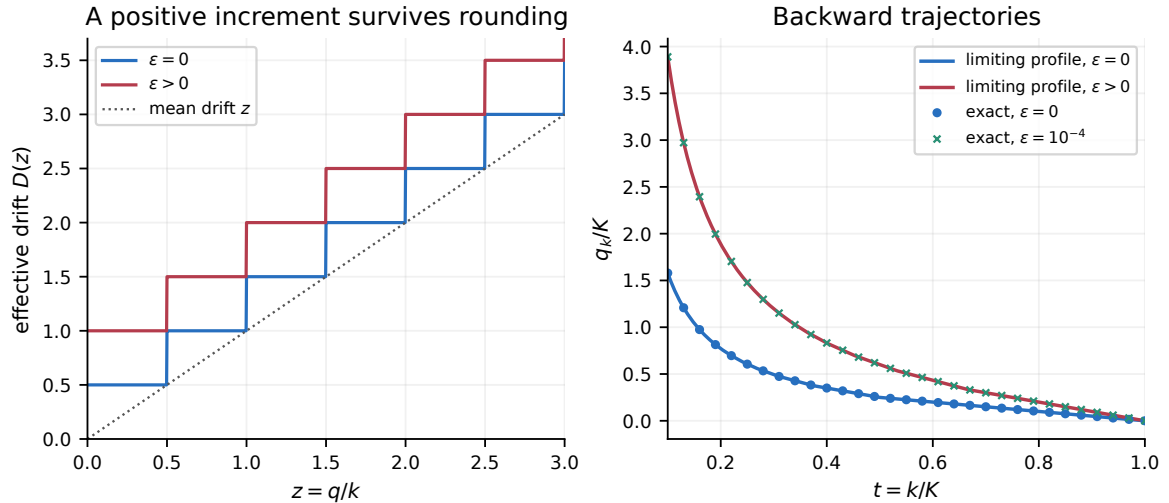


Figure 7: The ceiling drift and backward quotient profiles for the repeated-weight and active-zero examples. Markers use exact integer quotient arrays; smooth segments represent the explicit limiting solution. The difference between the limiting curves persists as the perturbation size tends to zero through positive values.

Drift	Lower endpoint	Upper endpoint
$C(z)$	0.318230318662	0.318389433821
$\frac{1}{2}C(z/2) + \frac{1}{2}C(3z/2)$	0.331814703827	0.331980611179
$\frac{1}{2}C(2z)$	0.159135049968	0.159174833730
$\frac{1}{2}(C(2z) + 1)$	0.392650009649	0.392748147617

Table 9: Rounded displays of exact rational finite-product certificate endpoints at $z = 1000$. These enclose the limiting constants, not the finite- K threshold ratios.

36.3 Certificates versus rounded numerical values

The certificate file stores the numerator and denominator of each endpoint in (27.16), evaluated through the finite product (30.2). These fractions are exact. The printed decimal approximations in table 9 are rounded displays of those endpoints, not claims of directed decimal rounding. In particular, inequalities should be checked against the stored fractions when every digit matters.

For the periodic weight products, the program uses 70-digit **Decimal** arithmetic. General Gamma-product displays use ordinary floating-point log-Gamma evaluations. Those evaluations are explicitly labeled as numerical diagnostics. The exact threshold recurrences and the exact rational certificate fractions do not depend on those floating-point values.

36.4 Reproducing the package

From the archive directory, run

```
python3 code/verify.py
```

to recreate the data and figures. Add `--no-figures` to run the arithmetic without NumPy or Matplotlib. The source and arithmetic use Python’s standard library; figure generation uses NumPy and Matplotlib. Compile the paper with

```
latexmk -pdf -interaction=nonstopmode periodic_rounding_extinction.tex
```

The archive includes the compiled PDF and every figure required by the LaTeX source, so reproduction of the figures is optional. The source audit and verification notes describe precisely which results were checked and which remain open.

[Added 7 October 2026, batch 110: this section describes the delivered archive. In this report the program is `code/03-schedules-verify.py`, its recorded outputs are `data/03-schedules-*` (the file named above as `data/verification.json` is `data/03-schedules-verification.json`), the figures are `figures/03-schedules-*`, and the source audit and verification notes are `03-schedules-SOURCE_AUDIT.txt` and `03-schedules-VERIFICATION_NOTES.txt`. The delivered PDF, the delivered `README.txt` and the checksum manifest are not shipped. The program writes `data/` and `figures/` under the directory given by `--output`, by default the directory above its own, with the delivered file names; run in place it would add unprefixed files to this report, so the report’s `README.md` runs it on a copy with an explicit output directory.]

37 Further research questions

The following questions are proposed continuations of this article. Except where an older conjecture is explicitly identified, their open status is relative to the results proved here and the sources examined, rather than a claim about every publication.

Research question 10 (Quantitative averaging and threshold errors). For a fixed finite periodic schedule with $\beta_k = \alpha_{k \bmod r} + O(k^{-1})$, obtain an explicit error bound in $T_w(K) = LKw_K + E(K)$ and in the full trajectory. How does the rate depend on the smallest positive slope and on the presence of active zero phases?

A proof must track the accumulated crossing errors and balance them against the singular initial tail. The empirical theorem provides no rate when the empirical law itself has no rate. For (31.1), the heuristic balance $\varepsilon q_k/k^2 \asymp 1$ with $q_k \asymp K^2/k$ suggests a scale $k \asymp \varepsilon^{1/3} K^{2/3}$. This is only a guide for a quantitative argument, not a proved boundary-layer estimate.

[Added 7 October 2026, batch 110: Part II proves such a bound in the one-type case of exact powers, $w_k = k^m$: $T_m(K) = L_m K^{m+1} + O_m(K^{(m+1)^2/(m+2)})$, with an explicit enclosure (theorems 12.1 and 17.1). Its drift sandwich (lemma 14.1) uses the power grid, so even general weights with the law $\delta_{(m,1)}$ are not covered, and every periodic schedule with more than one type stays open.]

Research question 11 (The classical bounded-error conjecture). Resolve $T_k(K) = K^2/\pi + O(K)$, equivalently $\tau_k(n) = \sqrt{\pi n} + O(1)$, and determine the appropriate counterparts for power or periodic schedules.

The proved classical error $O(K^{4/3})$ is the reliable benchmark [5]. OEIS A002491 records Knuth’s objection to the error summation in the later claimed sharpening [4]. The present article does not settle that issue. A recent preprint of Li studies the ordinary two-dimensional array with an $O((i+j+1)^{4/3})$ estimate [9]; it does not supply a periodic-weight error theorem used here.

[Added 7 October 2026, batch 110: this is Part I’s Research question 2 and Part II’s question 2; Part II’s remainder at $m = 1$ is the classical $O(K^{4/3})$. Open.]

Research question 12 (A fixed slope bound and strict increase). For prescribed m , B , and activity frequency ρ , determine the attainable interval of L under $0 \leq \beta_k \leq B$. What is the sharp range if every step is strictly increasing?

The spectrum theorem allows a different finite B for each construction, and its thinning argument introduces dormant steps. The lower bound (35.1) becomes relevant when ρ is fixed. Optimizing the ceiling average under a mean constraint is a distinct problem from optimizing an affine drift.

Research question 13 (Integer weights). Which constants can be realized when every w_k is a positive integer, with $w_k \sim Ak^m$? Which real-weight constructions survive a final rounding of the weights themselves?

The flat perturbation examples are deliberately real-valued. Rounding their weights to integers can erase the activity datum that causes the effect. Thus the integer restriction cannot be added to theorem 35.2 without another proof.

Research question 14 (Unbounded and signed local increments). Replace the uniform bound on β_k by an integrability or uniform-integrability condition, and determine whether a comparable law holds when some scaled increments are negative but the mean growth index is positive.

Rare large increments can be negligible in frequency and still affect the weighted trajectory. Negative phases remove the stepwise monotonicity used to prove transversality before averaging. Both modifications require new estimates; the current theorem does not cover them.

Research question 15 (Sparse large jumps). Find necessary or sufficient conditions on a density-zero exceptional set and its total logarithmic weight change that preserve the profile and constant when its scaled increments are not uniformly bounded.

The bounded case is [corollary 34.2](#). A successful extension must control both the exceptional quotient jumps and their contribution to the small-time weighted tail, rather than using density alone.

[Added 7 October 2026, batch 110: this is the part of Part I's Research question 5 that [corollary 34.2](#) leaves open. Together with Research questions 14 and 17, it is also what remains of Part I's Research question 4 outside the class of [theorem 27.2](#).]

Research question 16 (Quantitative recovery from approximate trajectories). Given noisy or finitely sampled observations of a backward profile, how accurately can its bounded augmented increment law be recovered? Under what separation or regularity assumptions are the positive slopes, their frequencies, and the active-zero mass stably identifiable?

[Theorem 35.4](#) settles exact identification from D for every bounded law in the article, and [proposition 35.3](#) identifies the scaling ambiguity between normalized profiles. Quantitative recovery is more delicate: obtaining D from a profile requires differentiation, and the Möbius formula uses increasingly many terms near zero slope. The scalar constant alone is insufficient; determining optimal bounds on a law from that single statistic is another question.

Research question 17 (Nonconvergent environments). For a schedule without a limiting augmented empirical law, characterize the cluster set of $T_w(K)/(Kw_K)$. How much of it is determined by empirical laws on macroscopic time windows, which may now depend on time?

The two-point construction in [theorem 32.1](#) isolates two homogeneous terminal environments. More general block schedules can have multiple environmental regions on the same horizon, suggesting a nonautonomous limiting equation whose coefficient law depends on t .

Research question 18 (Random schedules and fluctuations). For bounded random local increments, derive fluctuations of the threshold around its deterministic limit, under explicit mixing or independence assumptions. Separate environmental fluctuations from the arithmetic fluctuations caused by ceiling crossings.

The theorem is deterministic and applies to each realization with the required empirical convergence. It provides a leading law but does not establish a central limit theorem, a fluctuation scale, or independence of successive phases.

Research question 19 (Symbolic constants beyond commensurable slopes). For finitely many incommensurable positive slopes, study efficient certified evaluation of L , arithmetic properties of its defining product, and possible reductions to recognized special functions.

The integral and finite certificates remain valid, but there is no common cell for [theorem 30.1](#). The ordered union of distinct crossing sets replaces the finite Gamma product. This is an appropriate setting for symbolic and numerical investigation without asserting an unsupported closed form.

38 Proof dependencies and source audit

The main logical chain is exact survival duality, weight-ratio control, compactness with positive terminal drift, integrated ceiling averaging, a unique separated differential equation, and

small-time matching. The constant enclosures follow from the same differential equation, independently of finite-threshold experiments. The Gamma formulas additionally use the Gamma functional equation and its standard ratio asymptotic. The instability and spectrum results are explicit constructions within those proved hypotheses.

The original Erdős–Jabotinsky scans were checked for both authorship and the classical error statement [5]. The current OEIS entries were checked for their definitions and stated conjectures. In particular, the live A082528 entry still labels the general power-law statement as conjectural; the repository’s separate proof is not being represented as an accepted OEIS edit. The repository source identities used for the continuation are recorded in the accompanying `SOURCE_AUDIT.txt`. [Added 7 October 2026, batch 110: shipped as `03-schedules-SOURCE_AUDIT.txt`; the write confirmed its five blob identities (section 24.1). On 7 October 2026 the live A082528 entry still states the general- m law as a conjecture.]

The 2026 ordinary-array preprint [9] was inspected as related work. It repeats a stronger classical edge estimate in its historical discussion, while A002491 records the explicit Knuth objection to the earlier proof. We rely on neither that sharpening nor its repetition. The new averaging proof is self-contained, so no theorem of discontinuous differential equations is invoked without verification of its hypotheses.

Independent internal reviews checked the empirical-measure argument, the zero-slope distinction, the Gamma product, and the exact spectrum construction. Those reviews do not provide formal verification or external peer review. The computational claims are restricted to the ranges in section 36; asymptotic conclusions rely on proofs.

39 Notation reference

Table 10: The manuscript’s notation reference, as delivered
(caption added by the write).

Symbol	Meaning
w_k	Positive rounding weight at stage k , with $w_1 = 1$.
x_k	Value remaining after the stage- k rounding.
$\tau_w(n)$	First stage with zero value, from initial integer n .
$T_w(K)$	Least initial integer surviving through horizon K .
$q_k^{(K)}$	Least integer quotient at stage k that survives through K .
β_k	Scaled relative increment $k(w_k/w_{k-1} - 1)$.
σ_k	Indicator of a strictly increasing weight transition.
μ	Limiting empirical law of (β_k, σ_k) .
m, ρ, η	Mean slope, active frequency, and active-zero mass.
$C(u)$	Positive ceiling $\max(1, \lceil u \rceil)$.
$D(z)$	Average of $\sigma C(\beta z)$ with respect to μ .
$E(z)$	Average rounding residual $D(z) - mz$.
$H(z)$	Integral $\int_0^z (u + D(u))^{-1} du$.
$t(z)$	Inverse-profile time $e^{-H(z)}$.
$y(t)$	Backward quotient profile, with $y(t(z)) = zt(z)$.
L	Limit of $T_w(K)/(Kw_K)$, or of $zt(z)^{m+1}$.
$F(t)$	Normalized remaining mass $t^m y(t)/L$, with $F(0) = 1$.
ν_n	Probability measure recording removal time and increment state.

A Proof dependencies and source audit

A.1 Logical dependencies

The extinction proof proceeds through five independently checkable steps:

1. Exact stagewise survival duality gives the backward ceiling recurrence and the inverse threshold relation.
2. The integer increment bound and weighted rounding-error bound give compactness and force positive terminal drift.
3. Strict monotonicity of $my(t)/t$ controls the ceiling discontinuities, yielding a rigorous limit equation.
4. Solving its increment regimes gives the finite product; the beta integral evaluates its asymptotic constant.
5. A telescoping estimate controls the entire omitted initial part, allowing the two limits to be taken in the correct order.

The smooth-weight theorem reuses these steps after proving the local ratio and tail-sum consequences of its hypothesis. The forward-path theorem follows from a second use of the exact stagewise duality. No numerical estimate is an input to any of these proofs.

Two independent mathematical reviews checked the ceiling passage, initial-value convention, smooth-weight tail, inversion, profile endpoint argument, and block counterexample. The exact finite verification was generated by a separate implementation. These reviews improve confidence but are not a substitute for peer review or formal verification.

A.2 Sources reviewed and scope of the novelty statement

The principal source records are the OEIS definitions and conjectures for A073047, A082527, and A082528; the classical threshold entry A002491; Erdős and Jabotinsky’s original sieve paper; the Broline–Loeb preprint and the criticism attributed to Knuth in the threshold entry; Brown’s exposition; Lehericy’s 2023 proof for a different generalized rounding rule; and Li’s 2026 Tchoukaillon preprint. The bibliography provides direct links.

Targeted searches combined the exact sequence identifiers with power-rounding, asymptotic, Gamma, generalized sieve, and the classical authors’ names. Searches within ProveIt used the exact sequence identifiers and Tchoukaillon-related terms. No matching real-power theorem or formula for c_m was found in that review. This statement is intentionally bounded by the sources and search scope. A conjectural label remaining on an OEIS entry does not prove that no proof exists elsewhere.

The result should therefore be described precisely: this report supplies a complete proof of the real-parameter conjecture as recorded in A082528, identifies the square and cube constants, and proves the stated extensions. It makes no claim to have newly discovered the linear π law or the general increment-regime method. The numerical and symbolic evidence in the archive is reproducible and separately labeled.

B Suggested OEIS update text

The following concise mathematical statements are suitable for review before any submission. They have not been posted to OEIS.

[write] **Added 3 October 2026, batch 85.** This appendix is the manuscript's *draft* of OEIS update text, printed as delivered; it is not posted. Nothing has been submitted to OEIS: the package's `VERIFICATION_NOTES.txt` records that no OEIS updates were submitted, and the intake submitted nothing either. On 3 October 2026 A082528 still recorded the real- m law as a conjecture. Whether any of this text is ever proposed is a decision for a human editor after review of the proof; the repository does not submit it.

For A082527. The conjectured asymptotic constant is

$$c = 2\Gamma(2/3)^3 = 4.965917162430455473 \dots$$

Thus $a(n) \sim (2\Gamma(2/3)^3 n)^{1/3}$. This is the $m = 2$ case of the power-rounding extinction theorem.

For A082528. For every fixed real $m > 0$, if $x_1 = n$ and $x_k = k^m \lfloor x_{k-1}/k^m \rfloor$, the first zero satisfies

$$a(n, m) \sim \left(m\Gamma\left(\frac{m}{m+1}\right)^{m+1} n \right)^{1/(m+1)}.$$

In particular, the cubic constant is $3\Gamma(3/4)^4 = 6.764822981008760234 \dots$. A proof is given by the backward survival thresholds and their piecewise linear scaling limit.

For the linear cross-reference. The $m = 1$ specialization is the classical leading equivalent $a(n) \sim \sqrt{\pi n}$. It does not establish the stronger $O(1)$ error term in A073047.

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